Tetrahedra: puzzles and solutions

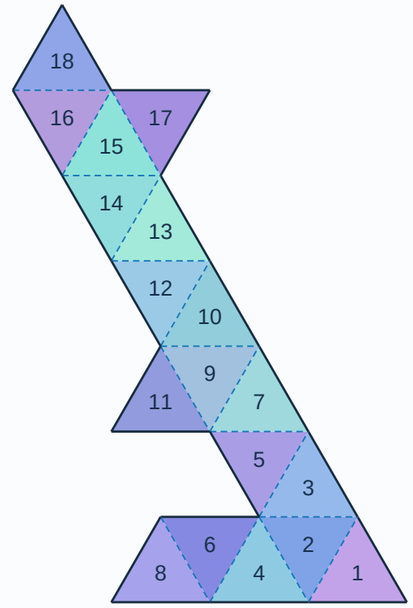
One through eight units | 11 puzzle nets | 523 catalog shapes

Try the puzzles. Check the solutions.

Part I - Puzzles. Fold each net into the targets shown. Cool net colors and independently randomized warm target colors give no panel-matching key.

Part II - Solutions. The same nets and targets, with matching panel colors.

Part III - Complete catalog. All 523 shapes, including shapes outside the selected puzzles. The table of contents and bookmarks link to every puzzle and solution.



A selected 18-triangle net for eight units

Cut solid black lines and any red internal lines. Fold blue dashed hinges. Only the Solutions section matches net colors to target faces. Downloadable SVGs and rotatable models accompany this PDF.

1 unit	1 shape	4 exterior triangles
2 units	1 shape	6 exterior triangles
3 units	1 shape	8 exterior triangles
4 units	3 shapes	10 exterior triangles
5 units	7 shapes	12 exterior triangles
6 units	23 shapes	14 exterior triangles
7 units	89 shapes	16 exterior triangles
8 units	398 shapes	18 exterior triangles

Seven- and eight-unit nets are useful sampled examples; no optimality is claimed. The five-unit cobra net is already known from Rick Mabry. These computations certify folded shapes, not collision-free folding motions. Templates omit tabs and thickness allowances.

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Checks and sources 73

The same shape ID appears in Puzzles, Solutions and the Catalog. The puzzle section intentionally hides the panel correspondence; the solution colors reveal it.

Part I - Puzzles

The same fixed cuts and hinges apply to every target shown for a net.

1 tetrahedra P1A: 1 targets

2 tetrahedra P2A: 1 targets

3 tetrahedra P3A: 1 targets

4 tetrahedra P4A: 3 targets

5 tetrahedra P5A: 7 targets

6 tetrahedra P6A: 19 targets; P6B: 14 targets

7 tetrahedra P7A: 40 targets; P7B: 36 targets

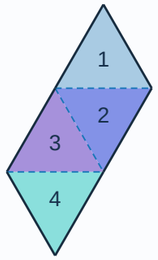
8 tetrahedra P8A: 101 targets; P8B: 98 targets

The first four nets are together on the next page. Each larger net begins with a large diagram and up to nine target shapes; continuation pages contain all its remaining targets. No unmatched shapes are mixed into a puzzle.

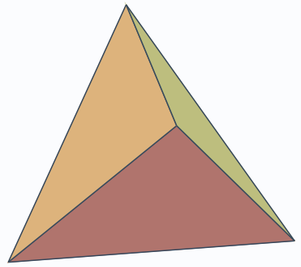
The cool net palette and warm target palette are randomized independently. Colors do not tell you which panel belongs on which face. Use the shape and hinges to solve the puzzle. Matching-color answers appear later in Solutions. Mirror pairs count as one shape; either paper side may face outward.

Puzzles - P1A-P4A

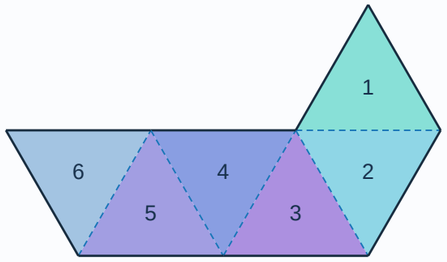
Cool nets / independent warm targets; no color key.



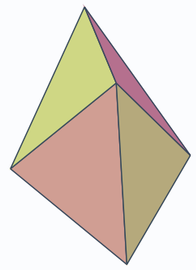
P1A - 1 unit



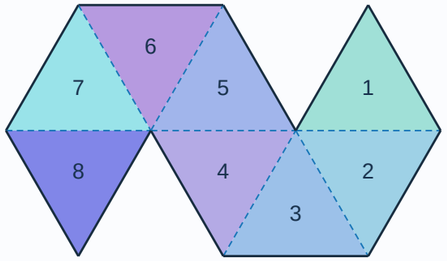
1-001



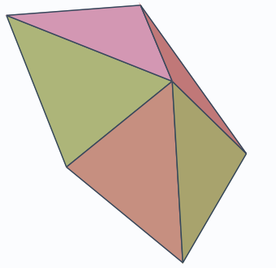
P2A - 2 units



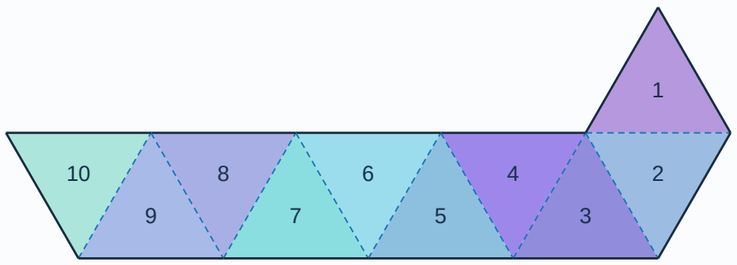
2-001



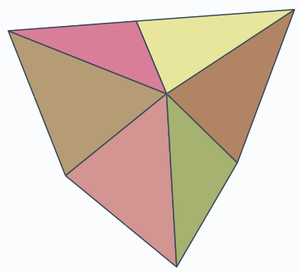
P3A - 3 units



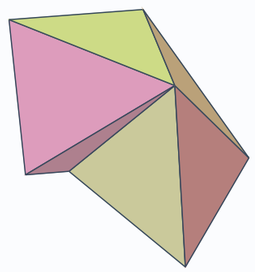
3-001



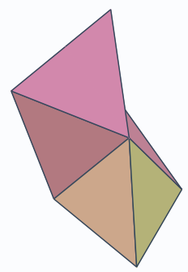
P4A - 4 units



4-001



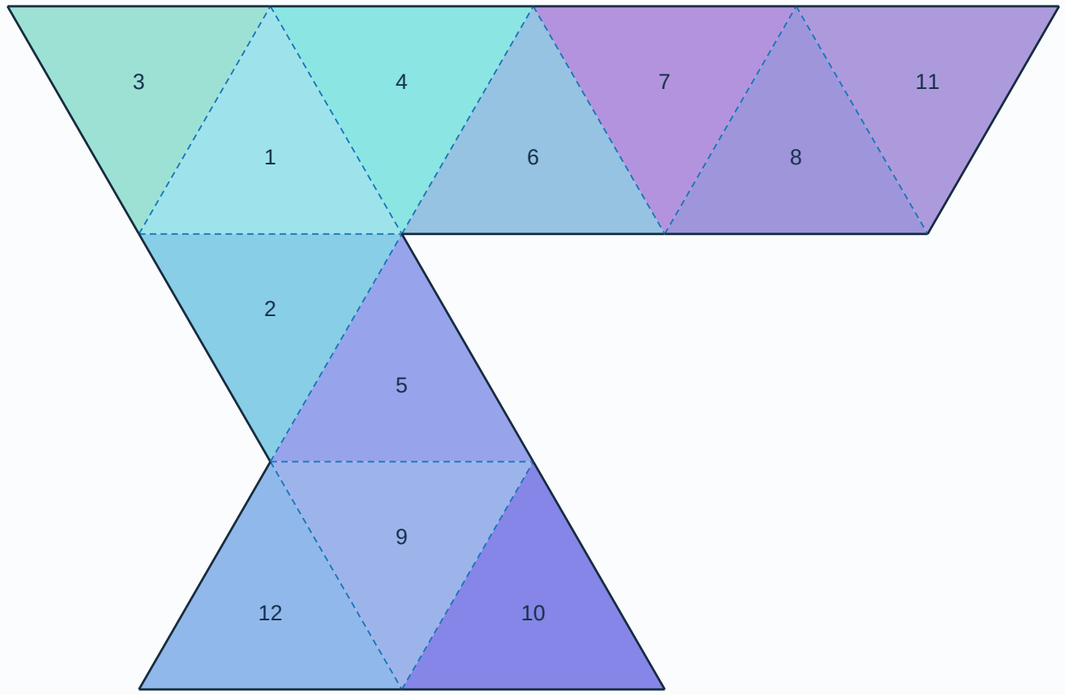
4-002

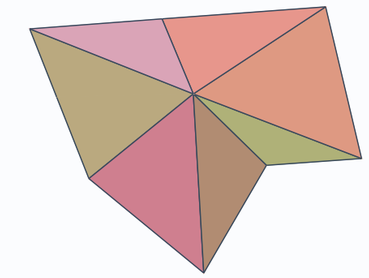


4-003

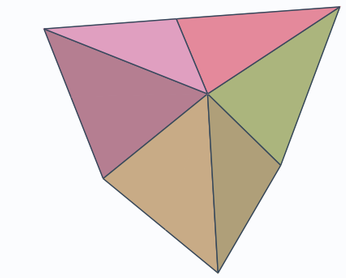
Puzzles P5A - 5 tetrahedra

7 matching shapes out of 7. Colors are independent; the folding correspondence is hidden.

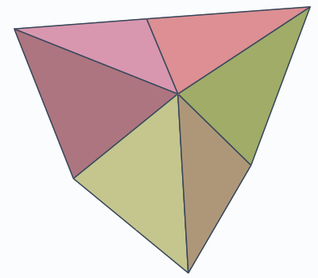




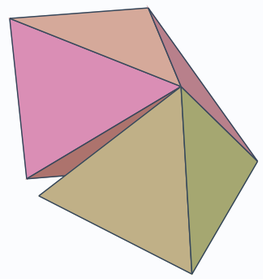
5-001



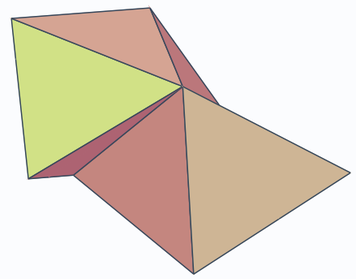
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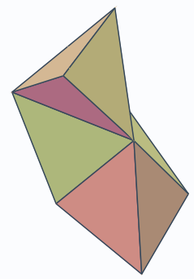
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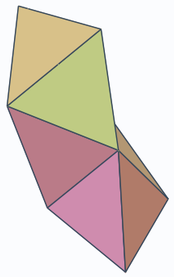
5-004



5-005



5-006



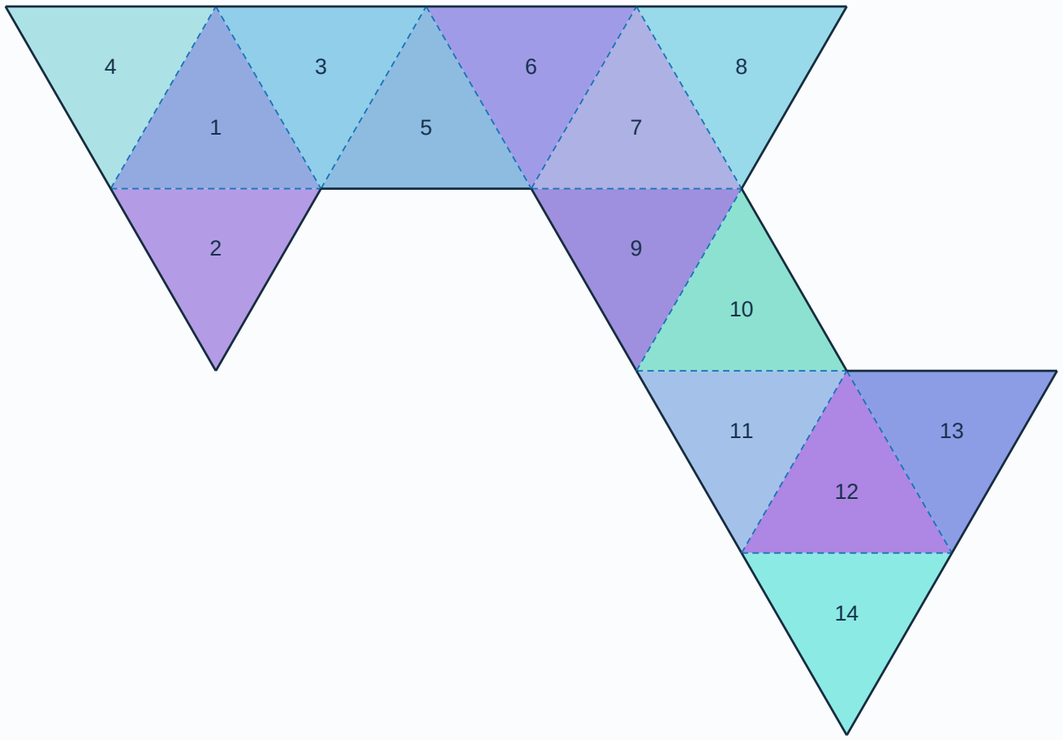
5-007

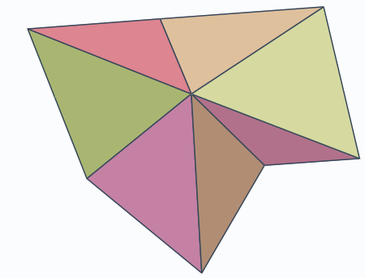
P5A 12 panels; 0 internal slits.

Known five-unit cobra net (Rick Mabry).

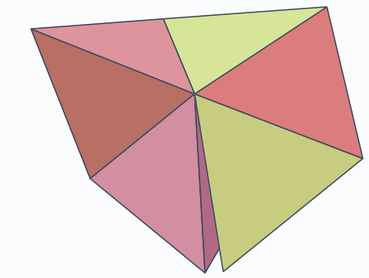
Puzzles P6A - 6 tetrahedra

19 matching shapes out of 23. Colors are independent; the folding correspondence is hidden.

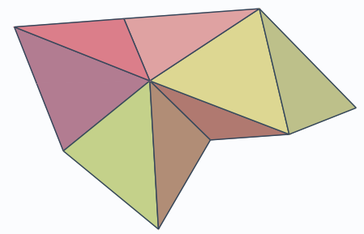




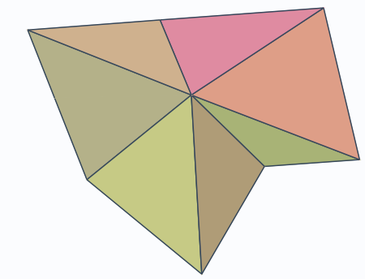
6-002



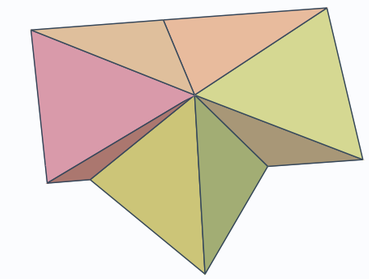
6-003



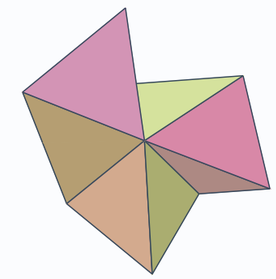
6-005



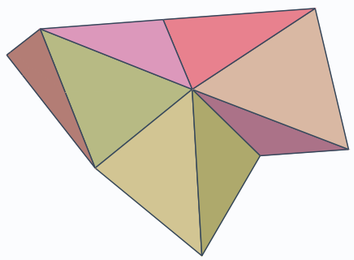
6-006



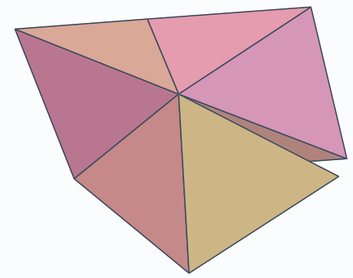
6-007



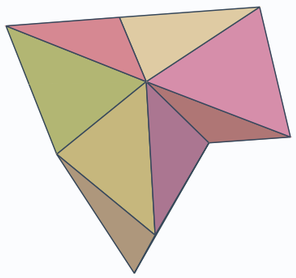
6-008



6-009



6-010



6-011

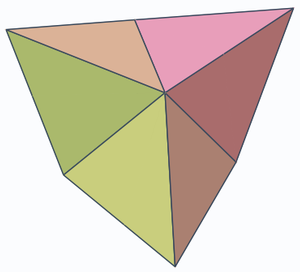
P6A 14 panels; 0 internal slits.

Six-unit Net A attains the checked 19-target maximum.

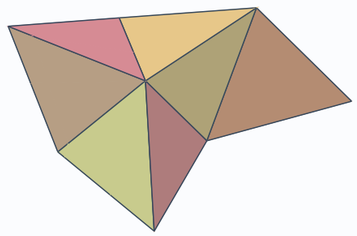
All remaining targets follow on the next page(s).

Puzzles P6A - 6 tetrahedra / targets continued

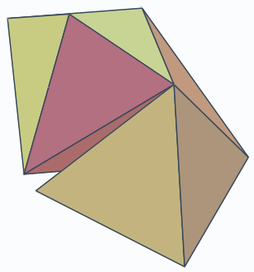
19 matching shapes out of 23. Colors are independent; the folding correspondence is hidden.



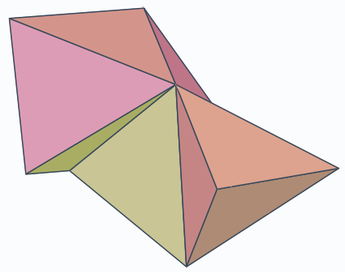
6-012



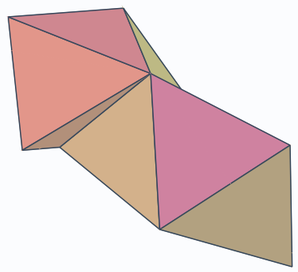
6-013



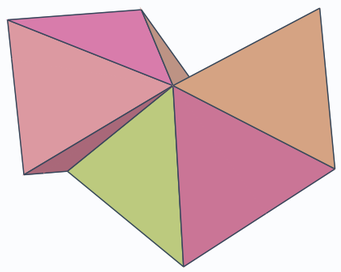
6-015



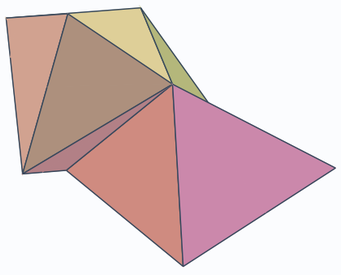
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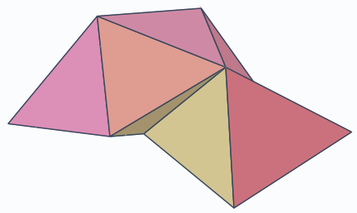
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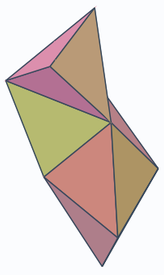
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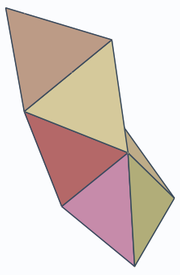
6-019



6-020



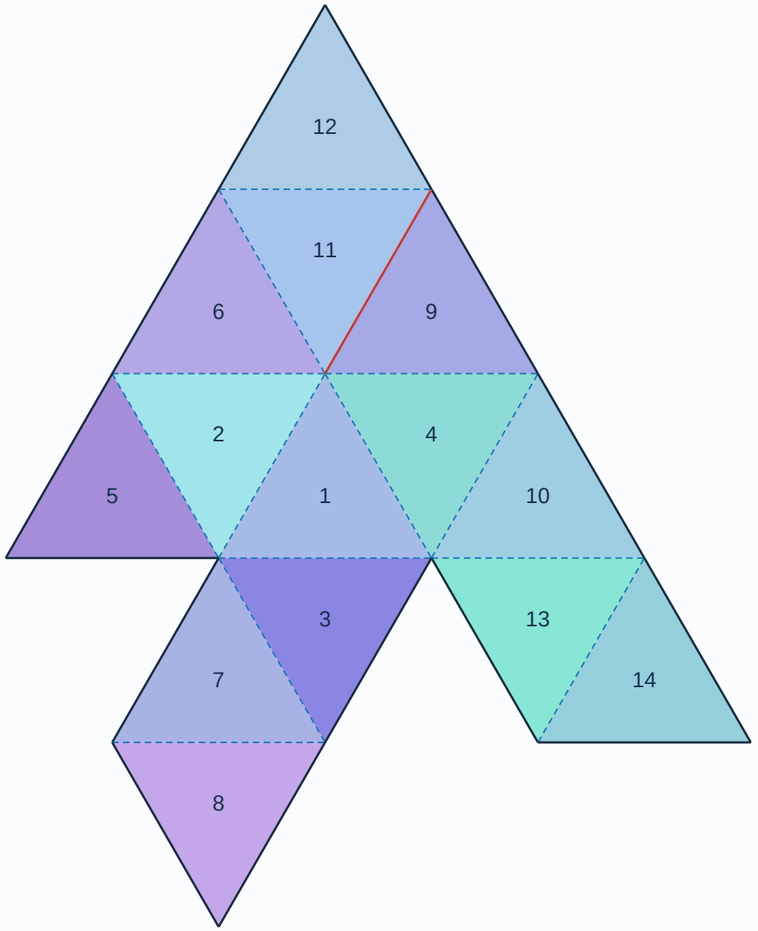
6-022

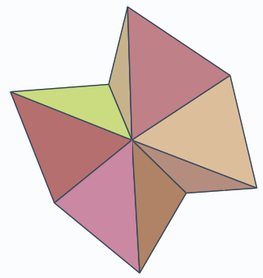


6-023

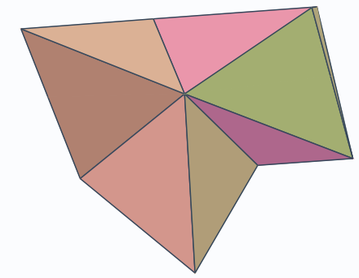
Puzzles P6B - 6 tetrahedra

14 matching shapes out of 23. Colors are independent; the folding correspondence is hidden.

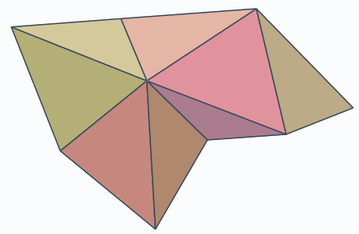




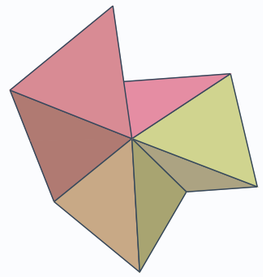
6-001



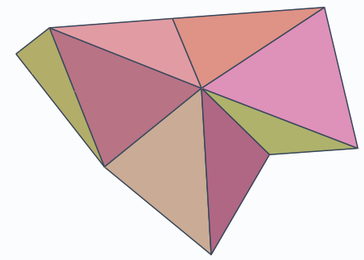
6-004



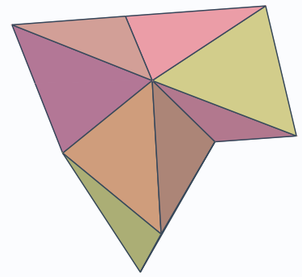
6-005



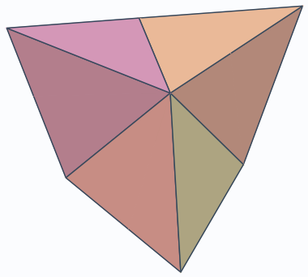
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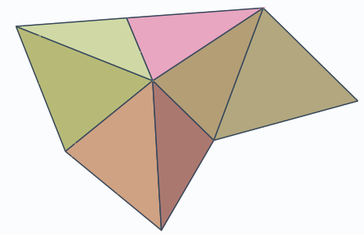
6-009



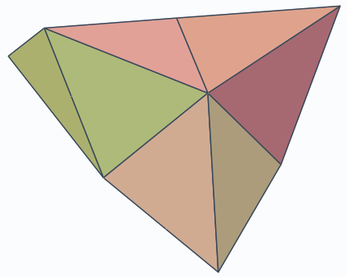
6-011



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6-013



6-014

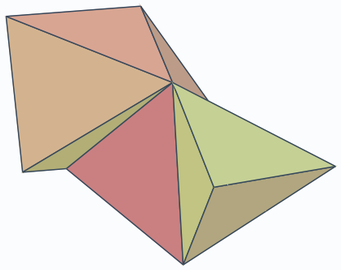
P6B 14 panels; 1 internal slit.

This companion supplies the four targets missing from P6A.

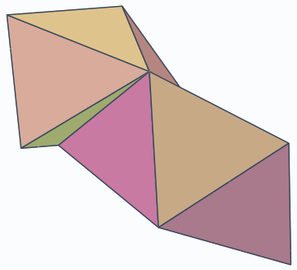
All remaining targets follow on the next page(s).

Puzzles P6B - 6 tetrahedra / targets continued

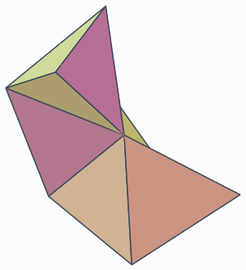
14 matching shapes out of 23. Colors are independent; the folding correspondence is hidden.



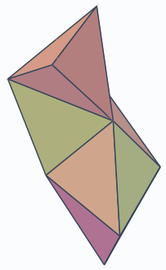
6-016



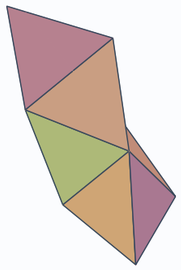
6-017



6-021



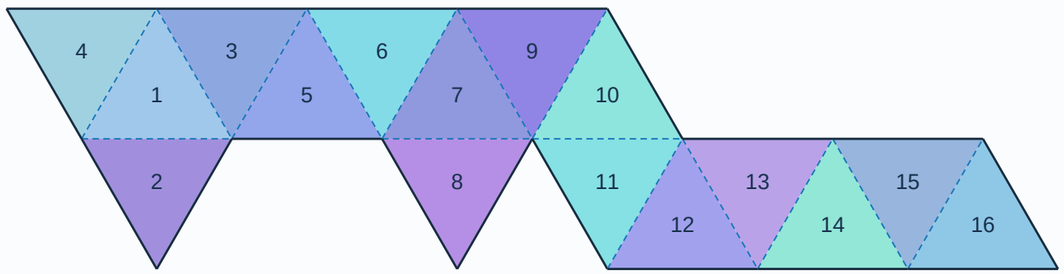
6-022

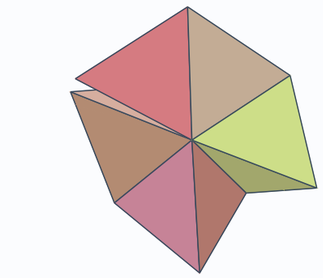


6-023

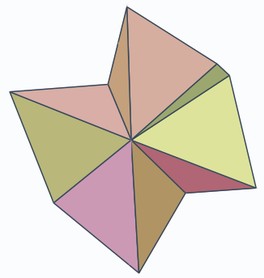
Puzzles P7A - 7 tetrahedra

40 matching shapes out of 89. Colors are independent; the folding correspondence is hidden.

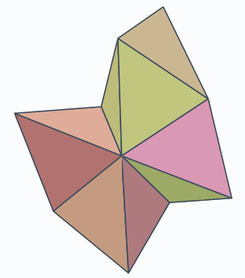




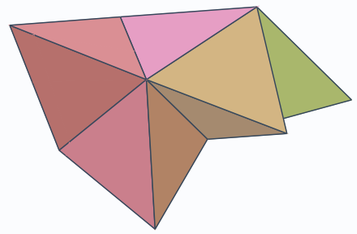
7-002



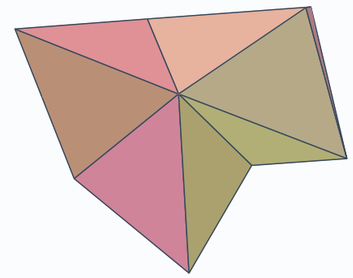
7-003



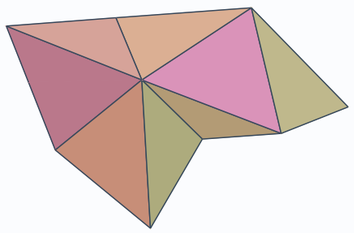
7-004



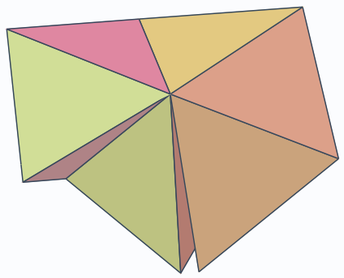
7-006



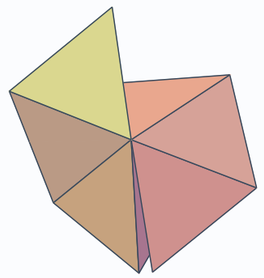
7-009



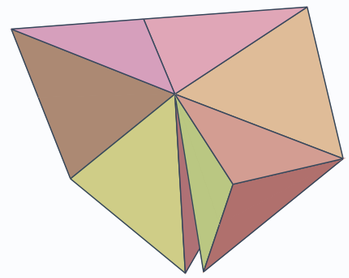
7-010



7-014



7-015



7-017

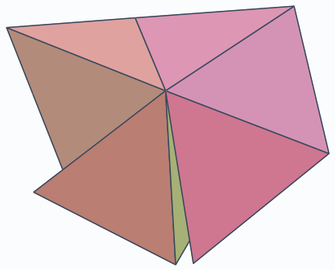
P7A 16 panels; 0 internal slits.

Selected from 500 sampled source trees; not proved optimal.

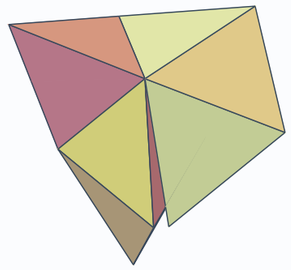
All remaining targets follow on the next page(s).

Puzzles P7A - 7 tetrahedra / targets continued

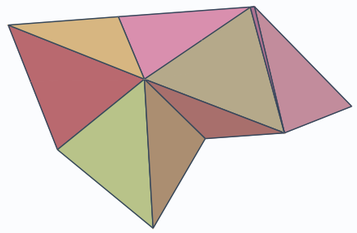
40 matching shapes out of 89. Colors are independent; the folding correspondence is hidden.



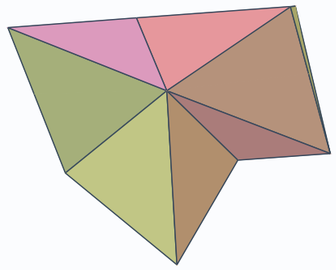
7-019



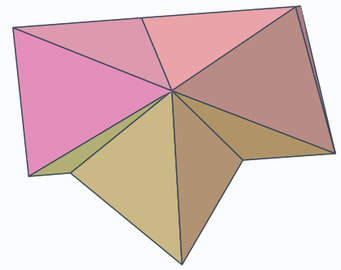
7-020



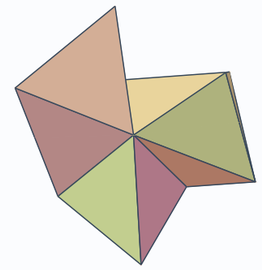
7-021



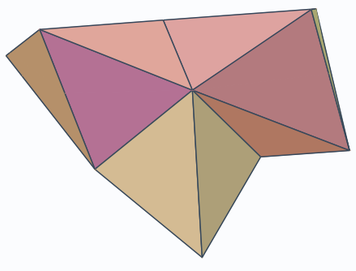
7-025



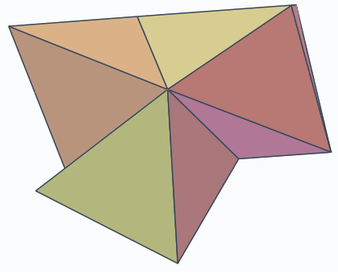
7-026



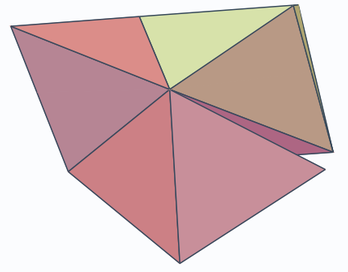
7-027



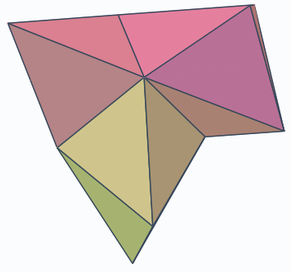
7-028



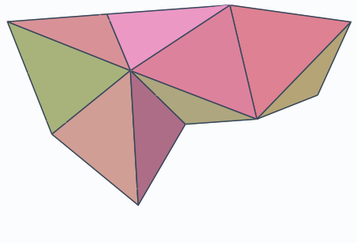
7-029



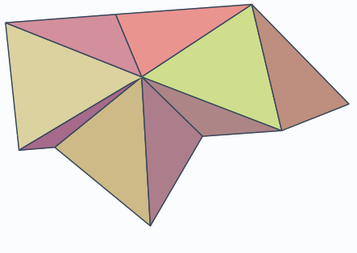
7-030



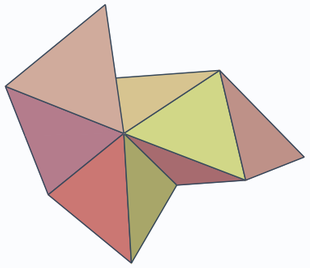
7-031



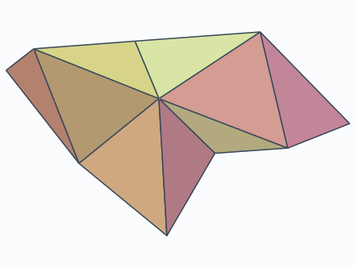
7-033



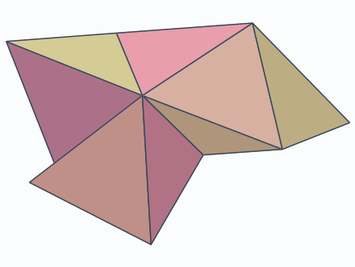
7-035



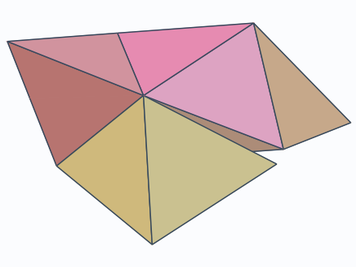
7-036



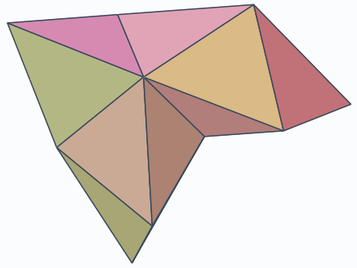
7-037



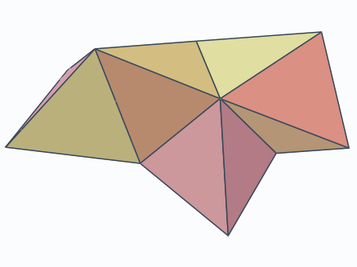
7-038



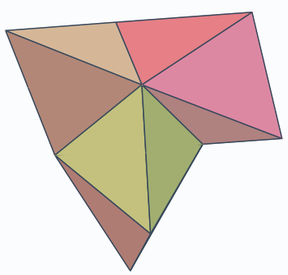
7-039



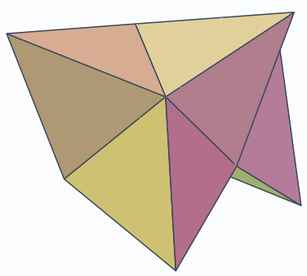
7-040



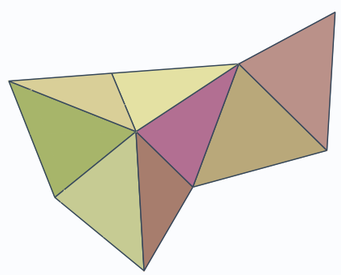
7-049



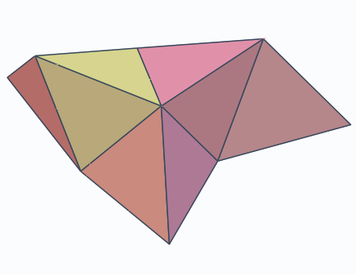
7-056



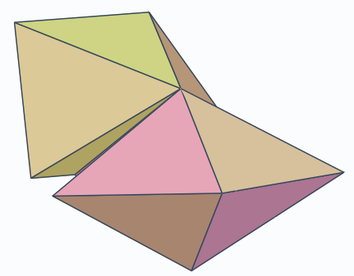
7-058



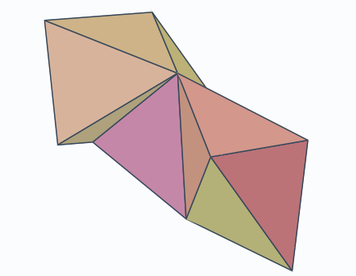
7-062



7-064



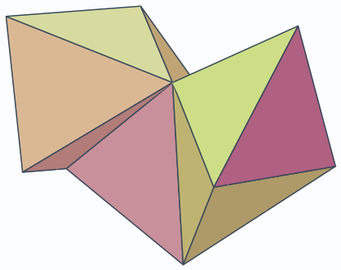
7-072



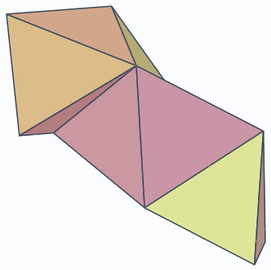
7-073

Puzzles P7A - 7 tetrahedra / targets continued

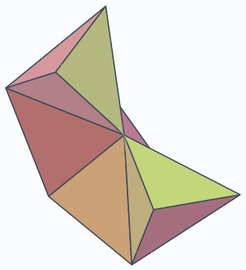
40 matching shapes out of 89. Colors are independent; the folding correspondence is hidden.



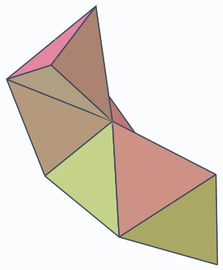
7-074



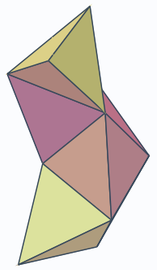
7-077



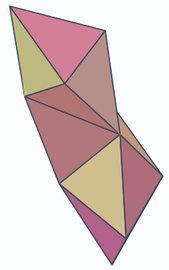
7-084



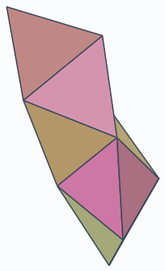
7-085



7-086



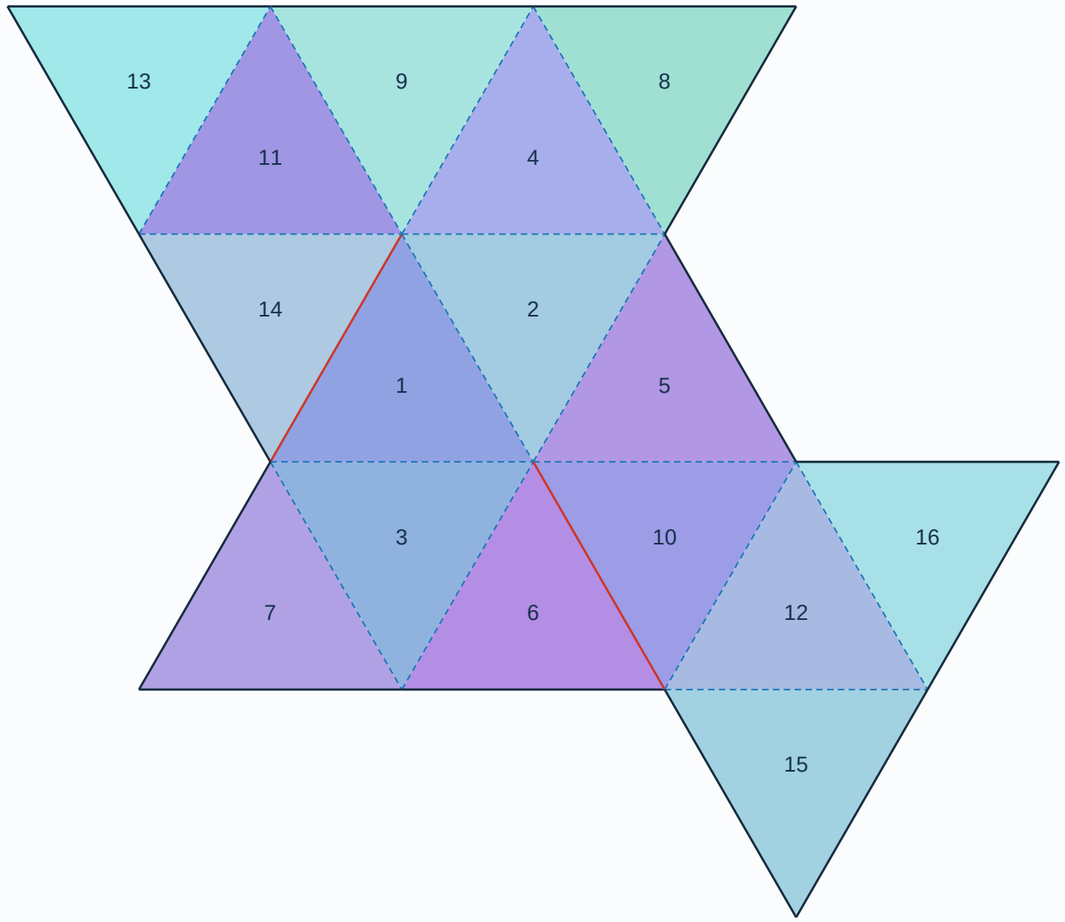
7-088

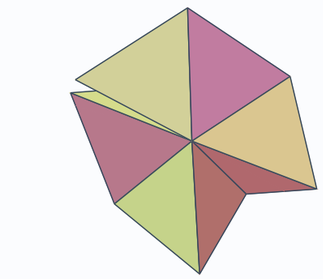


7-089

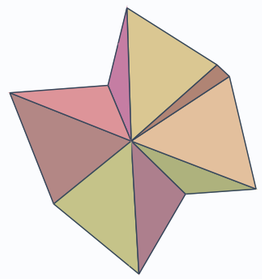
Puzzles P7B - 7 tetrahedra

36 matching shapes out of 89. Colors are independent; the folding correspondence is hidden.

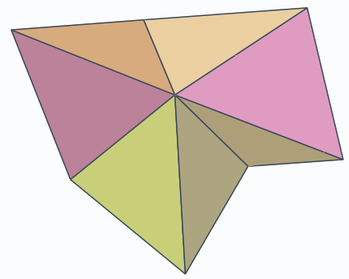




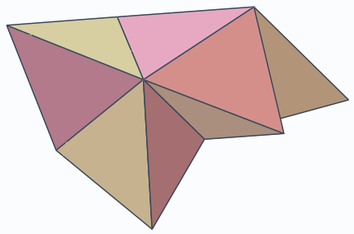
7-002



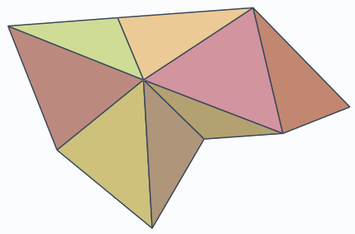
7-003



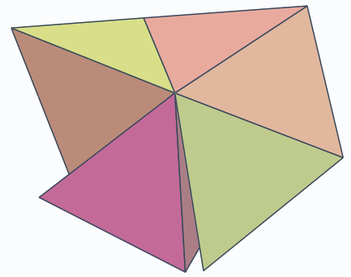
7-005



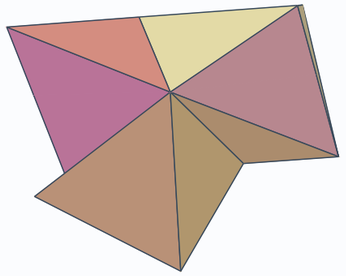
7-006



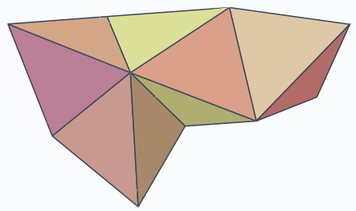
7-010



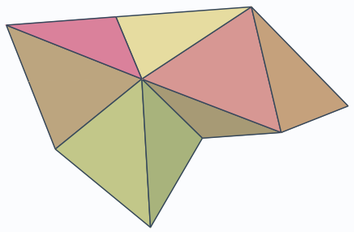
7-019



7-029



7-033



7-034

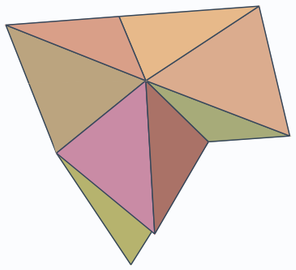
P7B 16 panels; 2 internal slits.

Selected from 500 sampled source trees; not proved optimal.

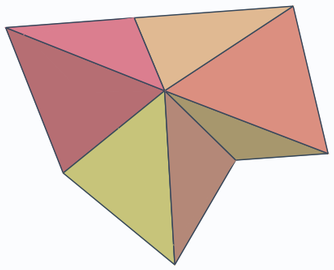
All remaining targets follow on the next page(s).

Puzzles P7B - 7 tetrahedra / targets continued

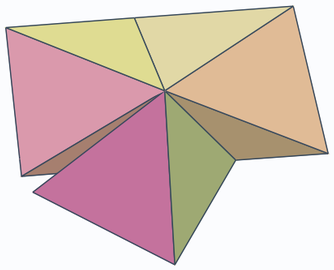
36 matching shapes out of 89. Colors are independent; the folding correspondence is hidden.



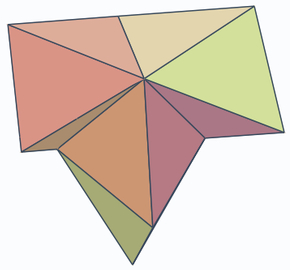
7-041



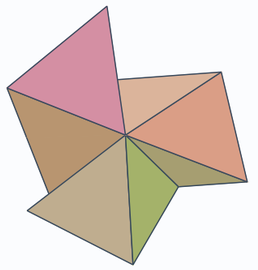
7-043



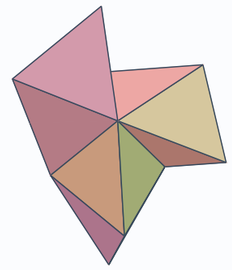
7-045



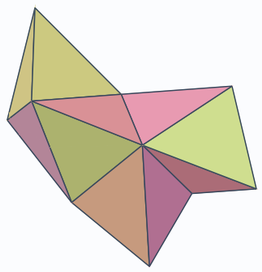
7-046



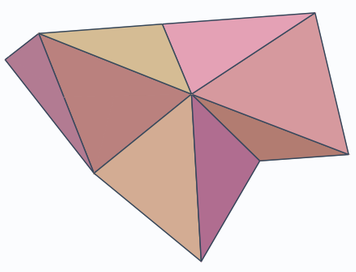
7-047



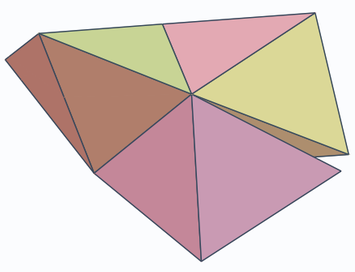
7-048



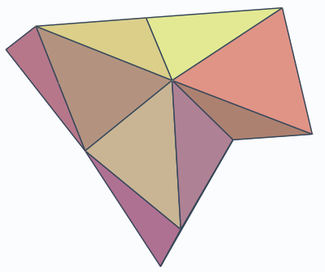
7-050



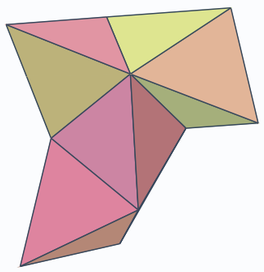
7-051



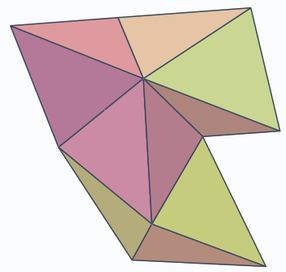
7-052



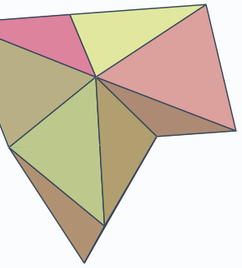
7-053



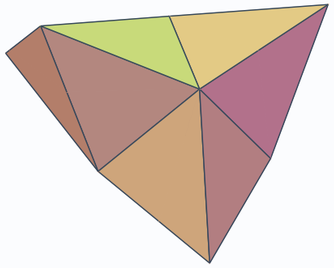
7-054



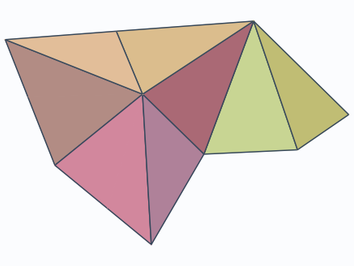
7-055



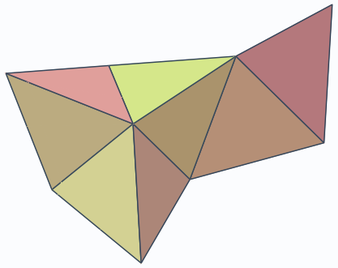
7-056



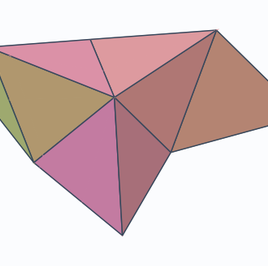
7-059



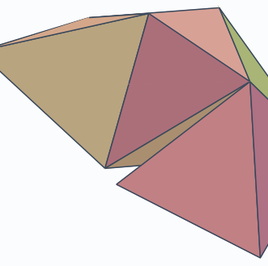
7-060



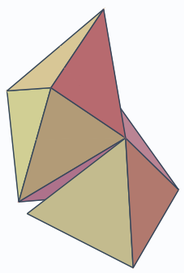
7-062



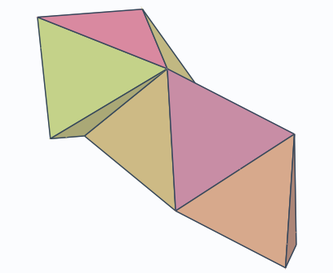
7-064



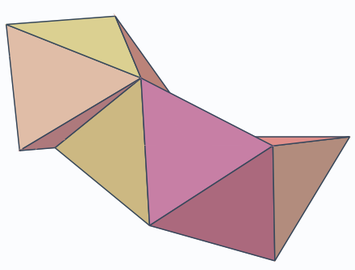
7-069



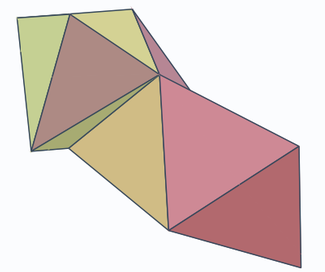
7-071



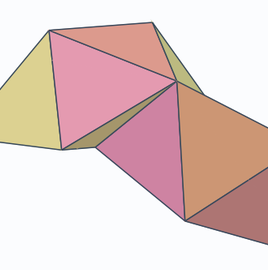
7-077



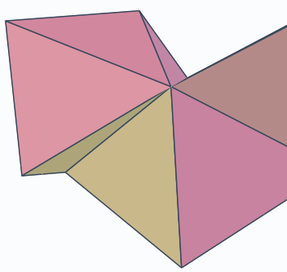
7-079



7-080



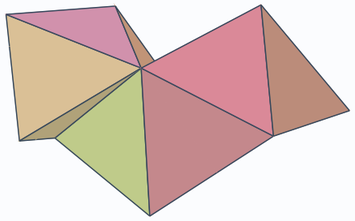
7-081



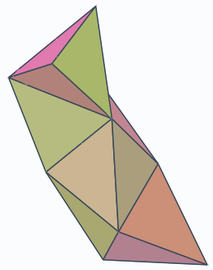
7-082

Puzzles P7B - 7 tetrahedra / targets continued

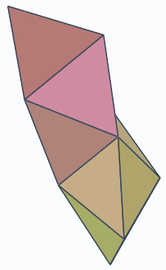
36 matching shapes out of 89. Colors are independent; the folding correspondence is hidden.



7-083



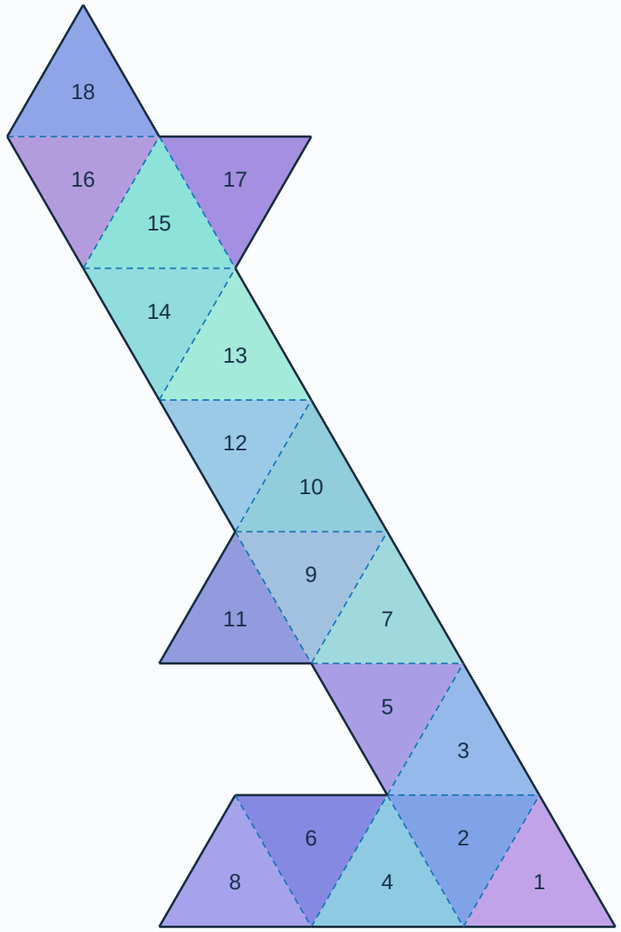
7-087



7-089

Puzzles P8A - 8 tetrahedra

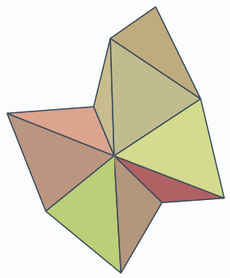
101 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



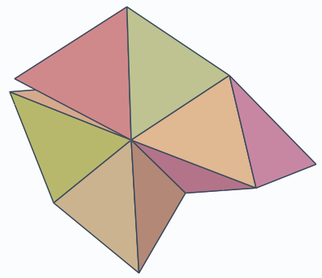
P8A 18 panels; 0 internal slits.

Selected from 500 sampled source trees; not proved optimal.

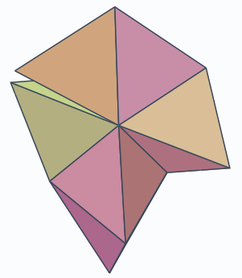
All remaining targets follow on the next page(s).



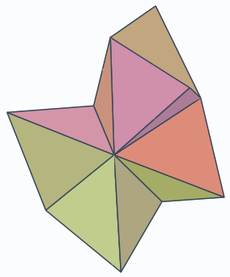
8-005



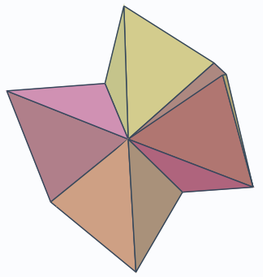
8-014



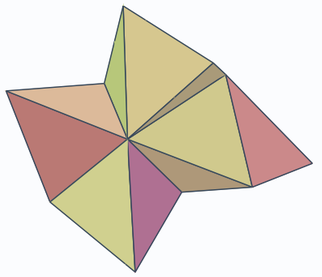
8-021



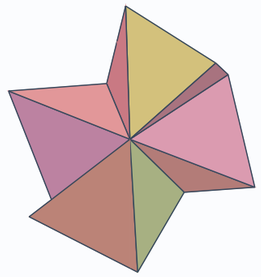
8-022



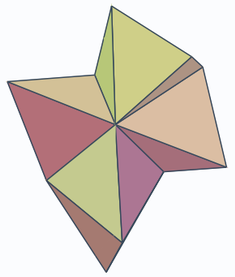
8-023



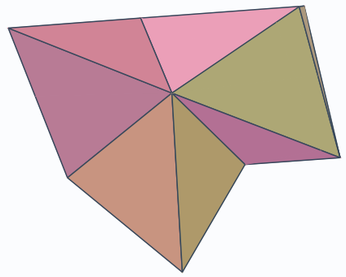
8-024



8-030



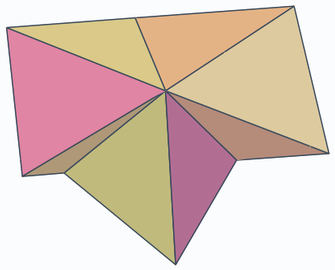
8-031



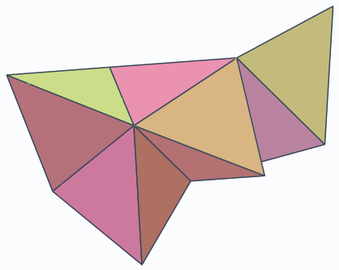
8-043

Puzzles P8A - 8 tetrahedra / targets continued

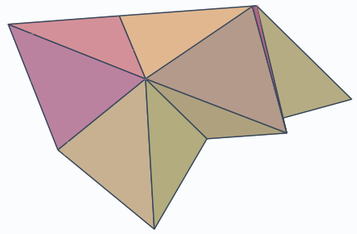
101 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



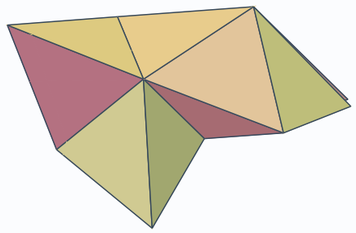
8-045



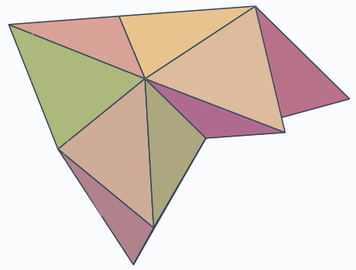
8-053



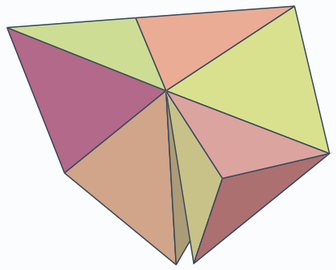
8-056



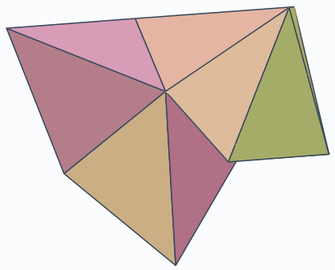
8-057



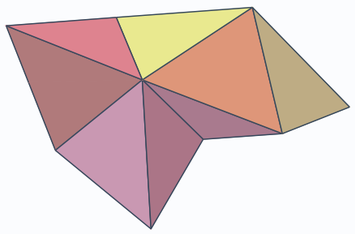
8-062



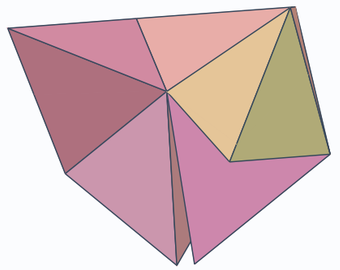
8-074



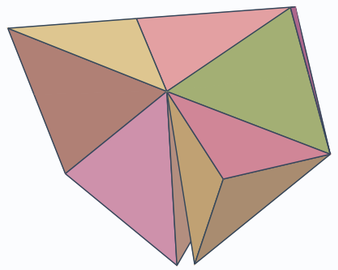
8-078



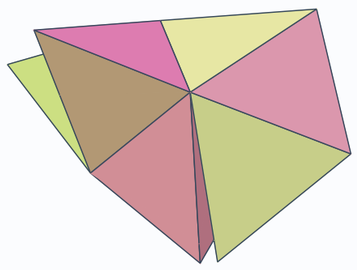
8-085



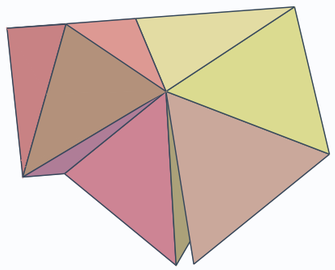
8-088



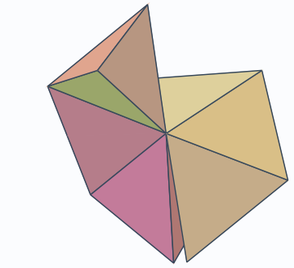
8-091



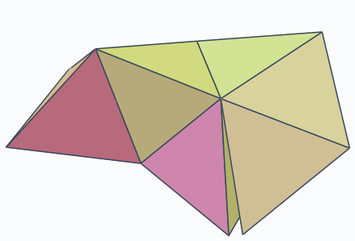
8-099



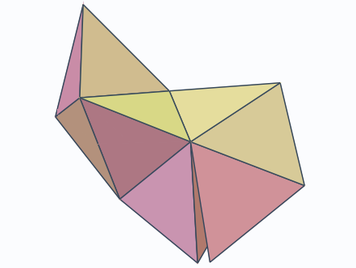
8-107



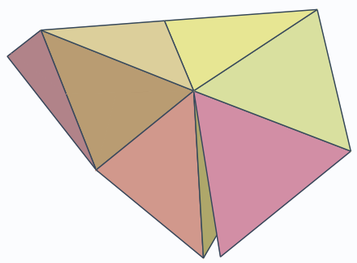
8-110



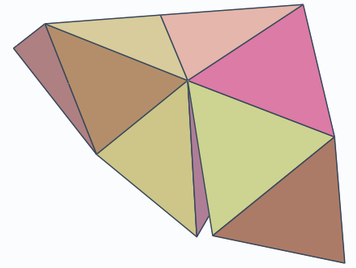
8-117



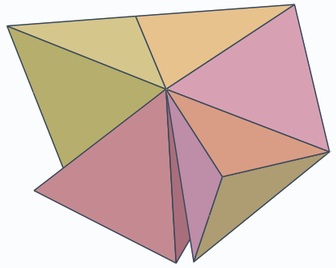
8-118



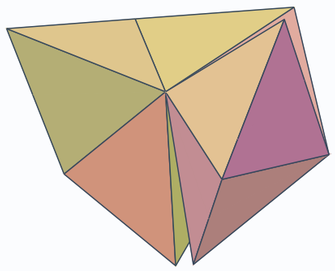
8-119



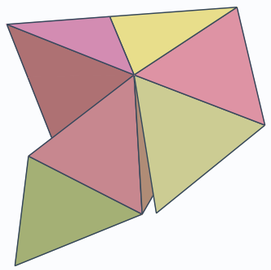
8-121



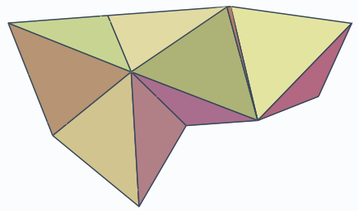
8-125



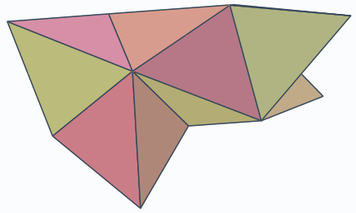
8-129



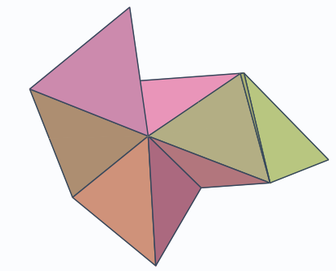
8-136



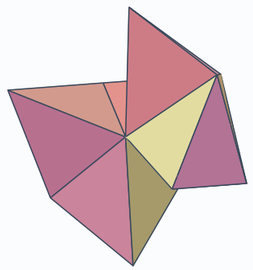
8-141



8-145



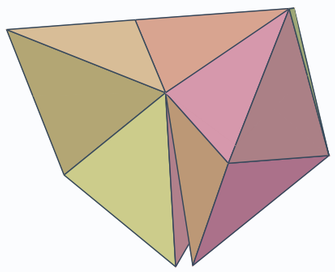
8-148



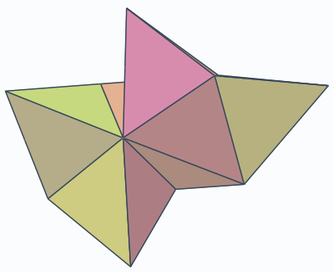
8-153

Puzzles P8A - 8 tetrahedra / targets continued

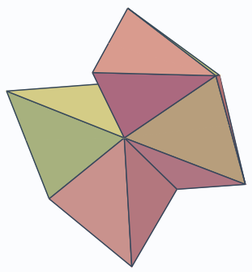
101 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



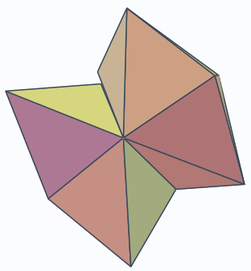
8-163



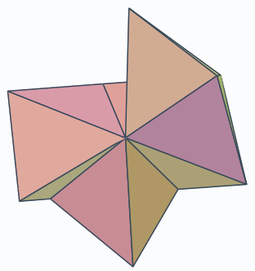
8-165



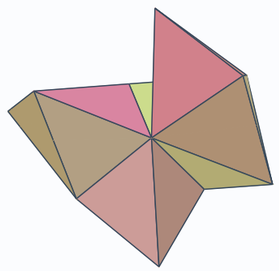
8-166



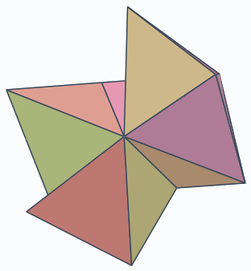
8-167



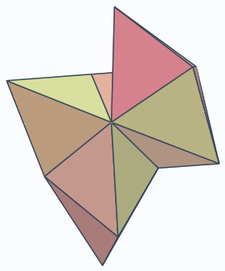
8-170



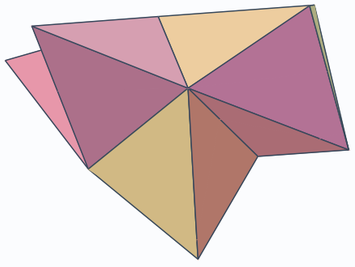
8-172



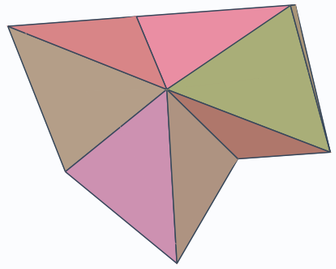
8-173



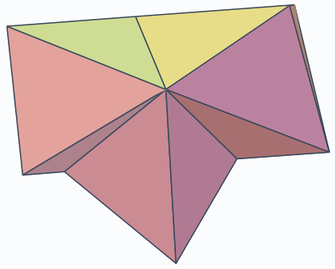
8-175



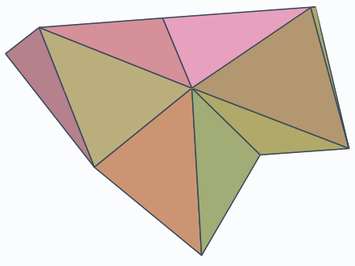
8-187



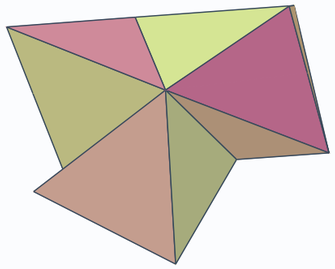
8-188



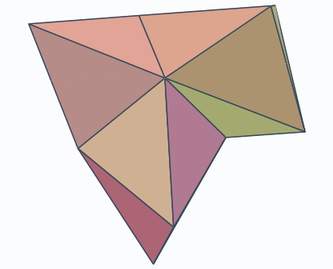
8-189



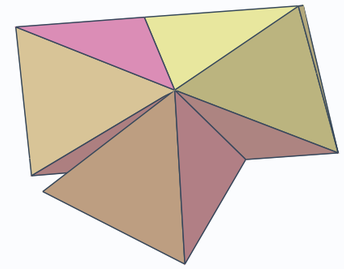
8-191



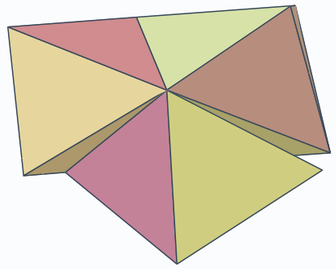
8-192



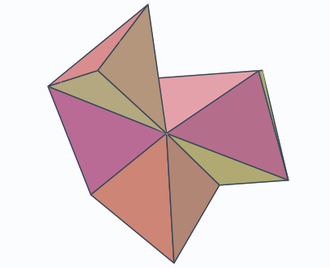
8-194



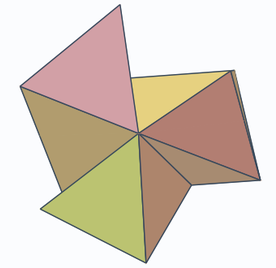
8-195



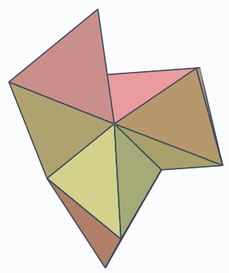
8-196



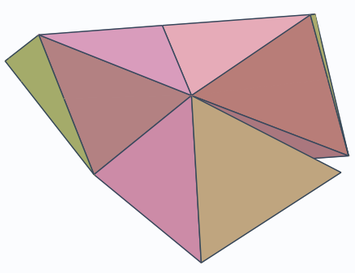
8-200



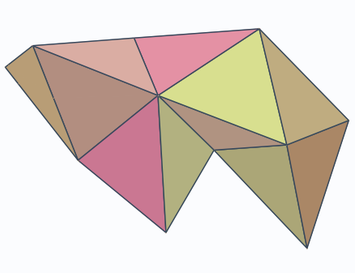
8-202



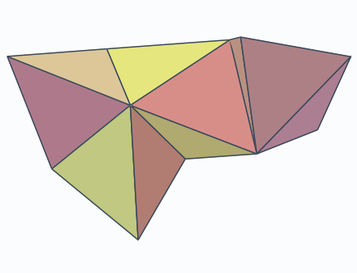
8-204



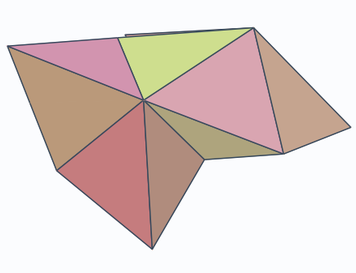
8-209



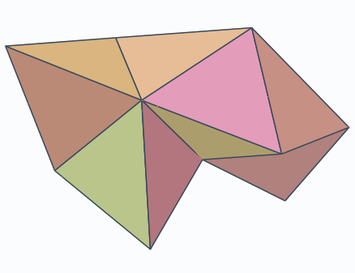
8-223



8-227



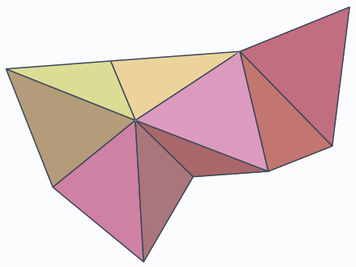
8-237



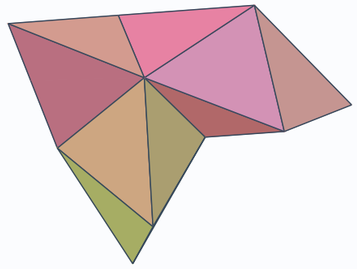
8-238

Puzzles P8A - 8 tetrahedra / targets continued

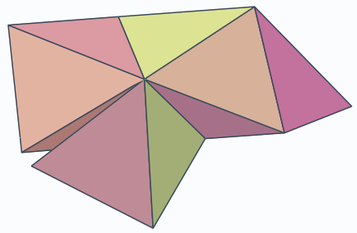
101 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



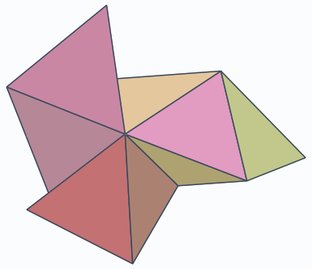
8-239



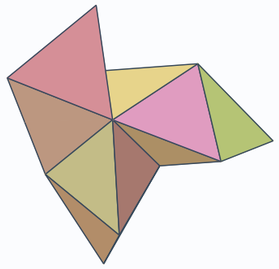
8-245



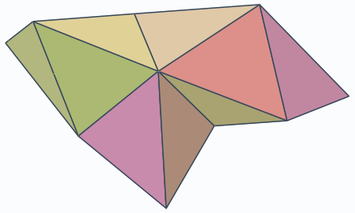
8-246



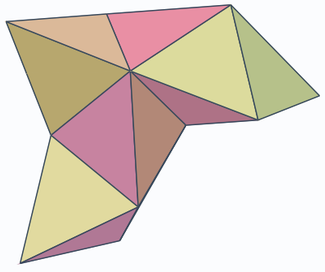
8-251



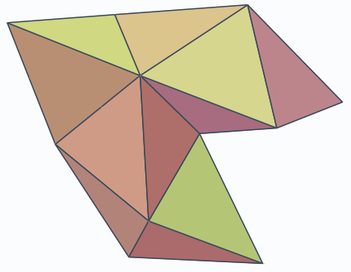
8-253



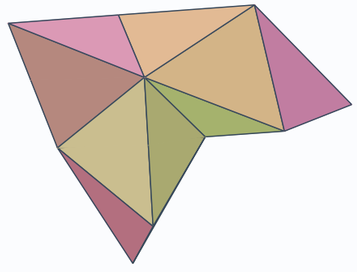
8-256



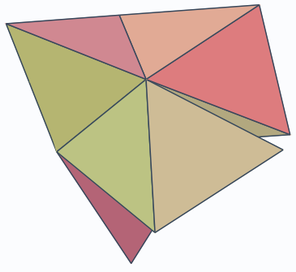
8-261



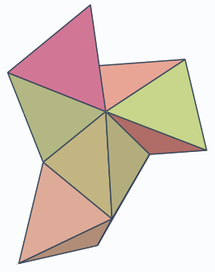
8-262



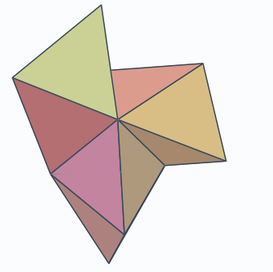
8-263



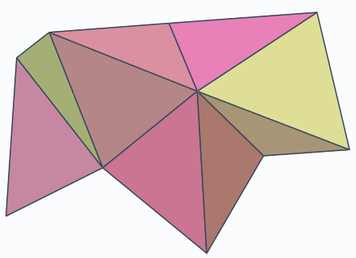
8-267



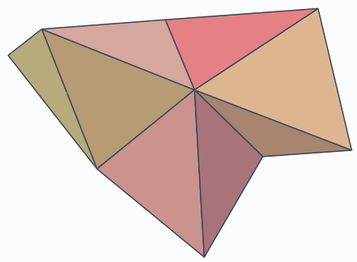
8-273



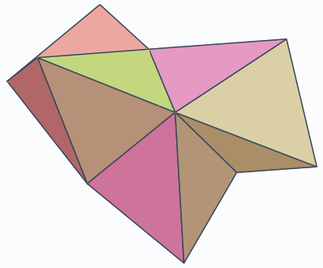
8-275



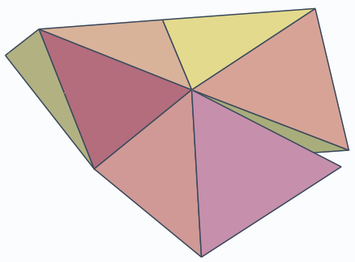
8-286



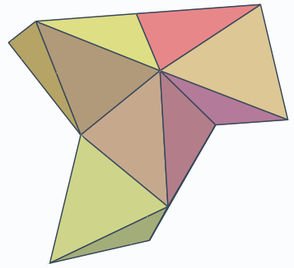
8-287



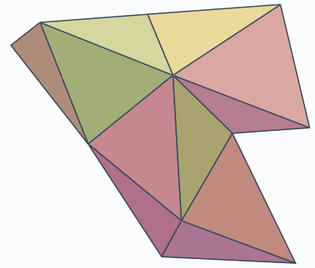
8-288



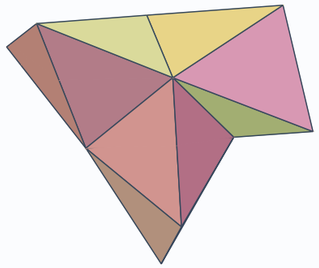
8-289



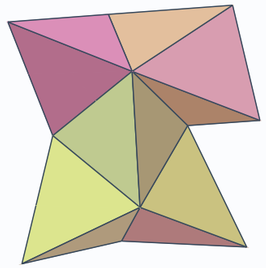
8-291



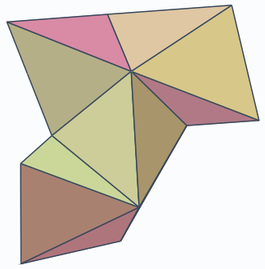
8-292



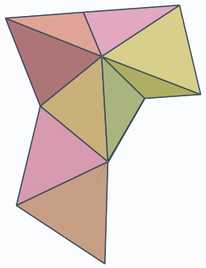
8-293



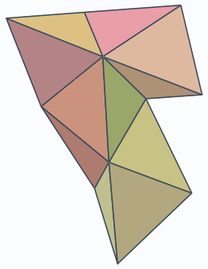
8-294



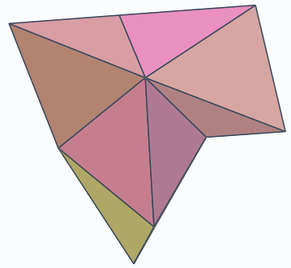
8-295



8-296



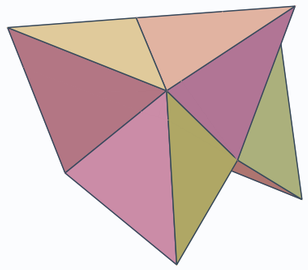
8-298



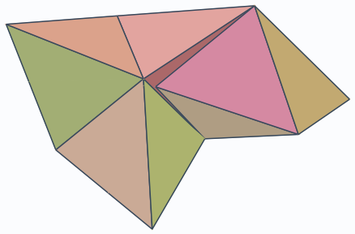
8-302

Puzzles P8A - 8 tetrahedra / targets continued

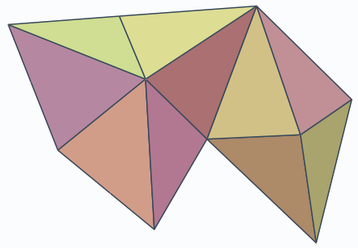
101 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



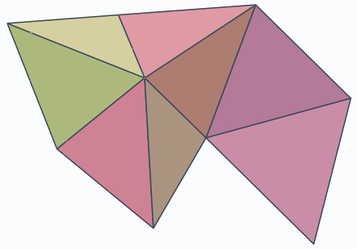
8-308



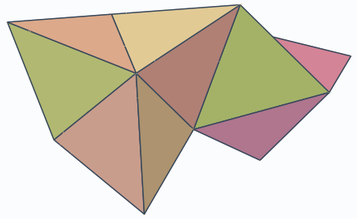
8-316



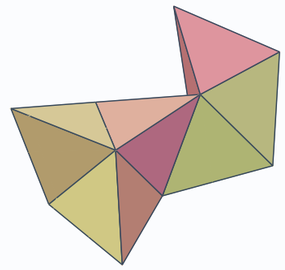
8-317



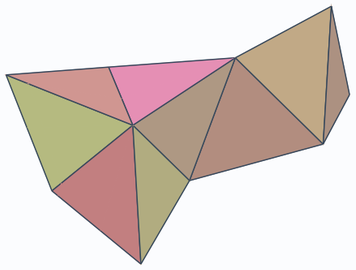
8-321



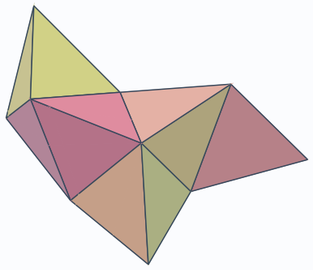
8-323



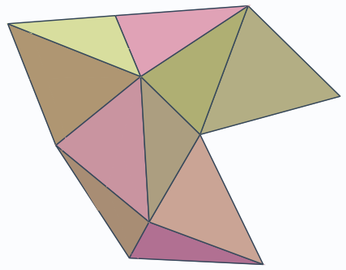
8-329



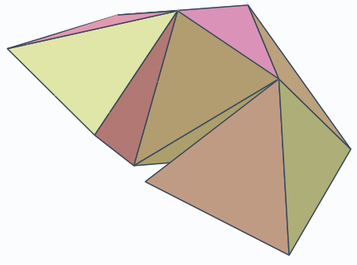
8-330



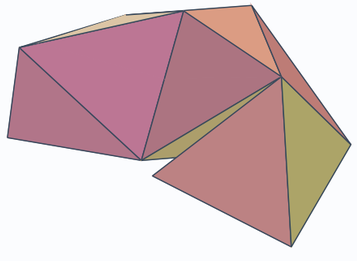
8-334



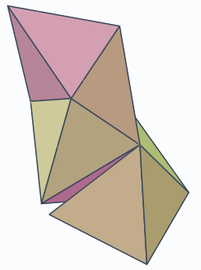
8-336



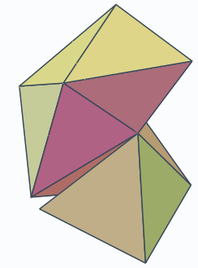
8-343



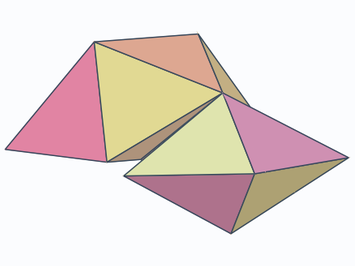
8-344



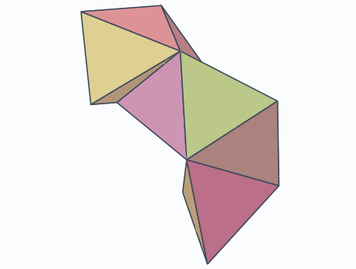
8-348



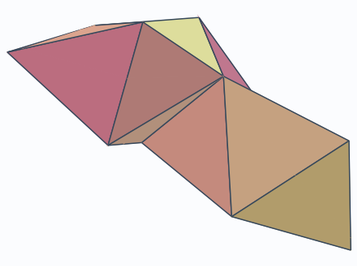
8-349



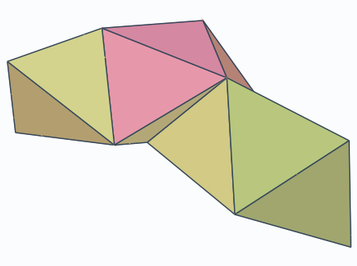
8-352



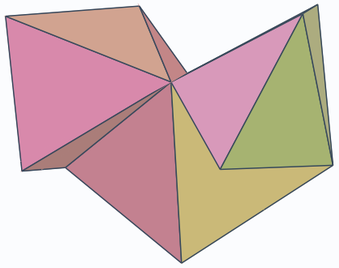
8-374



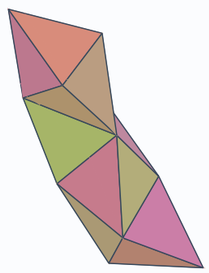
8-380



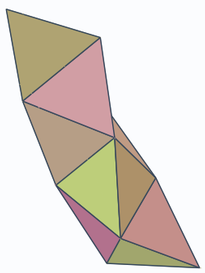
8-382



8-384



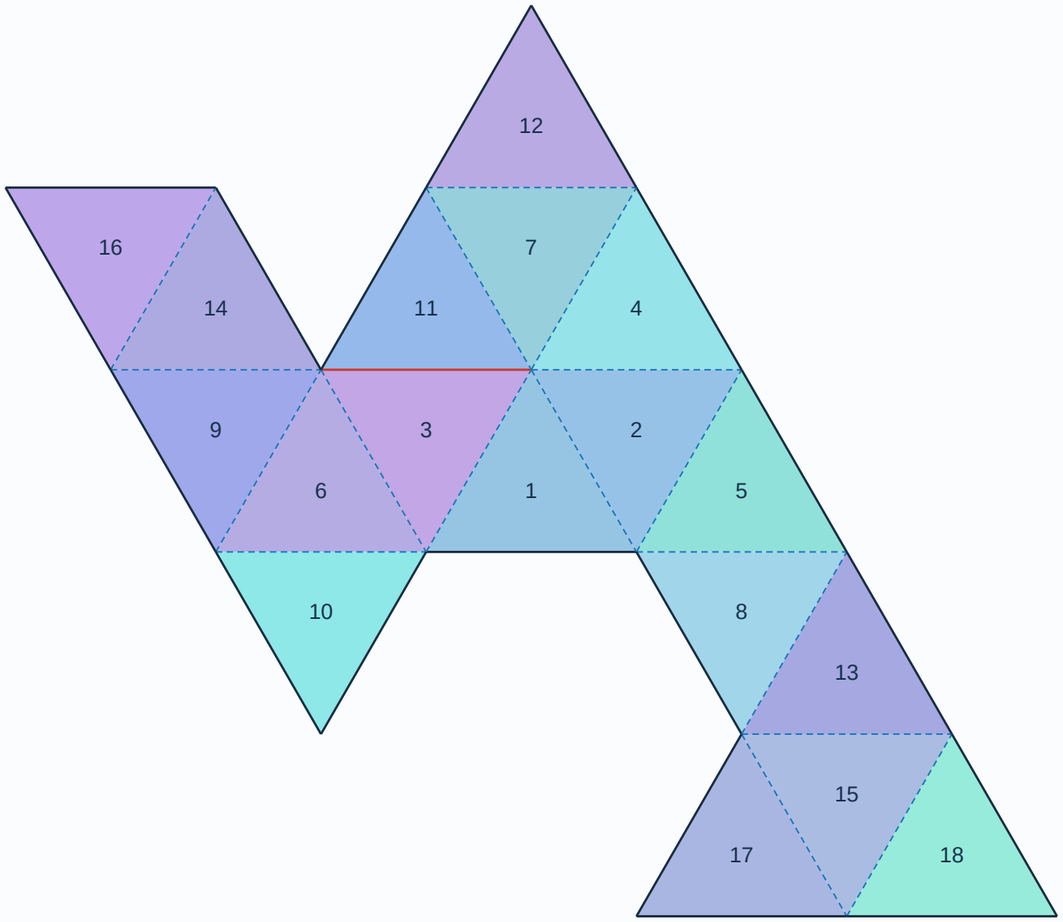
8-397

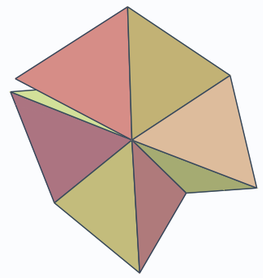


8-398

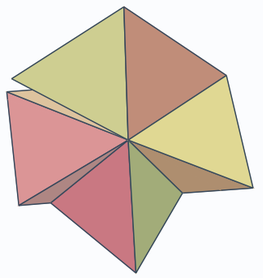
Puzzles P8B - 8 tetrahedra

98 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.

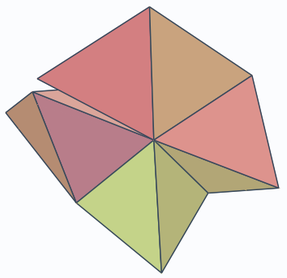




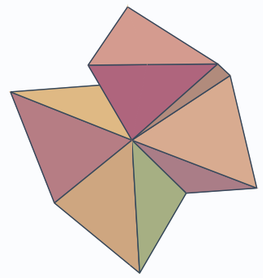
8-003



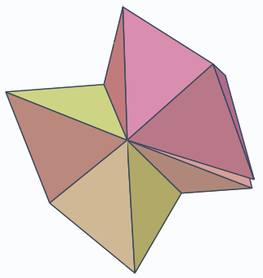
8-017



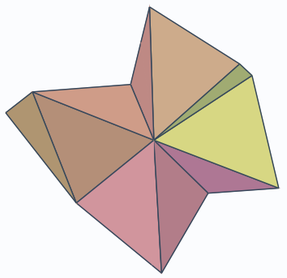
8-018



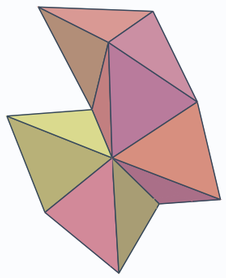
8-025



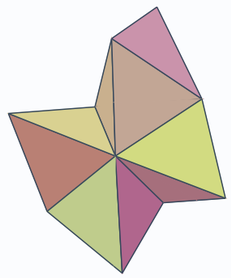
8-026



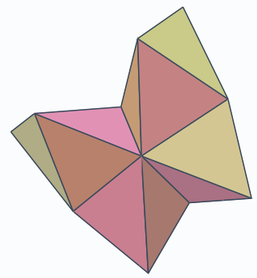
8-029



8-032



8-034



8-036

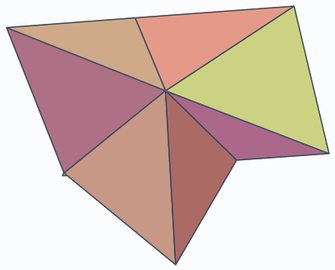
P8B 18 panels; 1 internal slit.

Selected from 500 sampled source trees; not proved optimal.

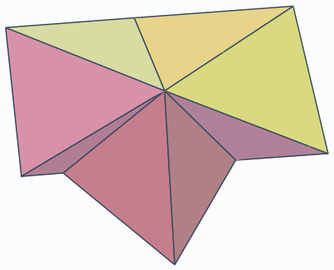
All remaining targets follow on the next page(s).

Puzzles P8B - 8 tetrahedra / targets continued

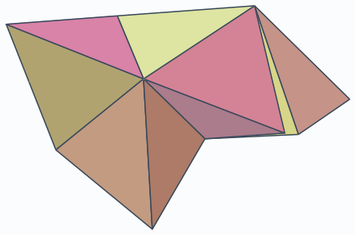
98 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



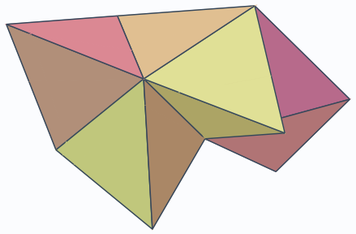
8-038



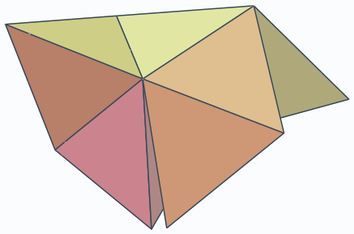
8-045



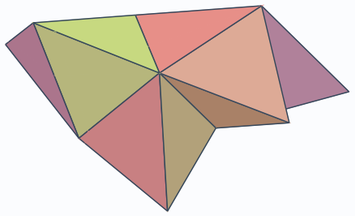
8-051



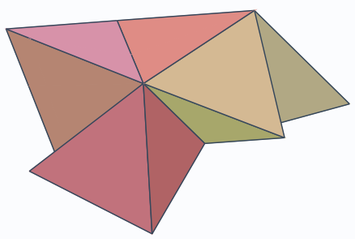
8-052



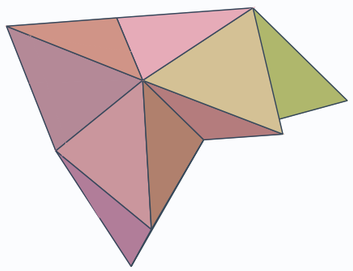
8-055



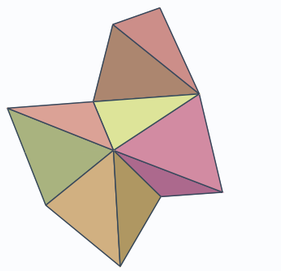
8-059



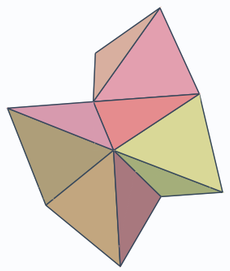
8-060



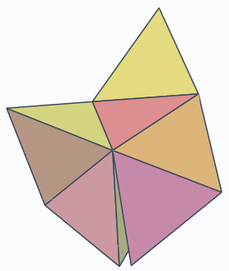
8-062



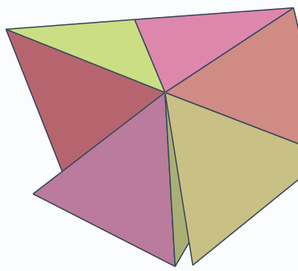
8-063



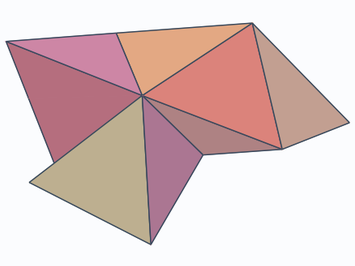
8-064



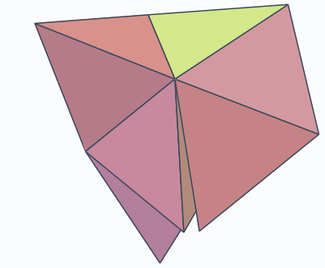
8-066



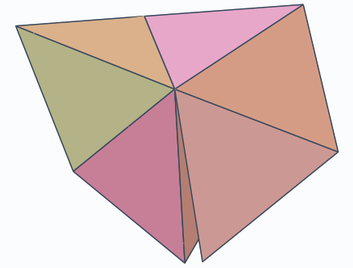
8-076



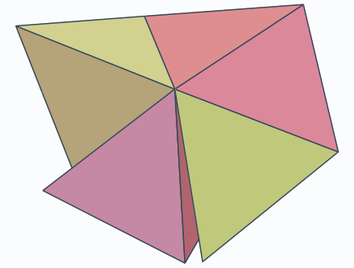
8-086



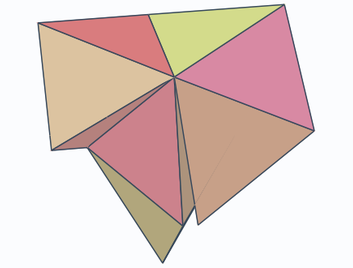
8-098



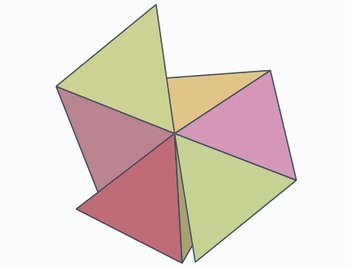
8-100



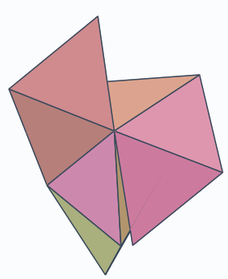
8-102



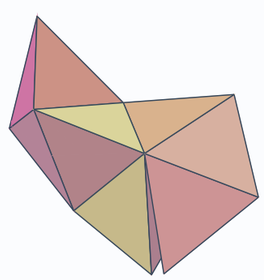
8-106



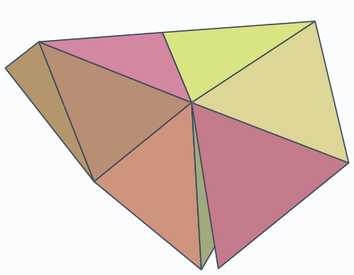
8-115



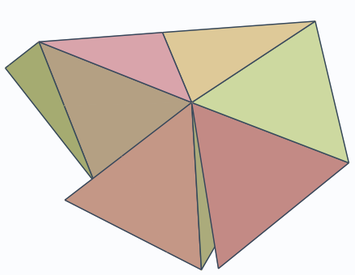
8-116



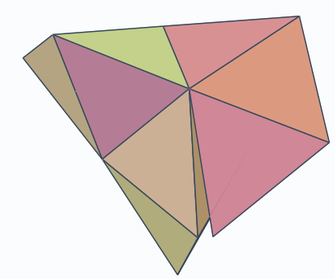
8-118



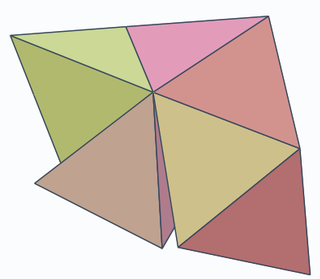
8-119



8-122



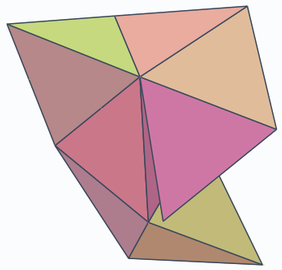
8-123



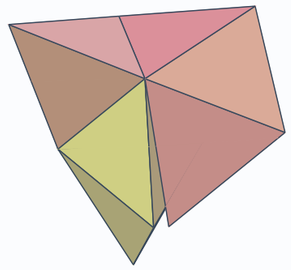
8-133

Puzzles P8B - 8 tetrahedra / targets continued

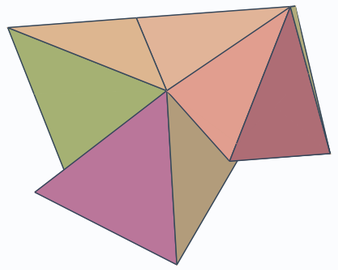
98 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



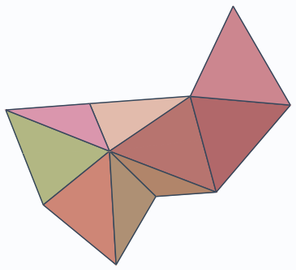
8-138



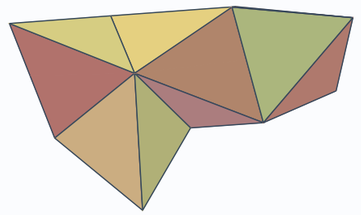
8-139



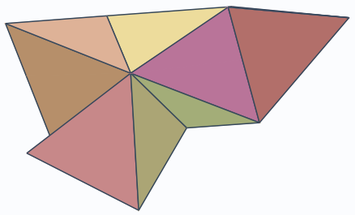
8-159



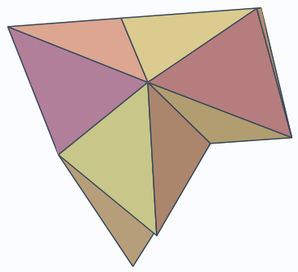
8-177



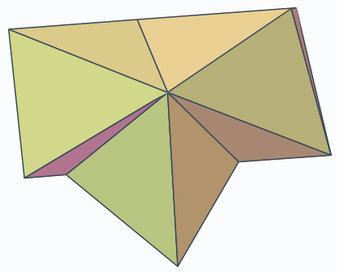
8-178



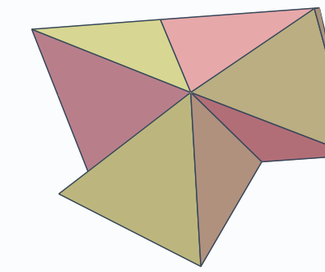
8-183



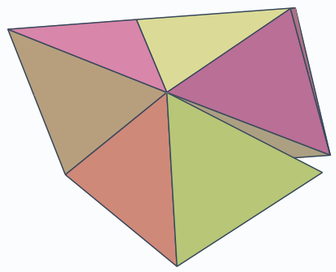
8-186



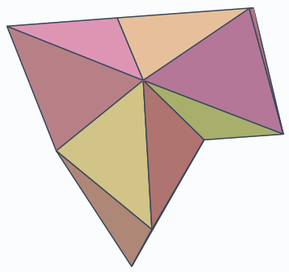
8-189



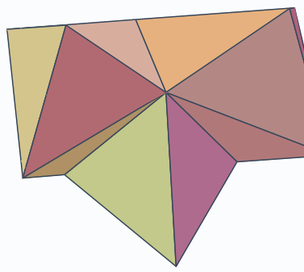
8-192



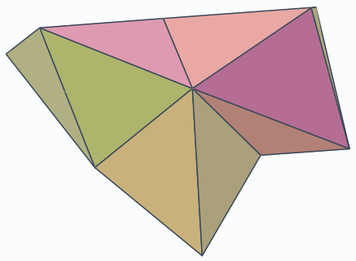
8-193



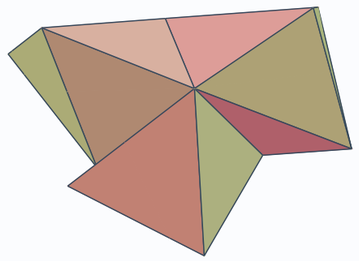
8-194



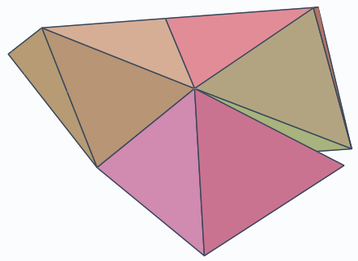
8-198



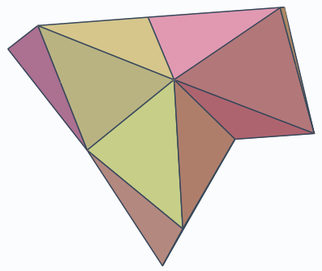
8-199



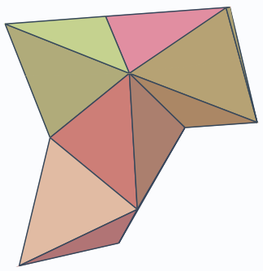
8-202



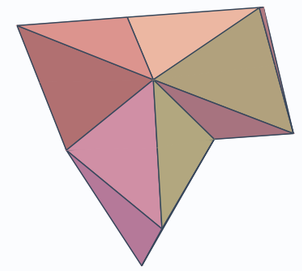
8-203



8-204



8-205



8-206

8-207

8-208

8-209

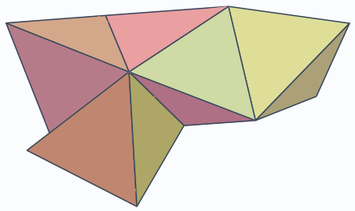
8-210

8-214

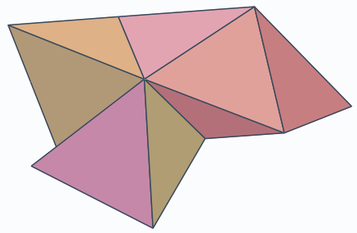
8-216

Puzzles P8B - 8 tetrahedra / targets continued

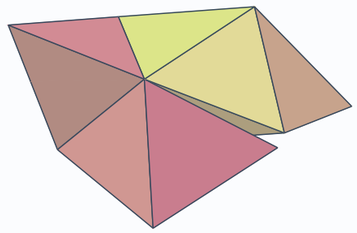
98 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



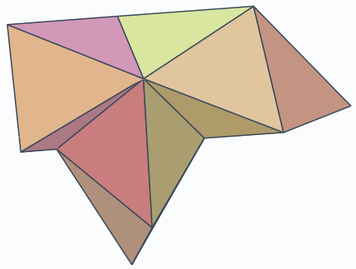
8-234



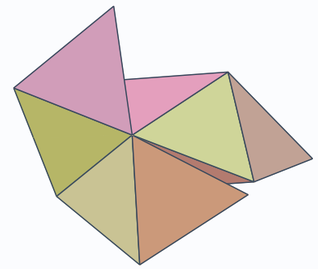
8-243



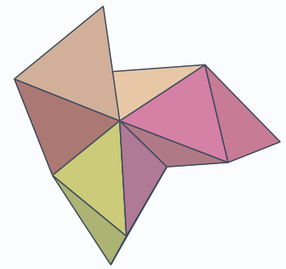
8-244



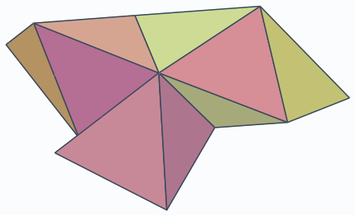
8-248



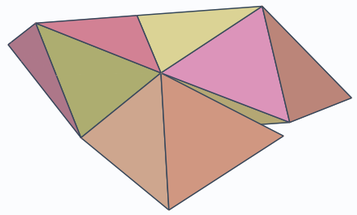
8-252



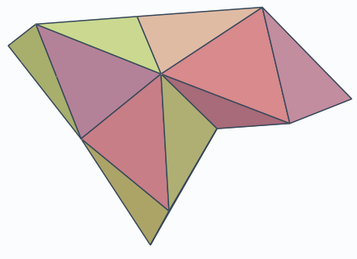
8-253



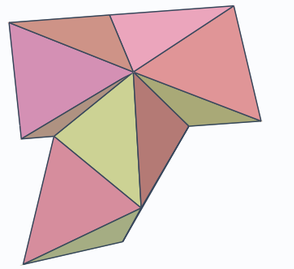
8-257



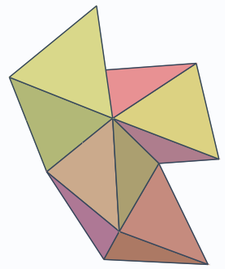
8-258



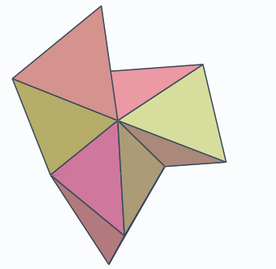
8-259



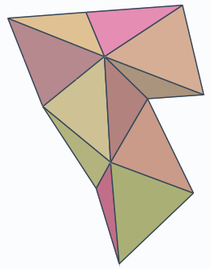
8-271



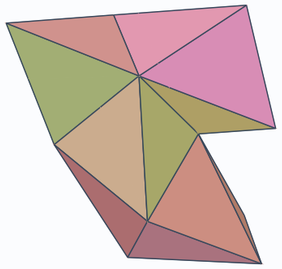
8-274



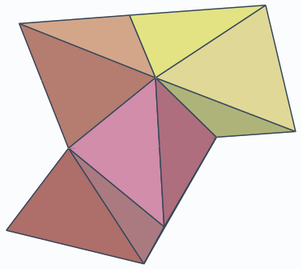
8-275



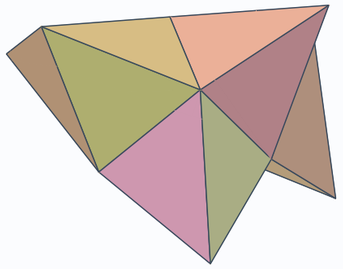
8-278



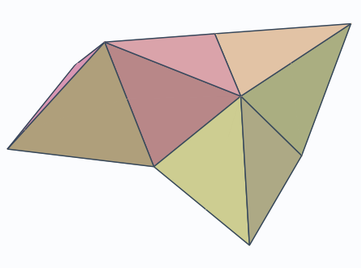
8-282



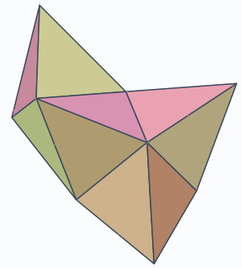
8-283



8-290



8-292



8-293

8-298

8-300

8-301

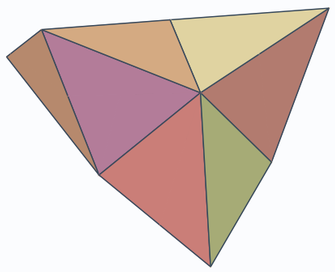
8-310

8-312

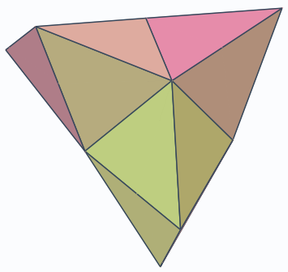
8-313

Puzzles P8B - 8 tetrahedra / targets continued

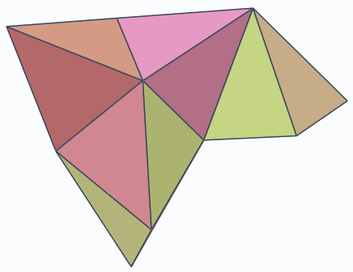
98 matching shapes out of 398. Colors are independent; the folding correspondence is hidden.



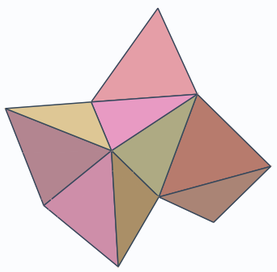
8-314



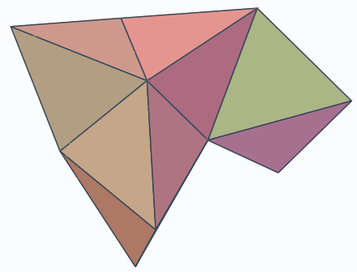
8-315



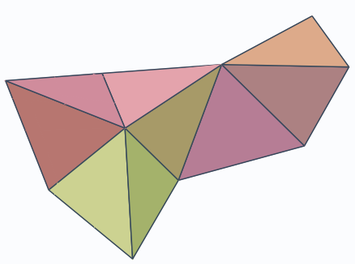
8-320



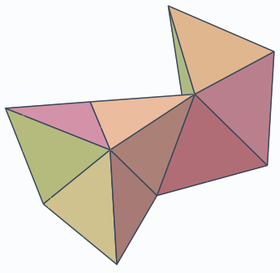
8-325



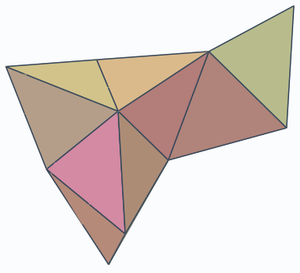
8-327



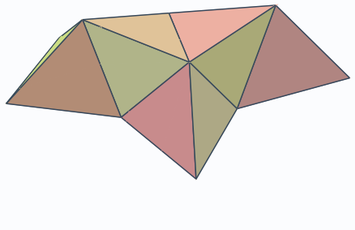
8-328



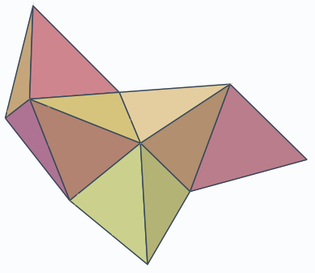
8-329



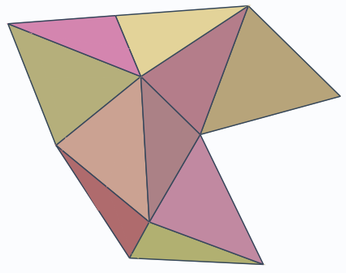
8-332



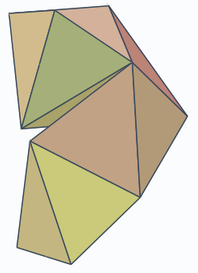
8-333



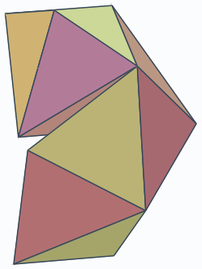
8-334



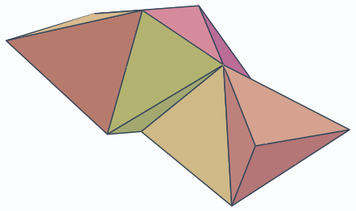
8-336



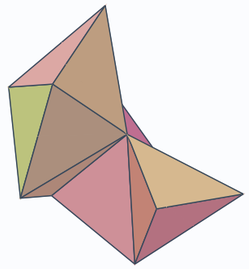
8-340



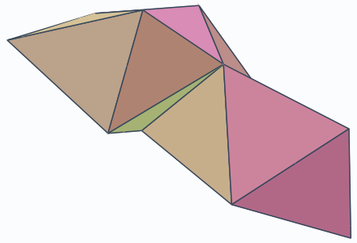
8-342



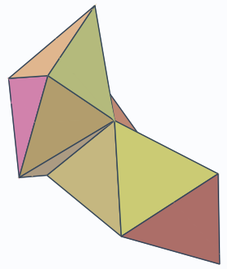
8-363



8-365



8-380



8-381

Part II - Solutions

The same fixed cuts and hinges apply to every target shown for a net.

1 tetrahedra P1A: 1 targets

2 tetrahedra P2A: 1 targets

3 tetrahedra P3A: 1 targets

4 tetrahedra P4A: 3 targets

5 tetrahedra P5A: 7 targets

6 tetrahedra P6A: 19 targets; P6B: 14 targets

7 tetrahedra P7A: 40 targets; P7B: 36 targets

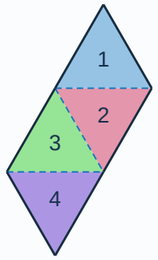
8 tetrahedra P8A: 101 targets; P8B: 98 targets

The first four nets are together on the next page. Each larger net begins with a large diagram and up to nine target shapes; continuation pages contain all its remaining targets. No unmatched shapes are mixed into a puzzle.

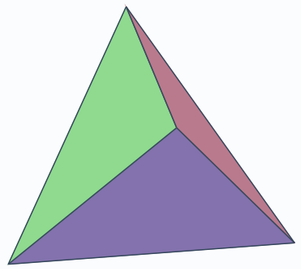
Colors now identify the same panels on the flat net and the folded target. This is the solution key. The same catalog shape may have different panel colors under a different puzzle net. Mirror pairs count as one shape; either paper side may face outward.

Solutions - P1A-P4A

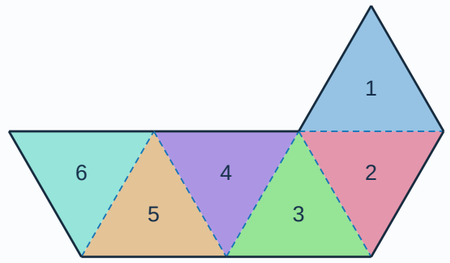
Matching colors show which net panels form the target faces.



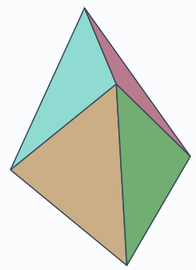
P1A - 1 unit



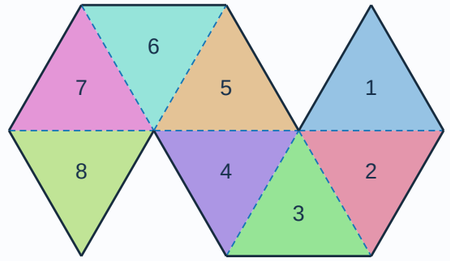
1-001



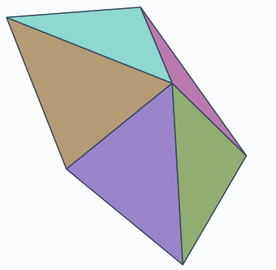
P2A - 2 units



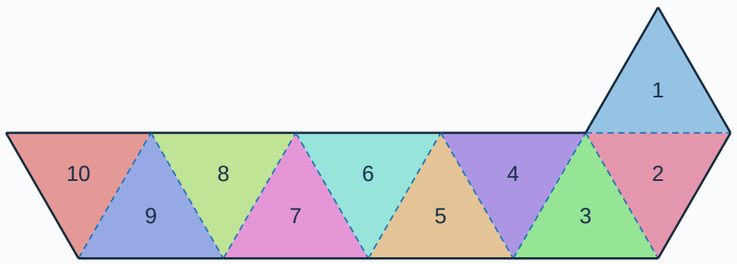
2-001



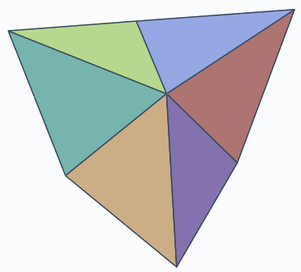
P3A - 3 units



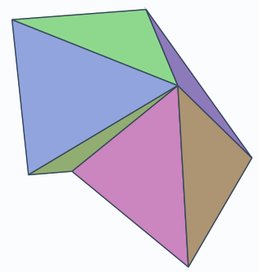
3-001



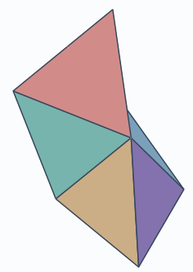
P4A - 4 units



4-001



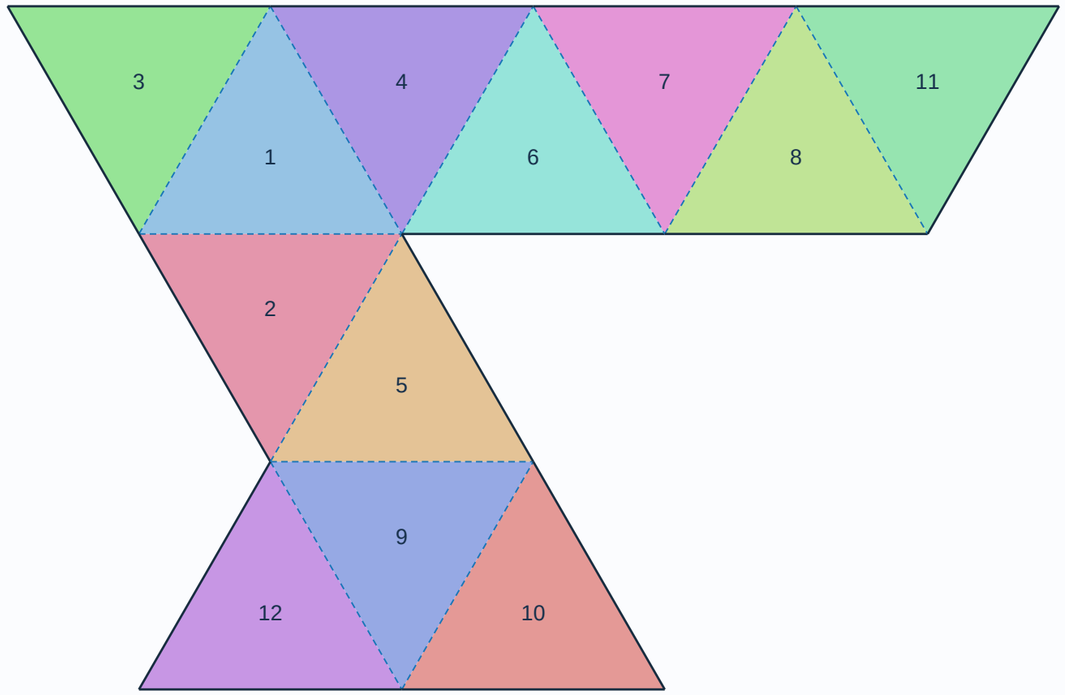
4-002

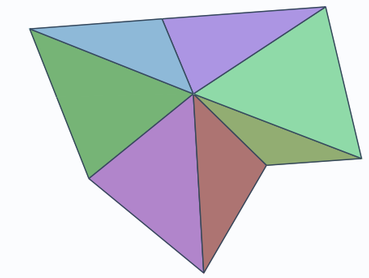


4-003

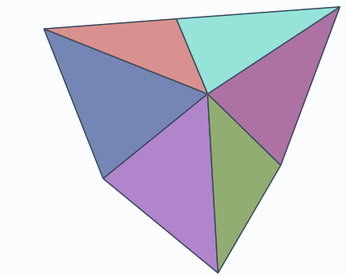
Solutions P5A - 5 tetrahedra

7 matching shapes out of 7. Matching colors identify the net panels on each target.

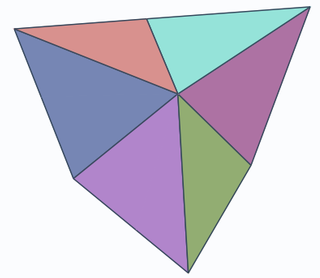




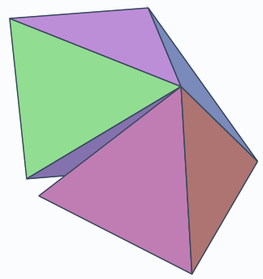
5-001



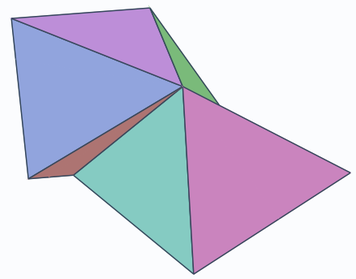
5-002



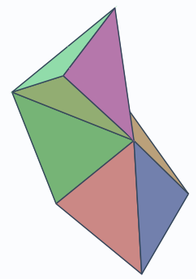
5-003



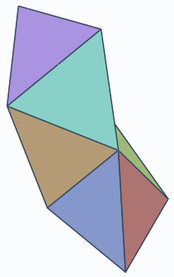
5-004



5-005



5-006



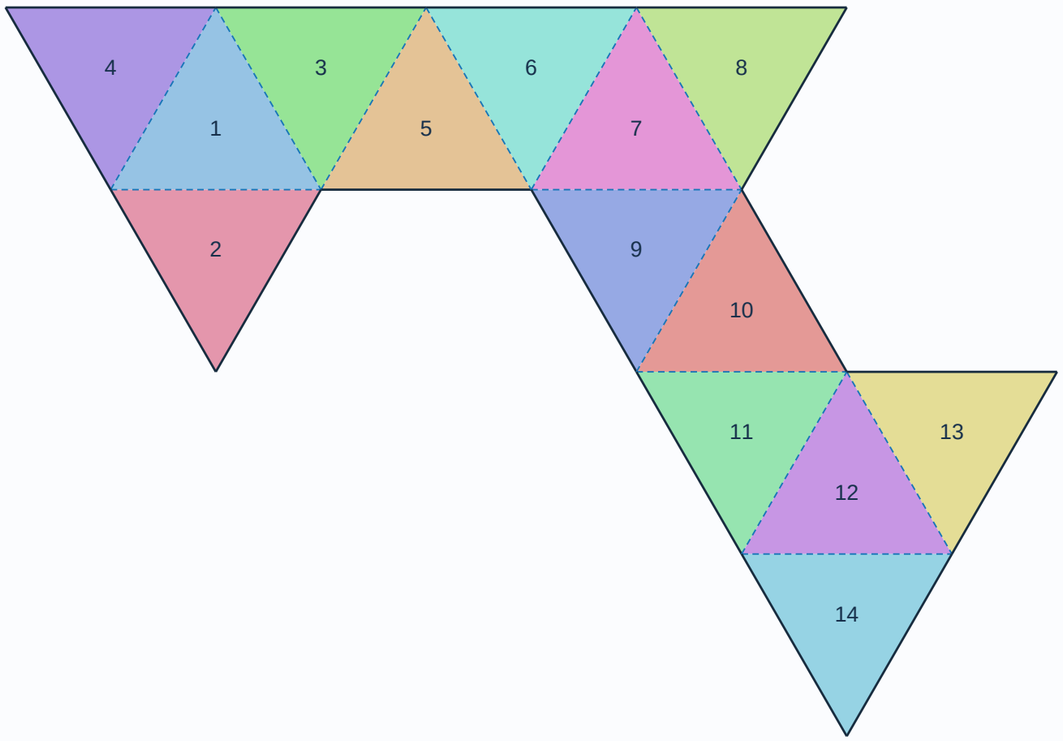
5-007

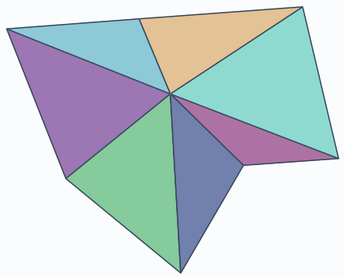
P5A 12 panels; 0 internal slits.

Known five-unit cobra net (Rick Mabry).

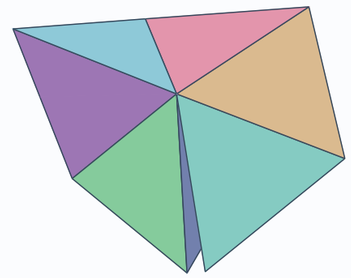
Solutions P6A - 6 tetrahedra

19 matching shapes out of 23. Matching colors identify the net panels on each target.

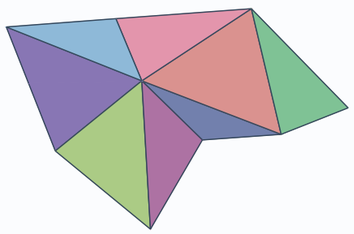




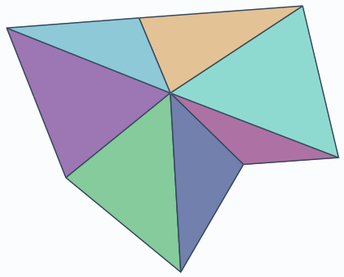
6-002



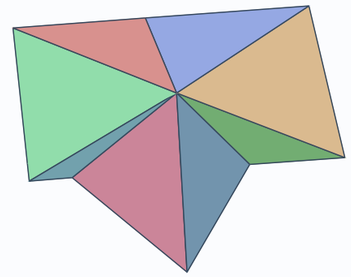
6-003



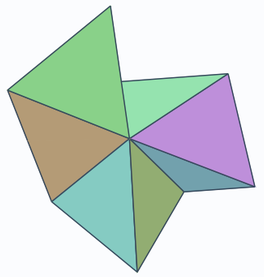
6-005



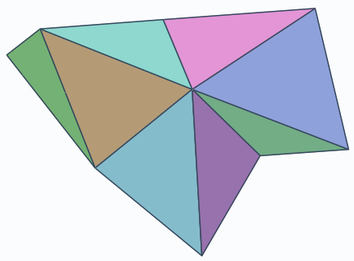
6-006



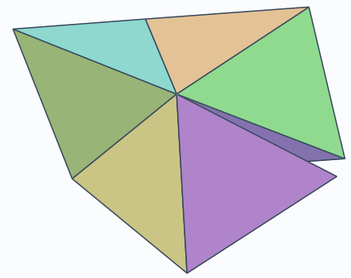
6-007



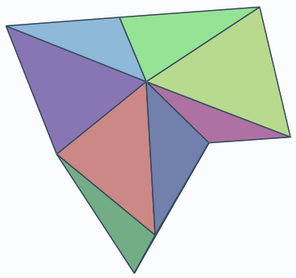
6-008



6-009



6-010



6-011

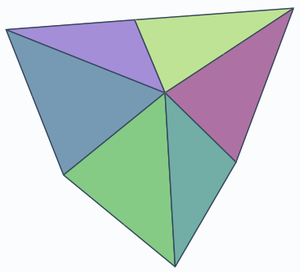
P6A 14 panels; 0 internal slits.

Six-unit Net A attains the checked 19-target maximum.

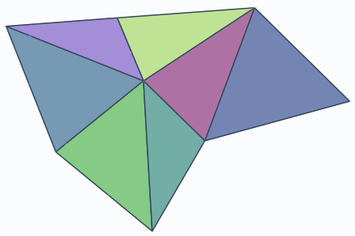
All remaining targets follow on the next page(s).

Solutions P6A - 6 tetrahedra / targets continued

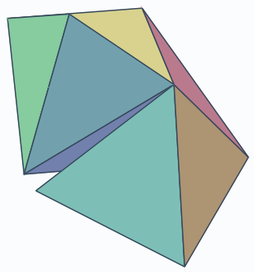
19 matching shapes out of 23. Matching colors identify the net panels on each target.



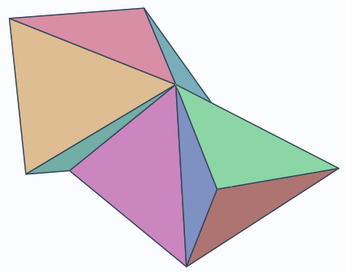
6-012



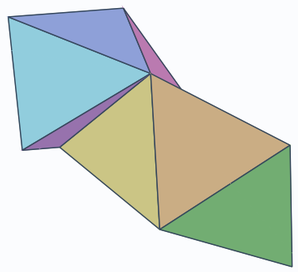
6-013



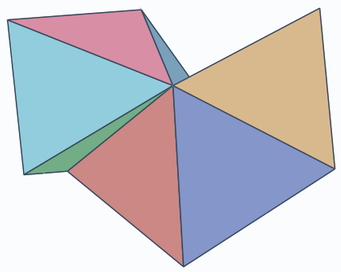
6-015



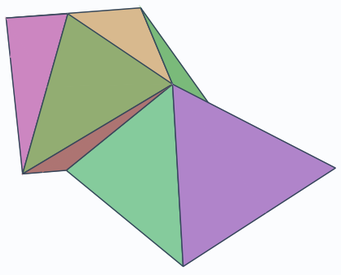
6-016



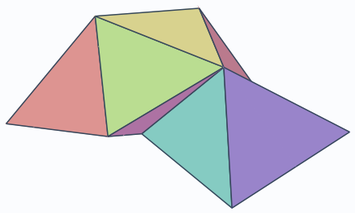
6-017



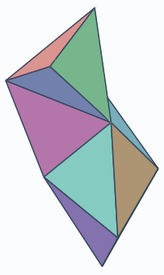
6-018



6-019



6-020



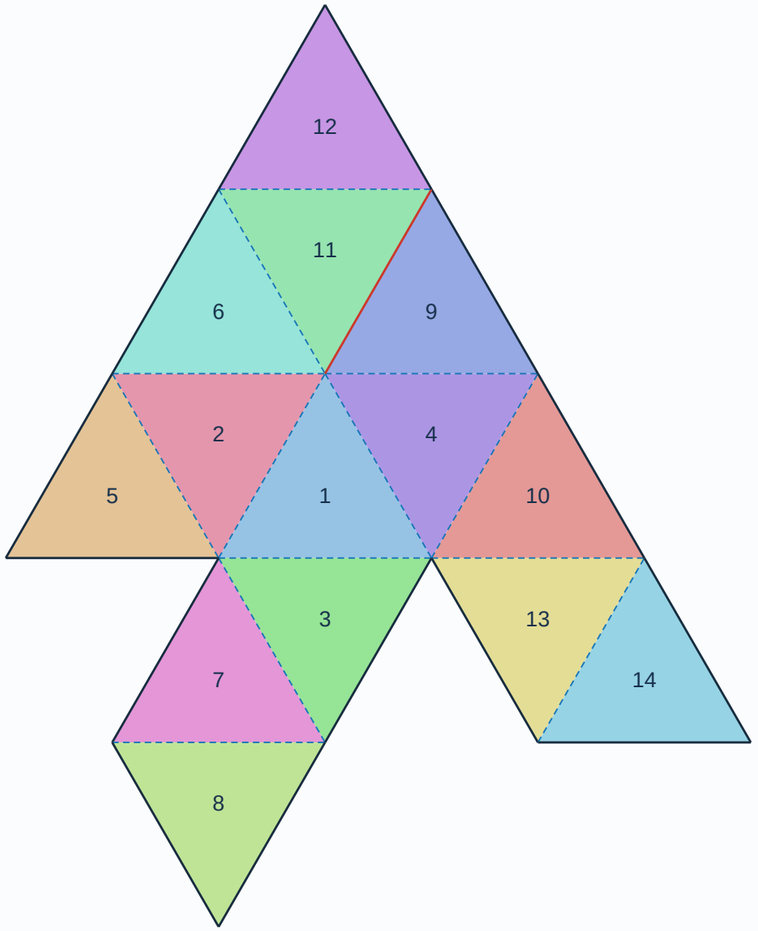
6-022



6-023

Solutions P6B - 6 tetrahedra

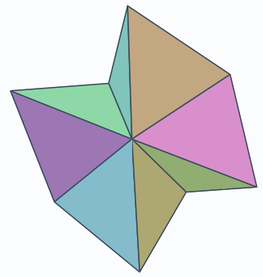
14 matching shapes out of 23. Matching colors identify the net panels on each target.



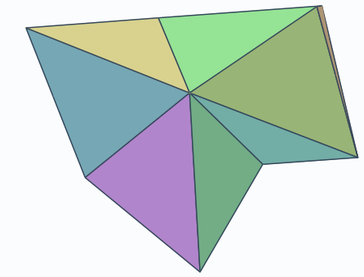
P6B 14 panels; 1 internal slit.

This companion supplies the four targets missing from P6A.

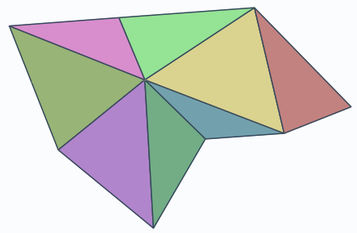
All remaining targets follow on the next page(s).



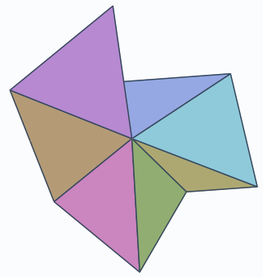
6-001



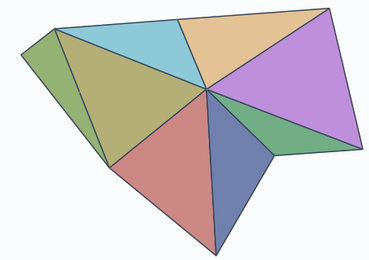
6-004



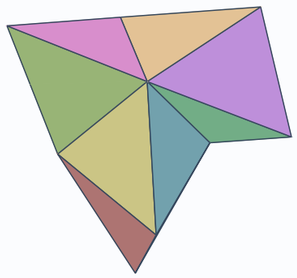
6-005



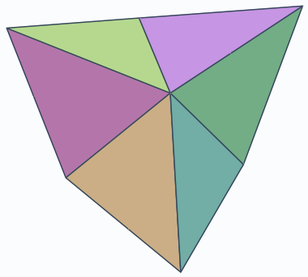
6-008



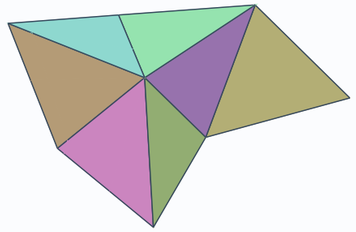
6-009



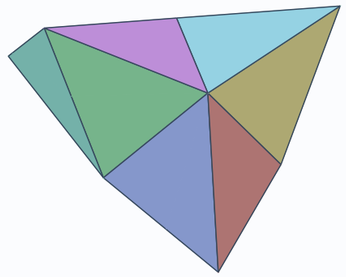
6-011



6-012



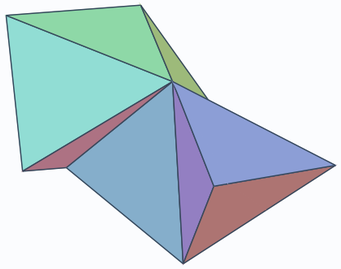
6-013



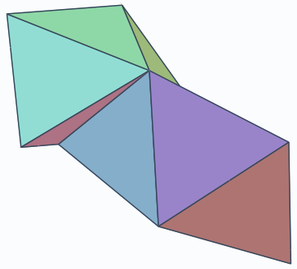
6-014

Solutions P6B - 6 tetrahedra / targets continued

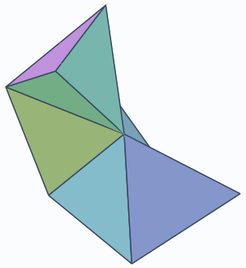
14 matching shapes out of 23. Matching colors identify the net panels on each target.



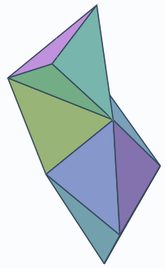
6-016



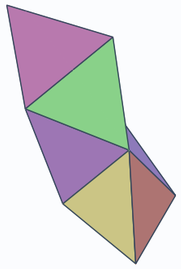
6-017



6-021



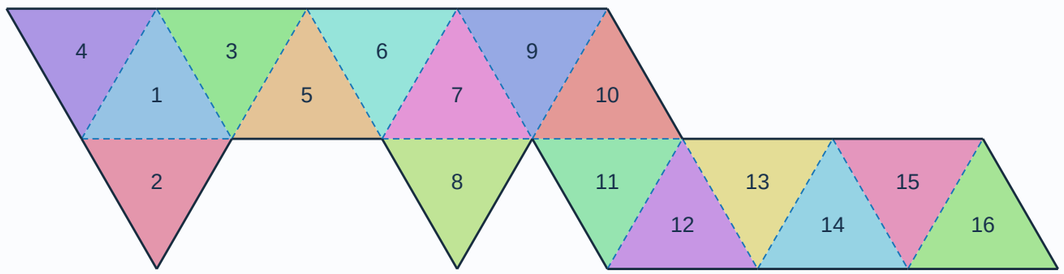
6-022



6-023

Solutions P7A - 7 tetrahedra

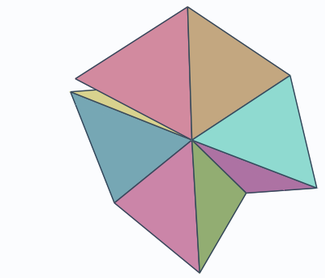
40 matching shapes out of 89. Matching colors identify the net panels on each target.



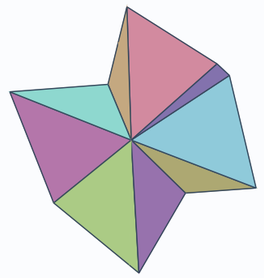
P7A 16 panels; 0 internal slits.

Selected from 500 sampled source trees; not proved optimal.

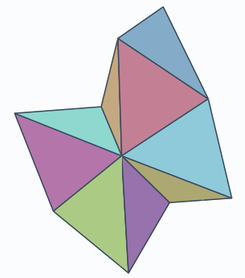
All remaining targets follow on the next page(s).



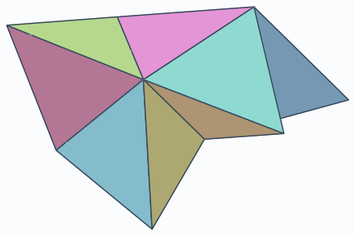
7-002



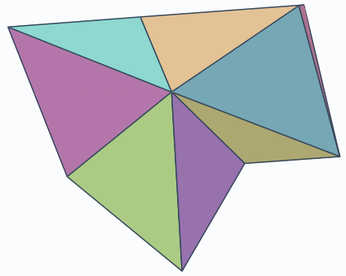
7-003



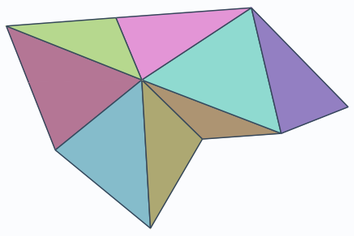
7-004



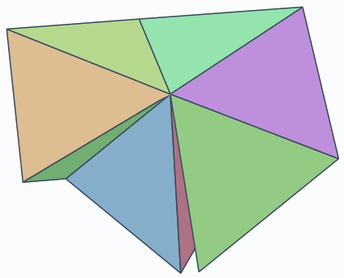
7-006



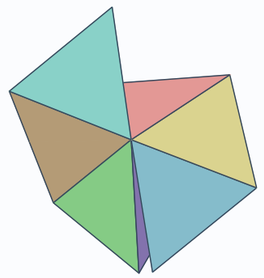
7-009



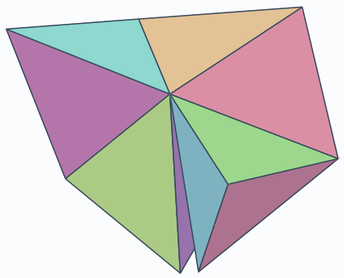
7-010



7-014



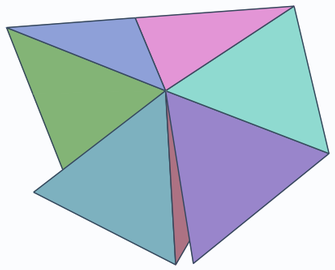
7-015



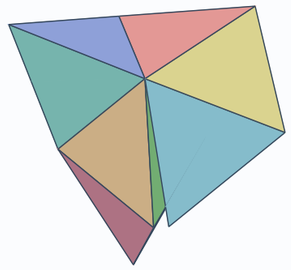
7-017

Solutions P7A - 7 tetrahedra / targets continued

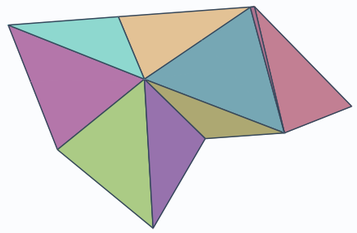
40 matching shapes out of 89. Matching colors identify the net panels on each target.



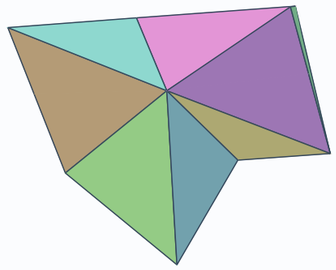
7-019



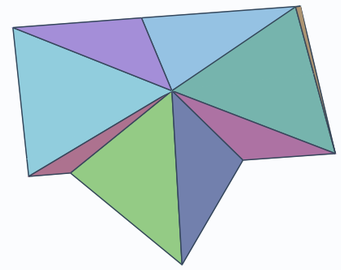
7-020



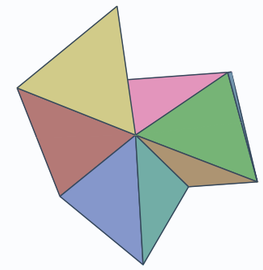
7-021



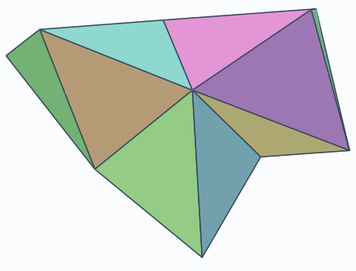
7-025



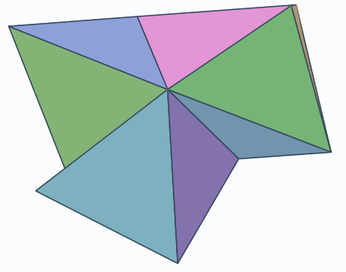
7-026



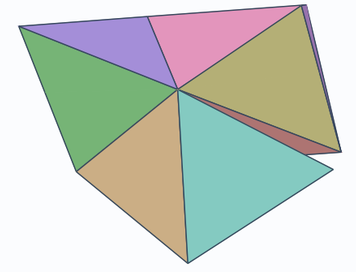
7-027



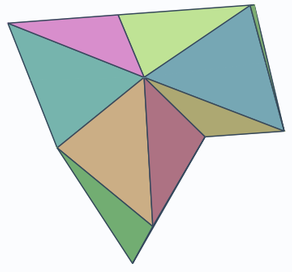
7-028



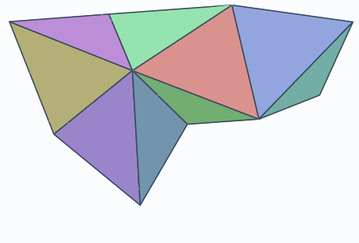
7-029



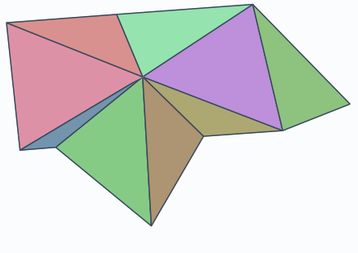
7-030



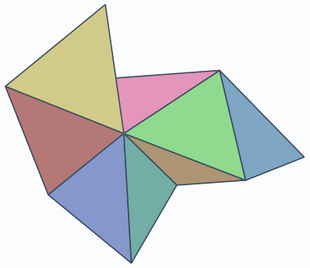
7-031



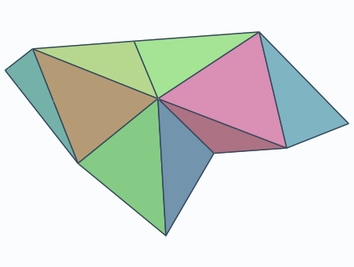
7-033



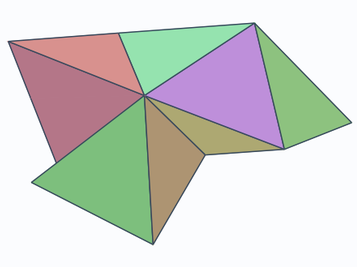
7-035



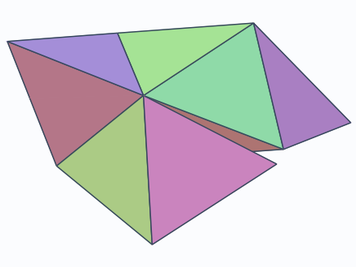
7-036



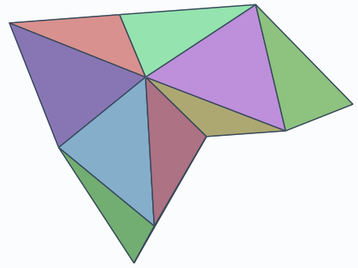
7-037



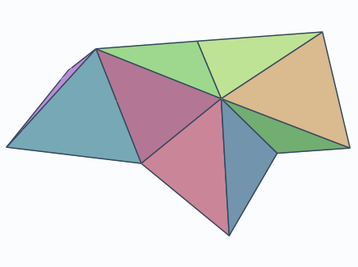
7-038



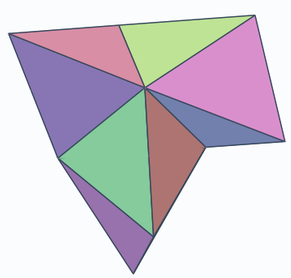
7-039



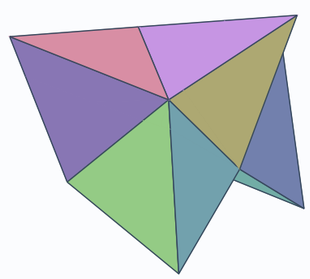
7-040



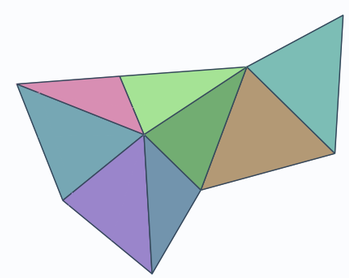
7-049



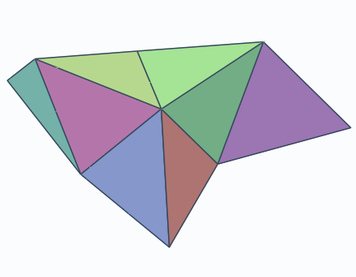
7-056



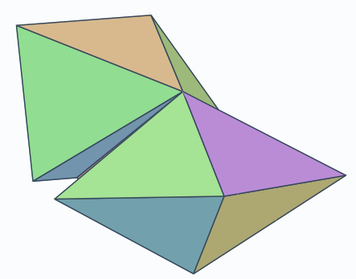
7-058



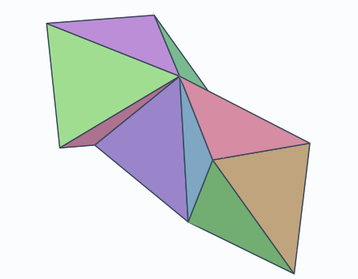
7-062



7-064



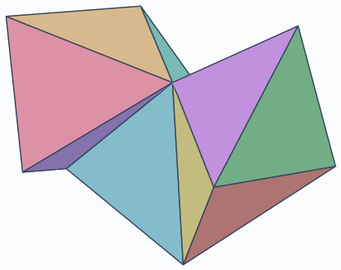
7-072



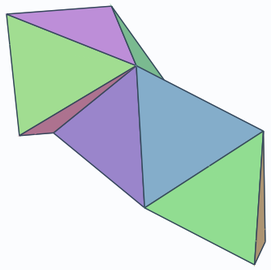
7-073

Solutions P7A - 7 tetrahedra / targets continued

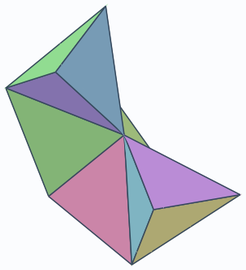
40 matching shapes out of 89. Matching colors identify the net panels on each target.



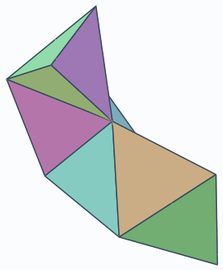
7-074



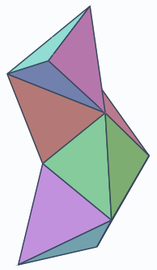
7-077



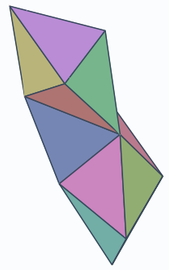
7-084



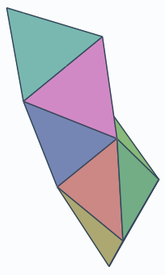
7-085



7-086



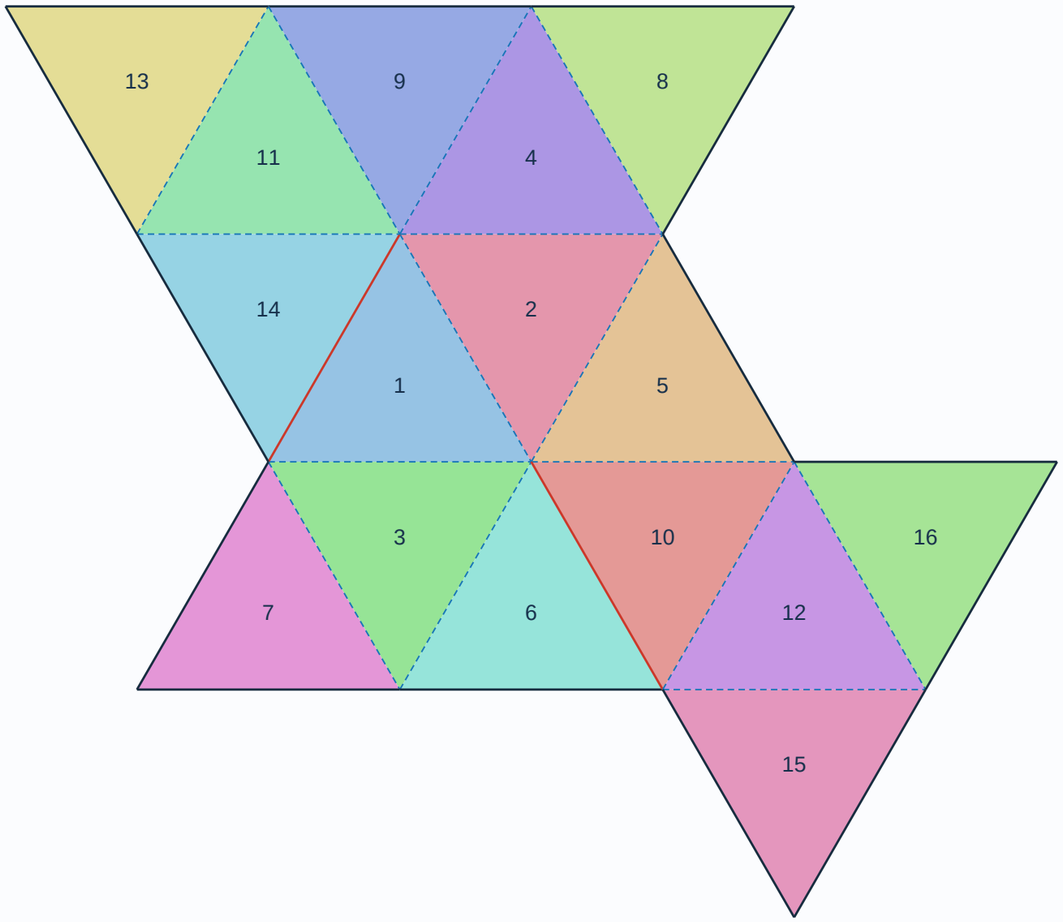
7-088

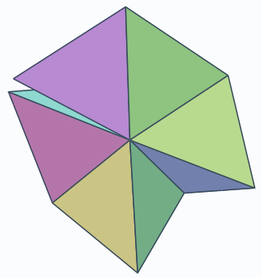


7-089

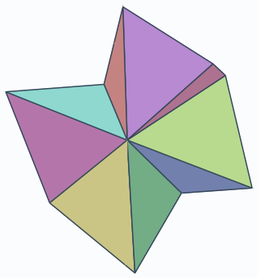
Solutions P7B - 7 tetrahedra

36 matching shapes out of 89. Matching colors identify the net panels on each target.

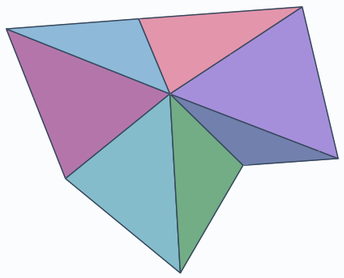




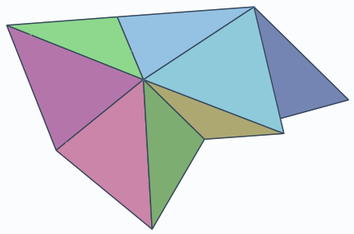
7-002



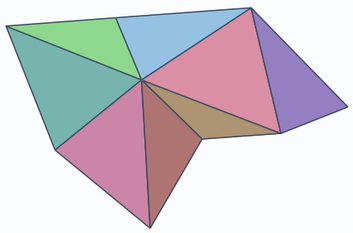
7-003



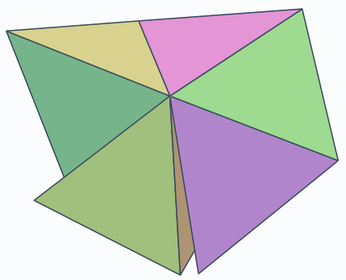
7-005



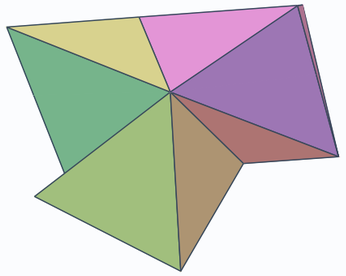
7-006



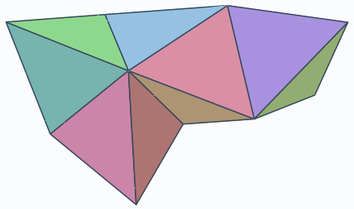
7-010



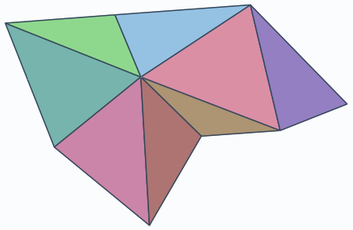
7-019



7-029



7-033



7-034

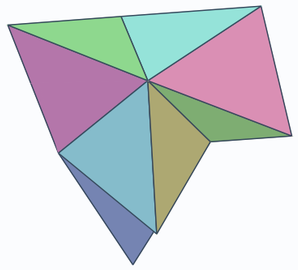
P7B 16 panels; 2 internal slits.

Selected from 500 sampled source trees; not proved optimal.

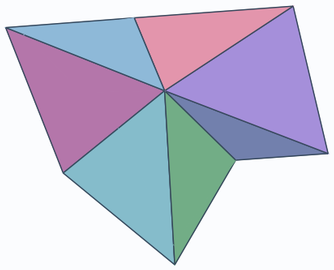
All remaining targets follow on the next page(s).

Solutions P7B - 7 tetrahedra / targets continued

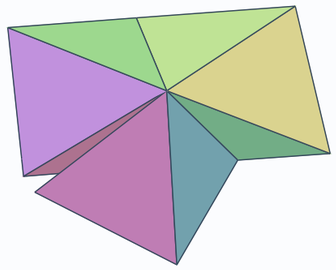
36 matching shapes out of 89. Matching colors identify the net panels on each target.



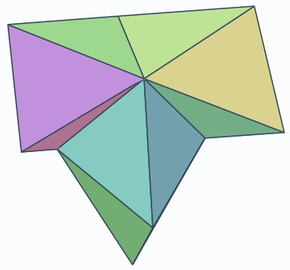
7-041



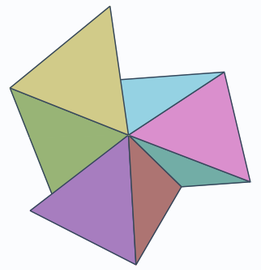
7-043



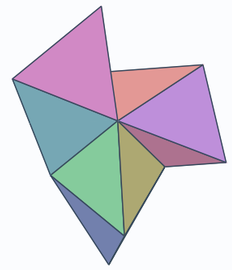
7-045



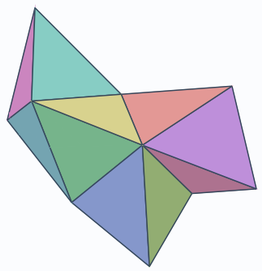
7-046



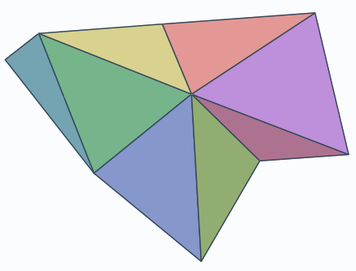
7-047



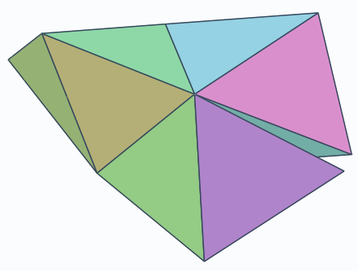
7-048



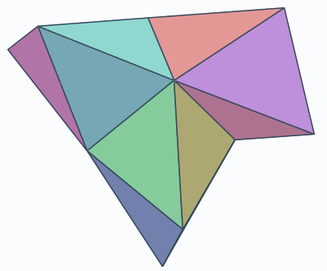
7-050



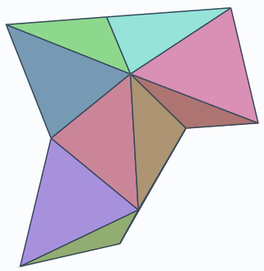
7-051



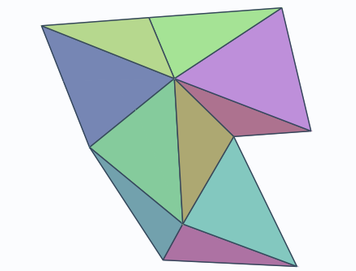
7-052



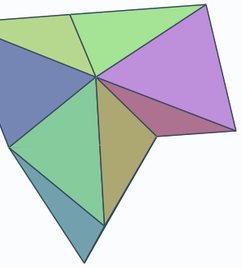
7-053



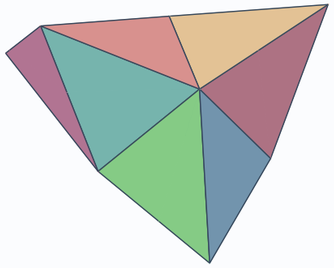
7-054



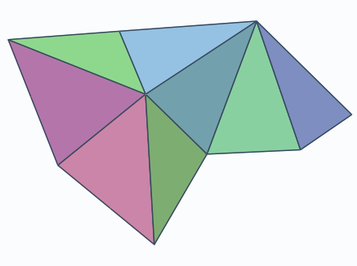
7-055



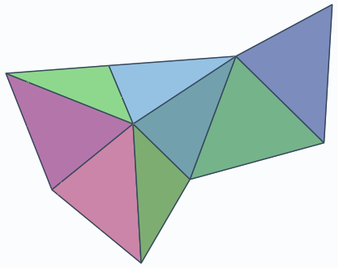
7-056



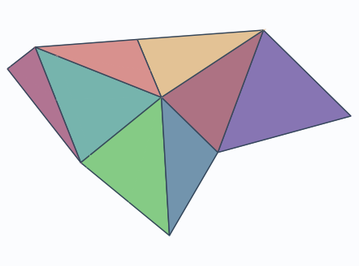
7-059



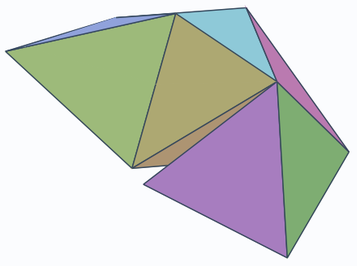
7-060



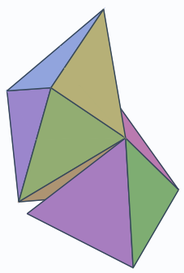
7-062



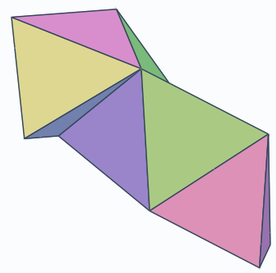
7-064



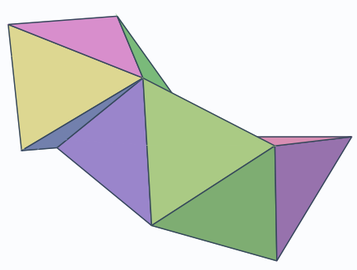
7-069



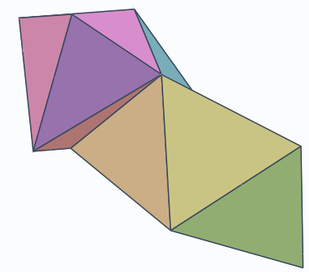
7-071



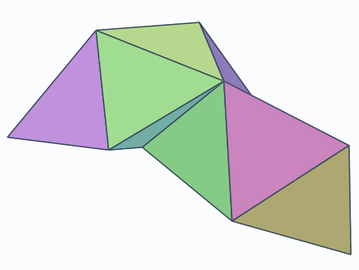
7-077



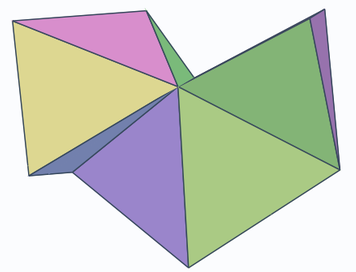
7-079



7-080



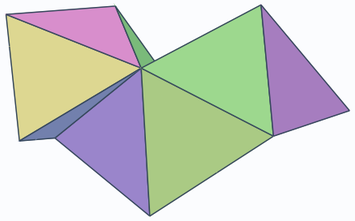
7-081



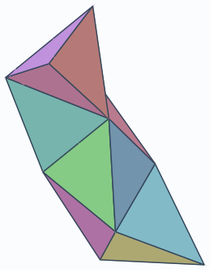
7-082

Solutions P7B - 7 tetrahedra / targets continued

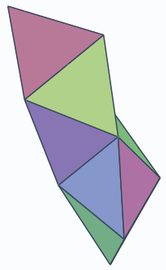
36 matching shapes out of 89. Matching colors identify the net panels on each target.



7-083



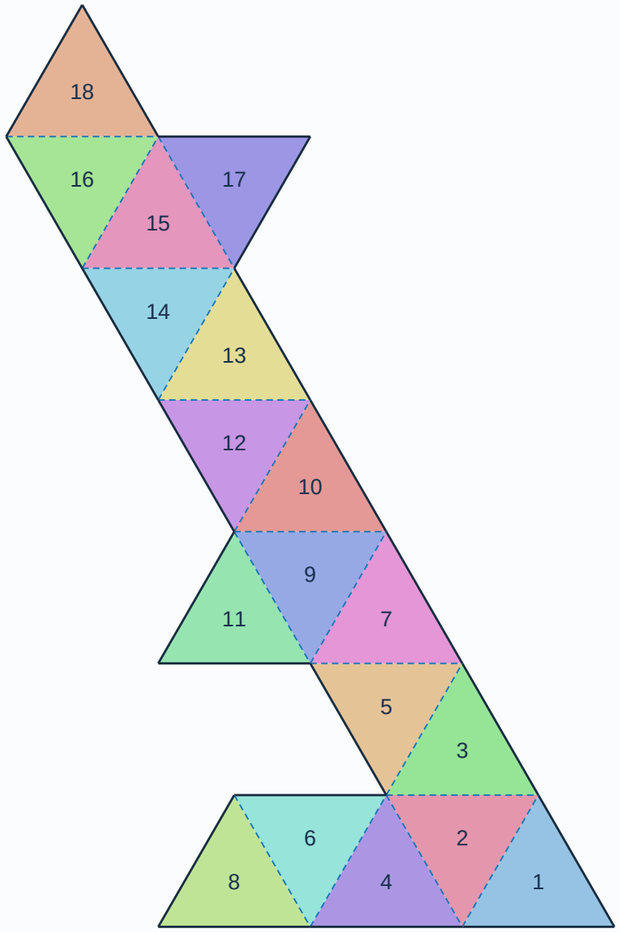
7-087



7-089

Solutions P8A - 8 tetrahedra

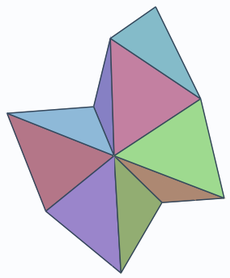
101 matching shapes out of 398. Matching colors identify the net panels on each target.



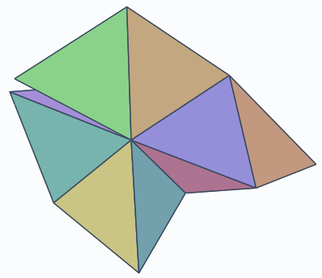
P8A 18 panels; 0 internal slits.

Selected from 500 sampled source trees; not proved optimal.

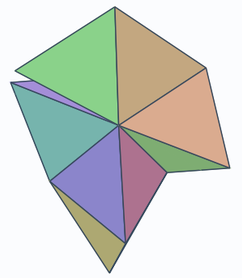
All remaining targets follow on the next page(s).



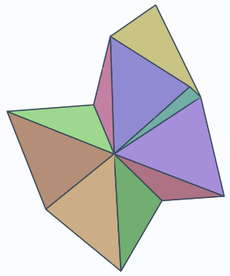
8-005



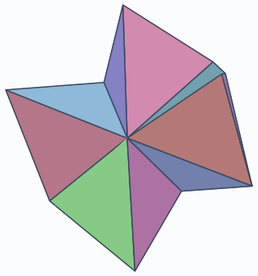
8-014



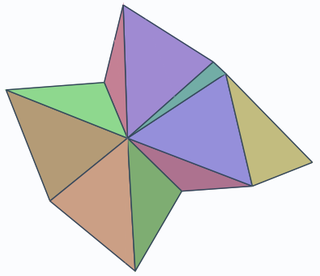
8-021



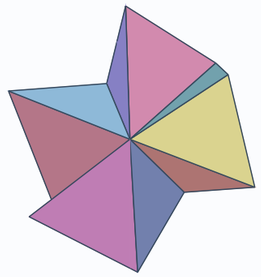
8-022



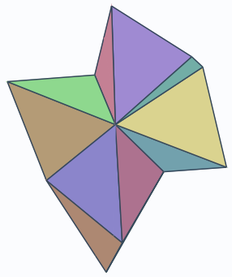
8-023



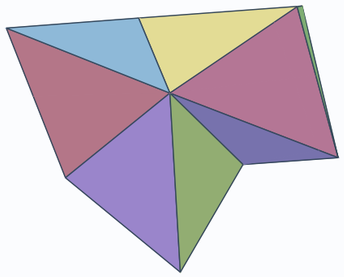
8-024



8-030



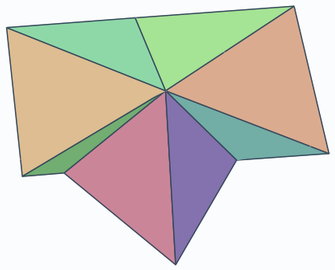
8-031



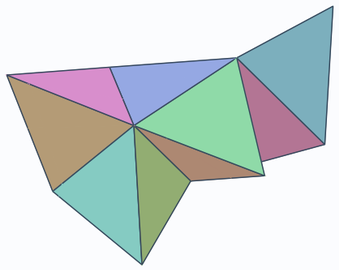
8-043

Solutions P8A - 8 tetrahedra / targets continued

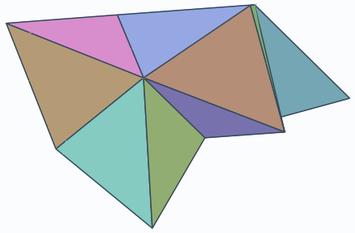
101 matching shapes out of 398. Matching colors identify the net panels on each target.



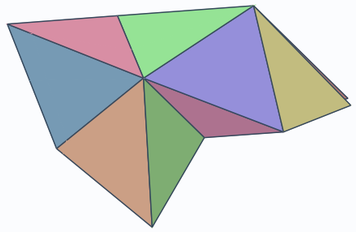
8-045



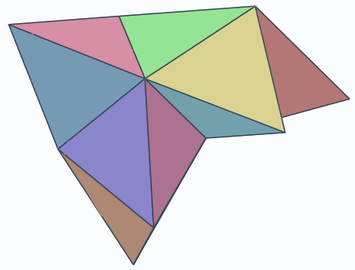
8-053



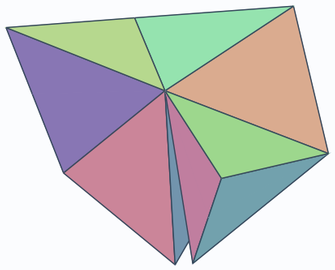
8-056



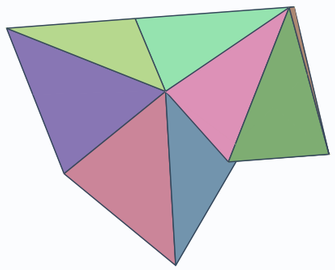
8-057



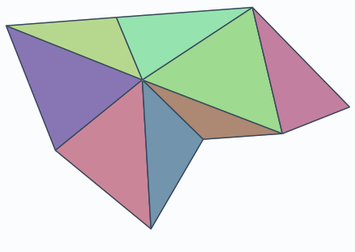
8-062



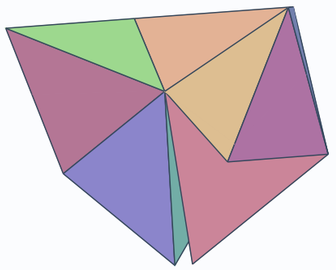
8-074



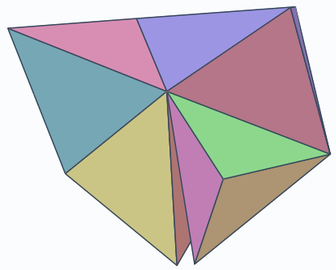
8-078



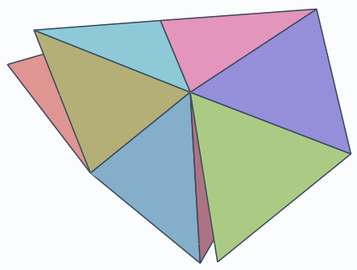
8-085



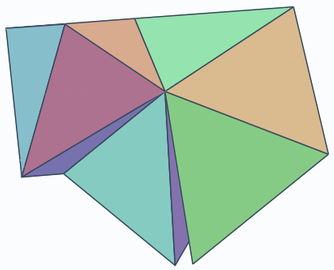
8-088



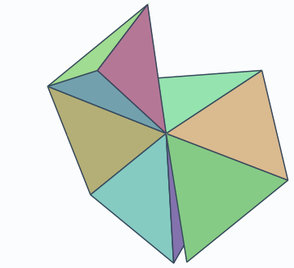
8-091



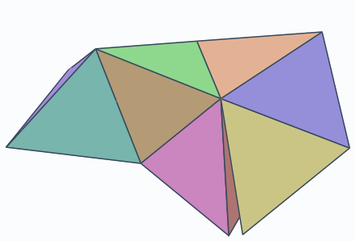
8-099



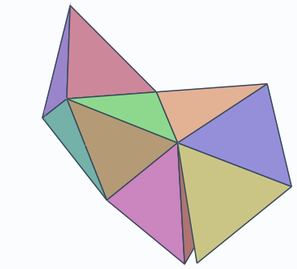
8-107



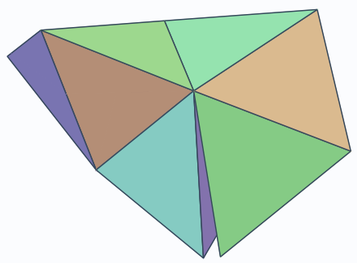
8-110



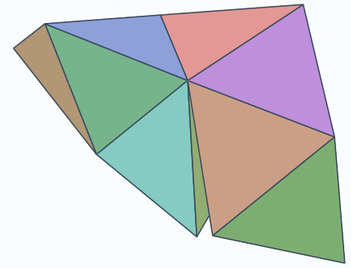
8-117



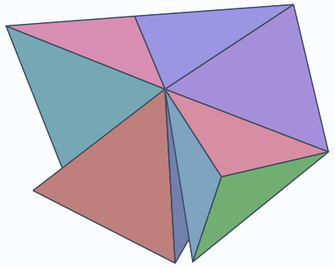
8-118



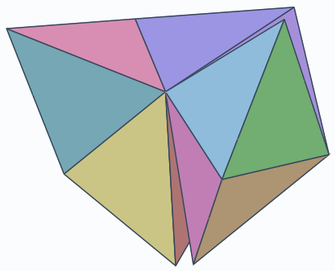
8-119



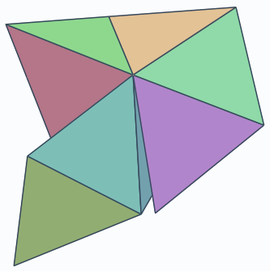
8-121



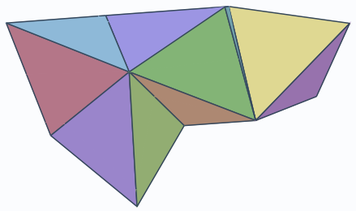
8-125



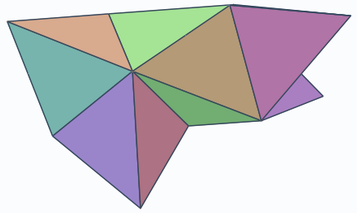
8-129



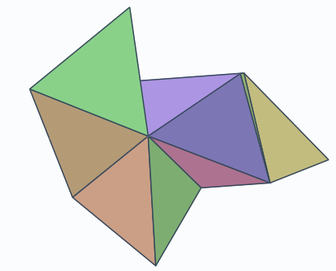
8-136



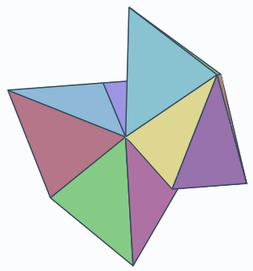
8-141



8-145



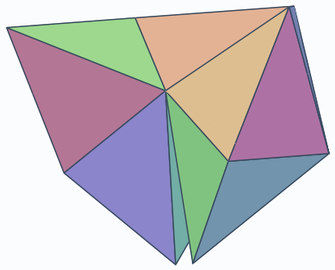
8-148



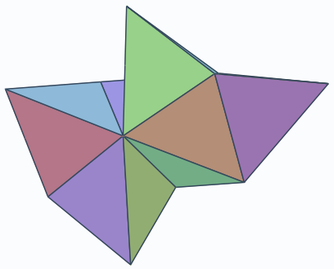
8-153

Solutions P8A - 8 tetrahedra / targets continued

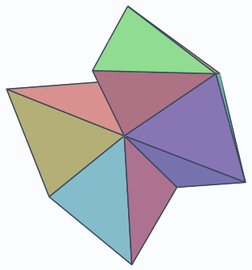
101 matching shapes out of 398. Matching colors identify the net panels on each target.



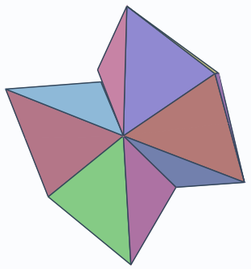
8-163



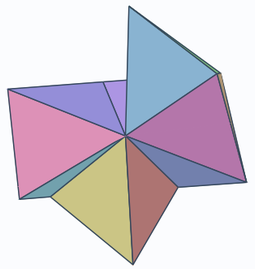
8-165



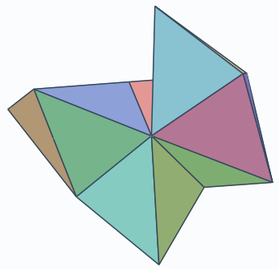
8-166



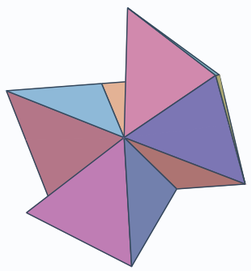
8-167



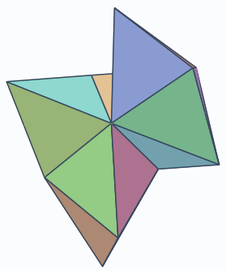
8-170



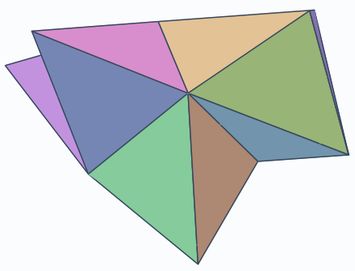
8-172



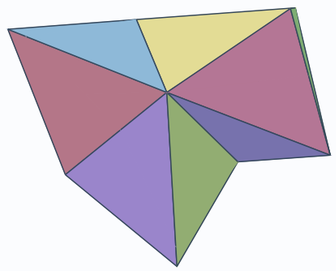
8-173



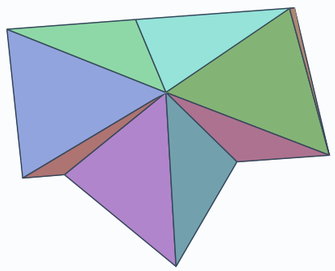
8-175



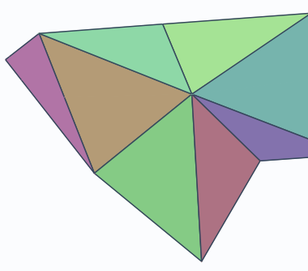
8-187



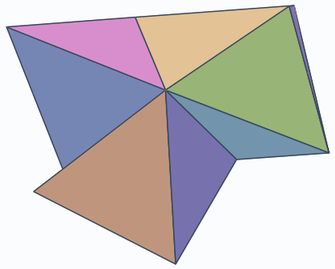
8-188



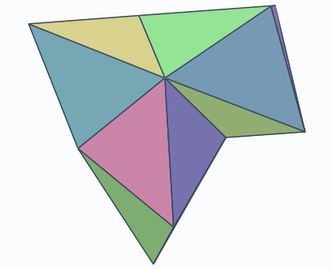
8-189



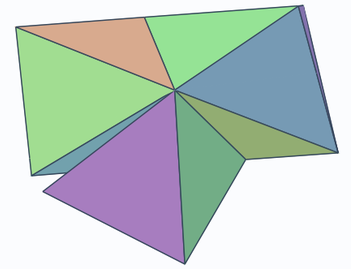
8-191



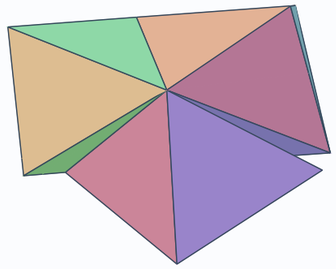
8-192



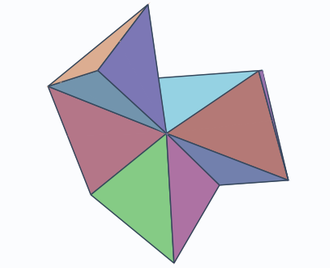
8-194



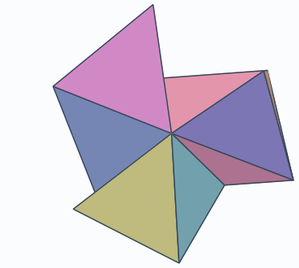
8-195



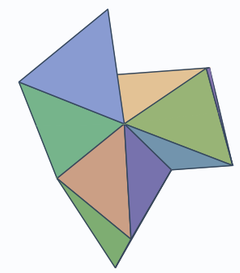
8-196



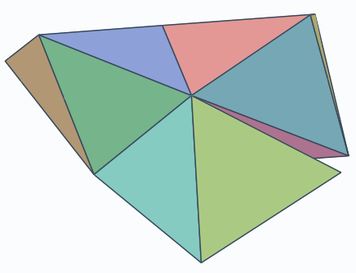
8-200



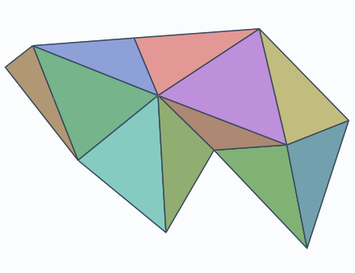
8-202



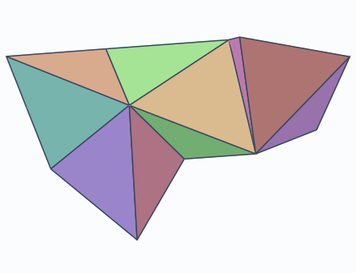
8-204



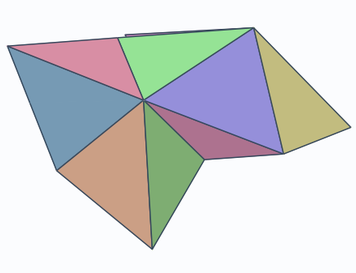
8-209



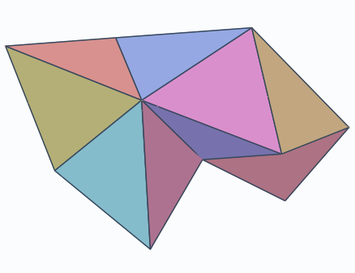
8-223



8-227



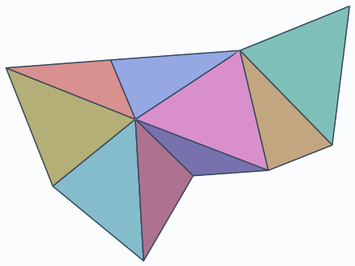
8-237



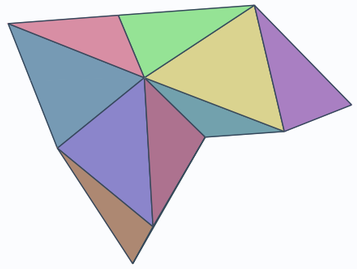
8-238

Solutions P8A - 8 tetrahedra / targets continued

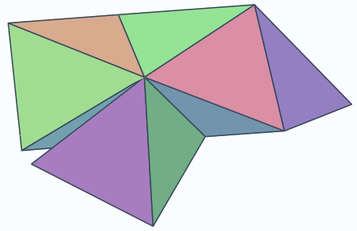
101 matching shapes out of 398. Matching colors identify the net panels on each target.



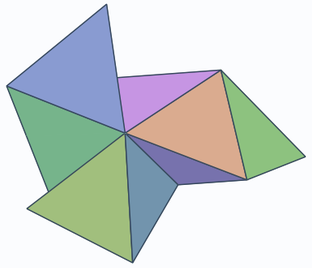
8-239



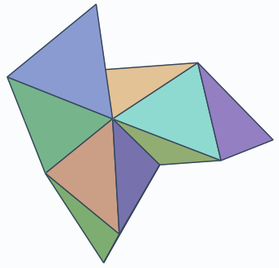
8-245



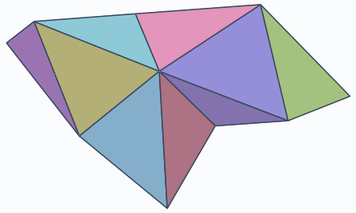
8-246



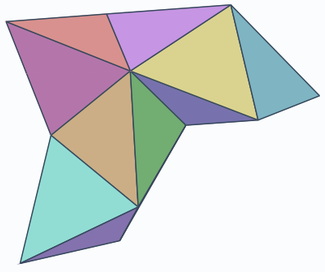
8-251



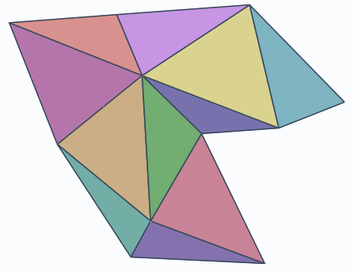
8-253



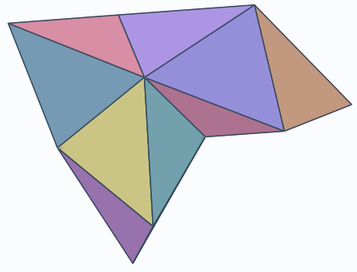
8-256



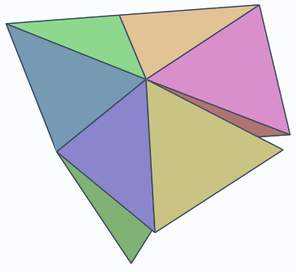
8-261



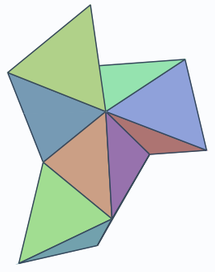
8-262



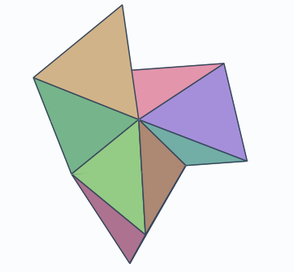
8-263



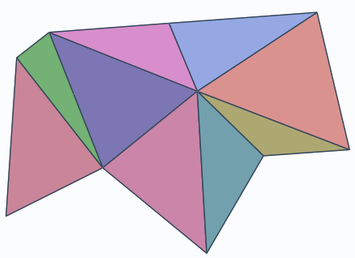
8-267



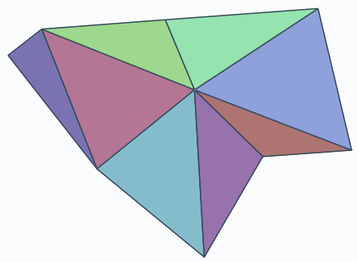
8-273



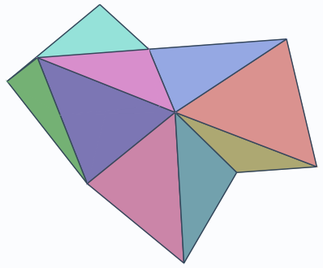
8-275



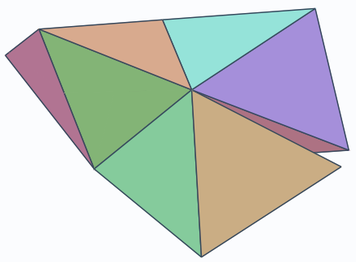
8-286



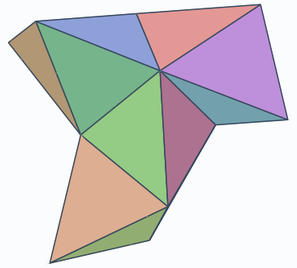
8-287



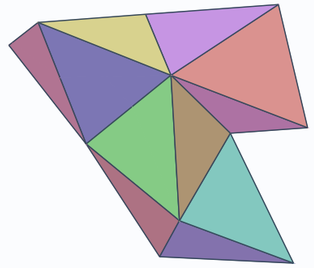
8-288



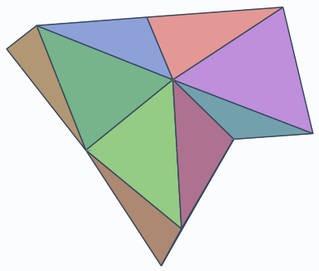
8-289



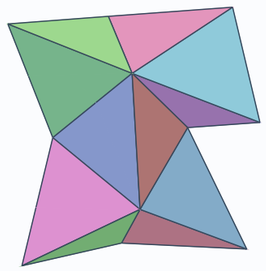
8-291



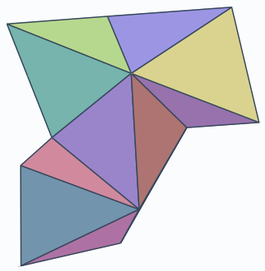
8-292



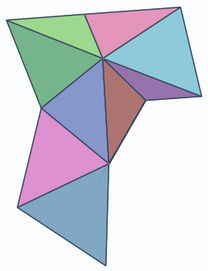
8-293



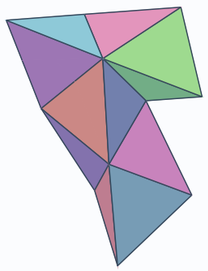
8-294



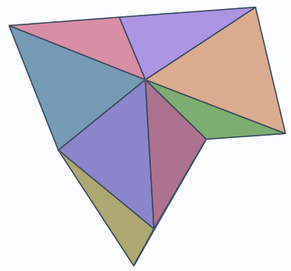
8-295



8-296



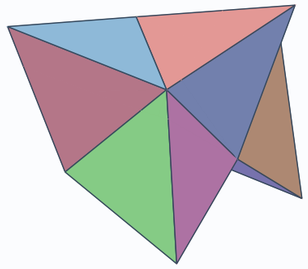
8-298



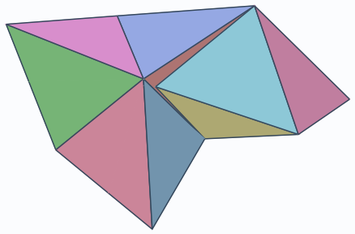
8-302

Solutions P8A - 8 tetrahedra / targets continued

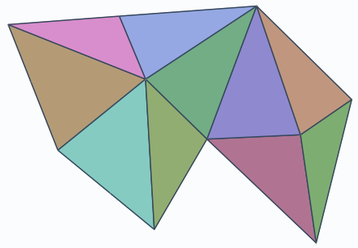
101 matching shapes out of 398. Matching colors identify the net panels on each target.



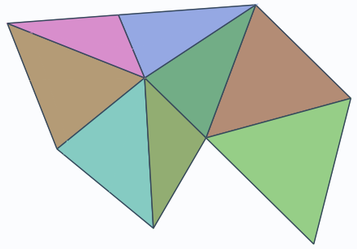
8-308



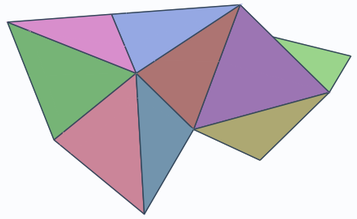
8-316



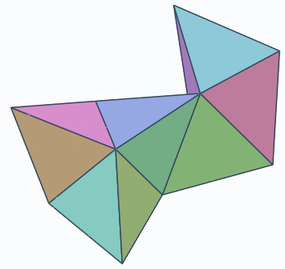
8-317



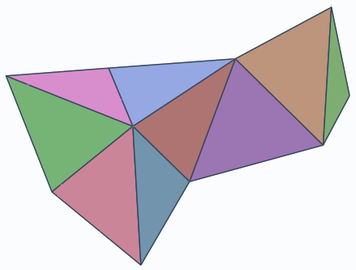
8-321



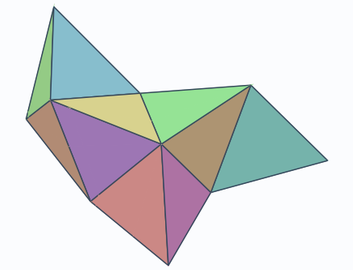
8-323



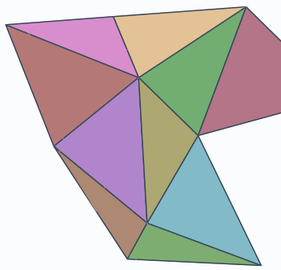
8-329



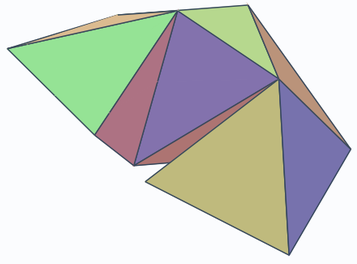
8-330



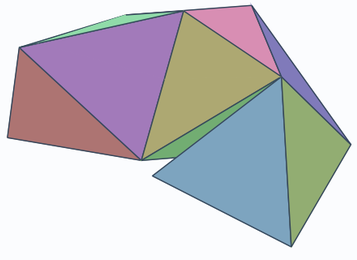
8-334



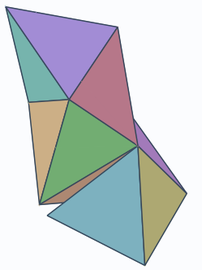
8-336



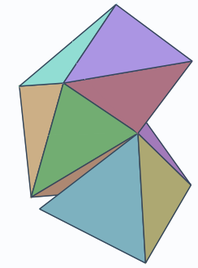
8-343



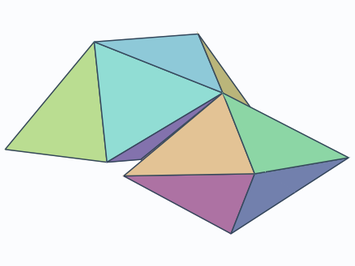
8-344



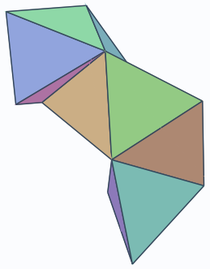
8-348



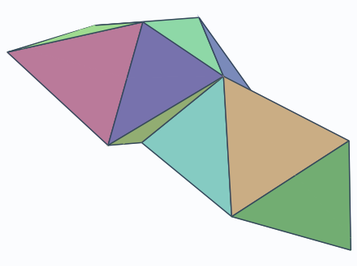
8-349



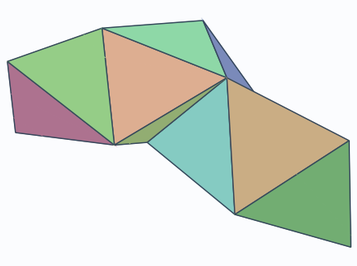
8-352



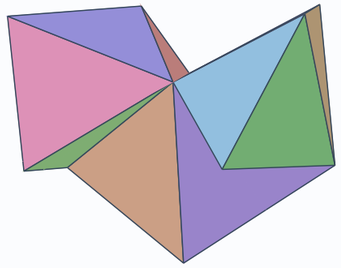
8-374



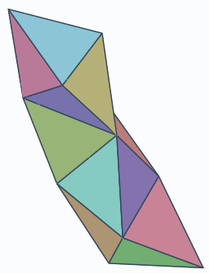
8-380



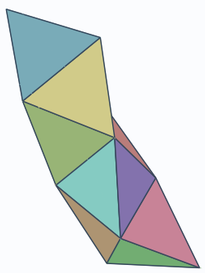
8-382



8-384



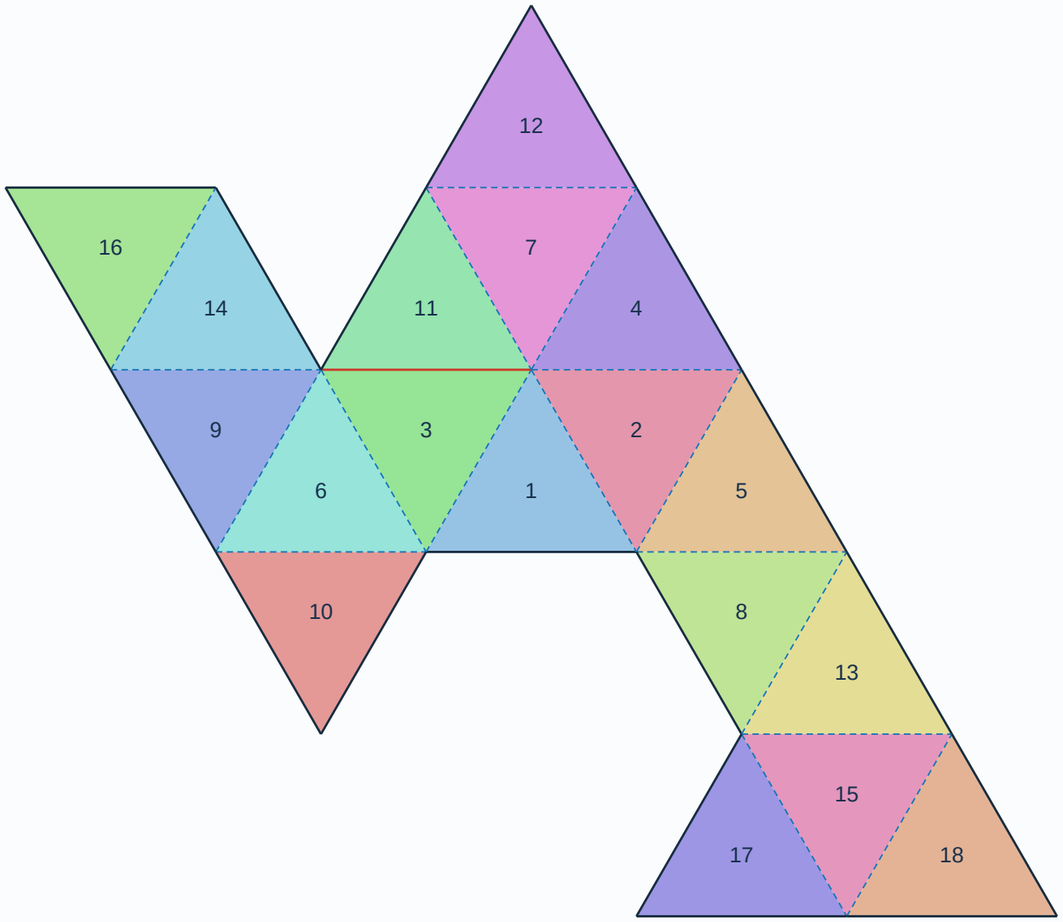
8-397



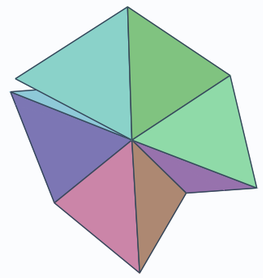
8-398

Solutions P8B - 8 tetrahedra

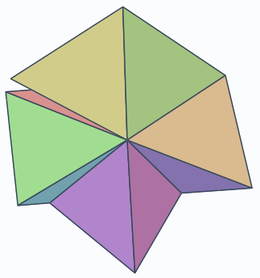
98 matching shapes out of 398. Matching colors identify the net panels on each target.



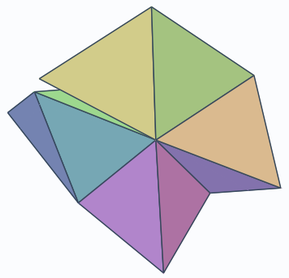
P8B 18 panels; 1 internal slit.



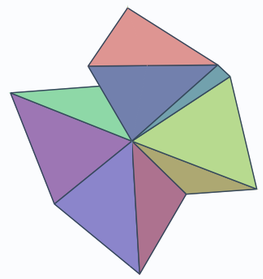
8-003



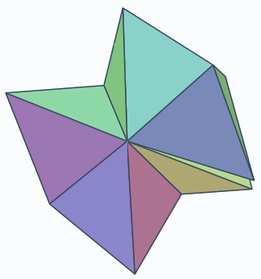
8-017



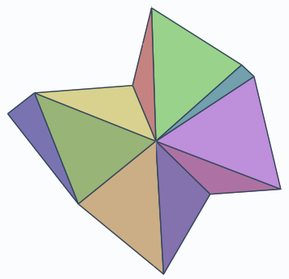
8-018



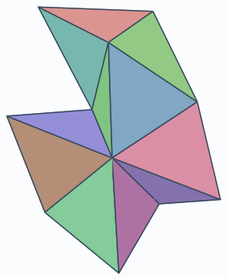
8-025



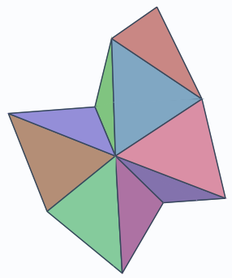
8-026



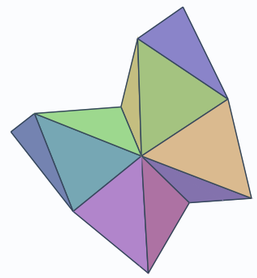
8-029



8-032



8-034



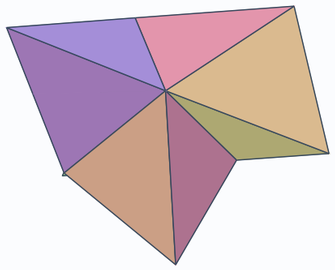
8-036

Selected from 500 sampled source trees; not proved optimal.

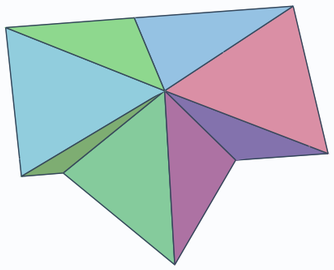
All remaining targets follow on the next page(s).

Solutions P8B - 8 tetrahedra / targets continued

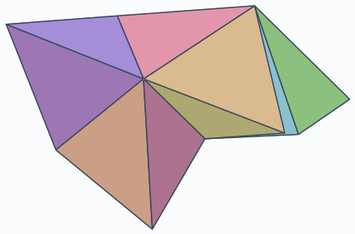
98 matching shapes out of 398. Matching colors identify the net panels on each target.



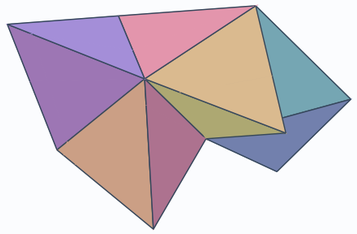
8-038



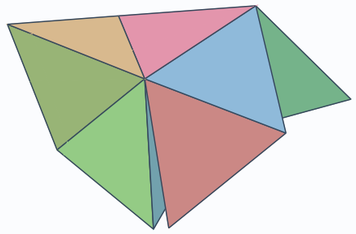
8-045



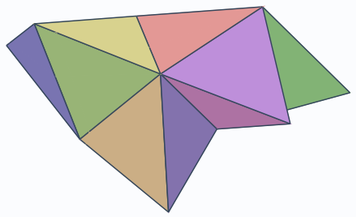
8-051



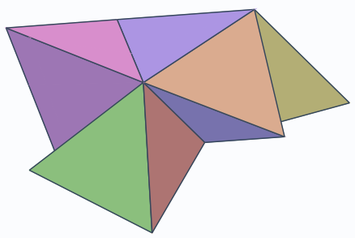
8-052



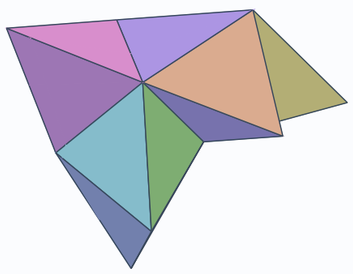
8-055



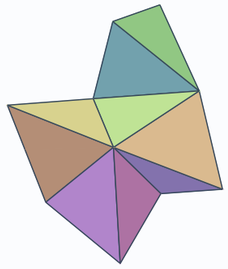
8-059



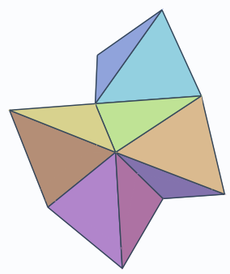
8-060



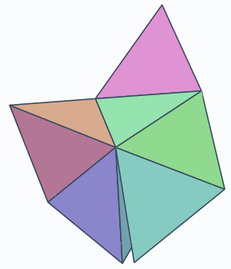
8-062



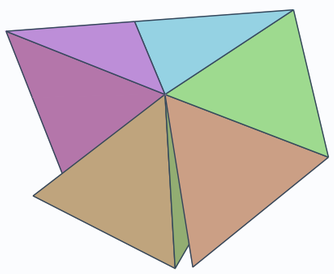
8-063



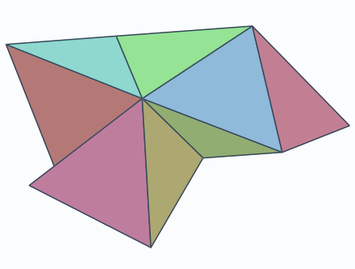
8-064



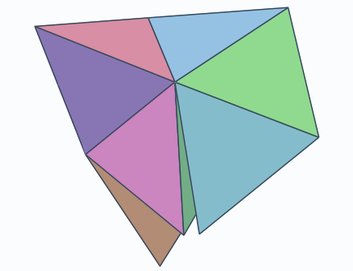
8-066



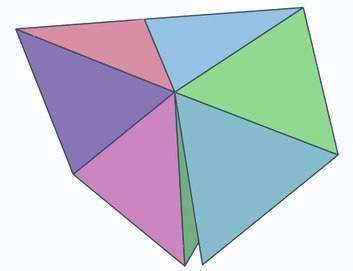
8-076



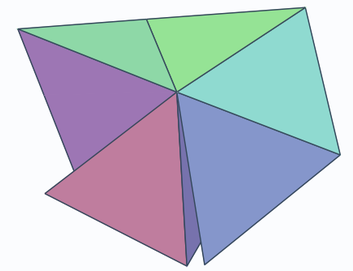
8-086



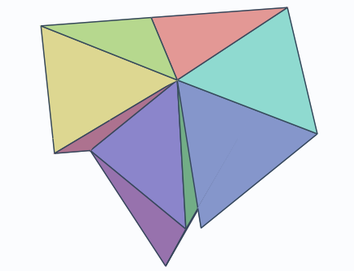
8-098



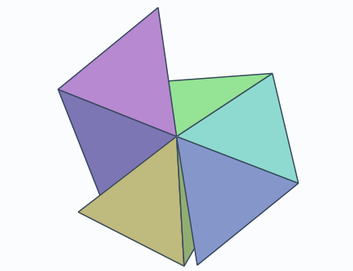
8-100



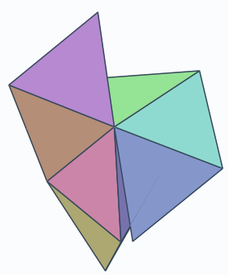
8-102



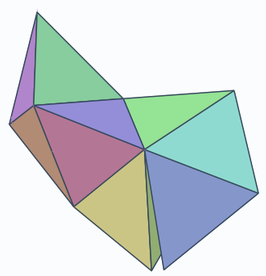
8-106



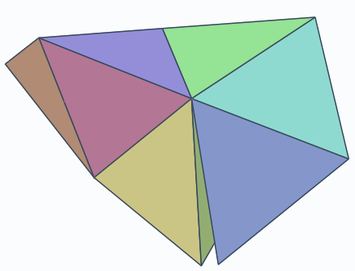
8-115



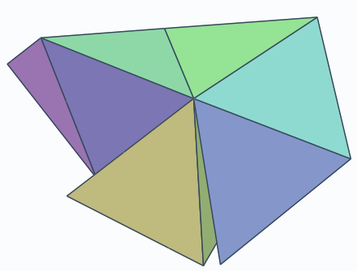
8-116



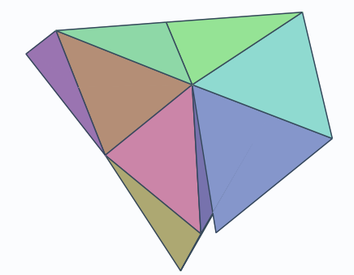
8-118



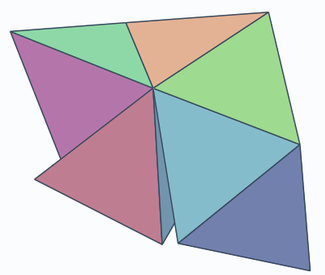
8-119



8-122



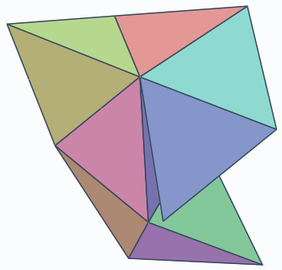
8-123



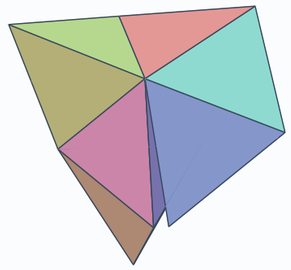
8-133

Solutions P8B - 8 tetrahedra / targets continued

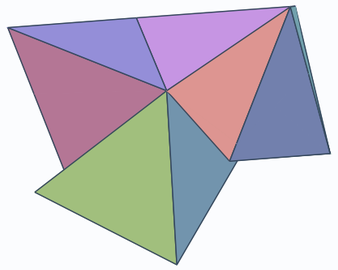
98 matching shapes out of 398. Matching colors identify the net panels on each target.



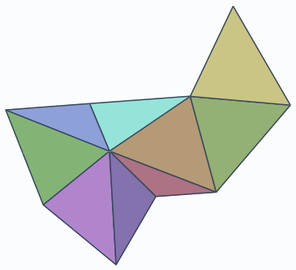
8-138



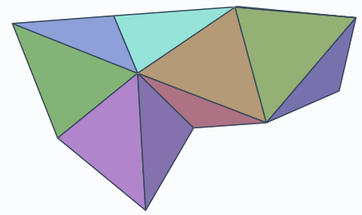
8-139



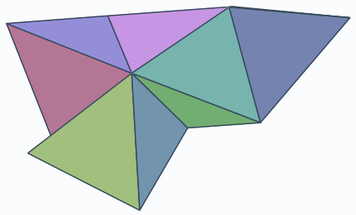
8-159



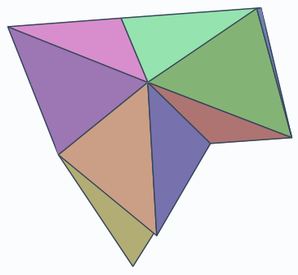
8-177



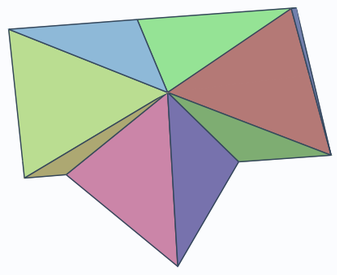
8-178



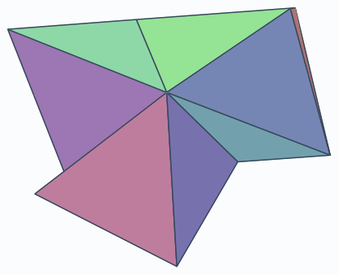
8-183



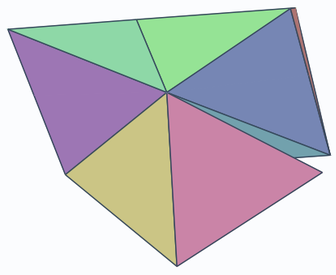
8-186



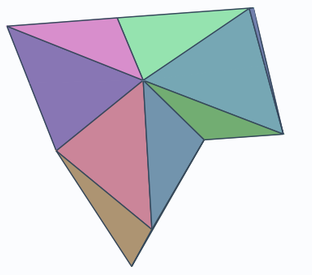
8-189



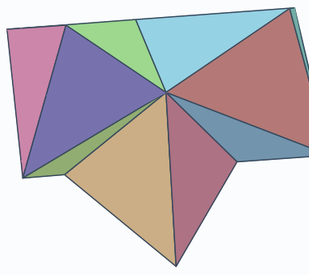
8-192



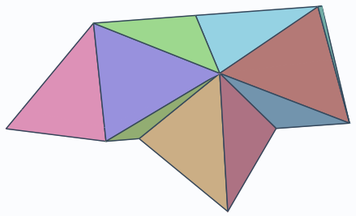
8-193



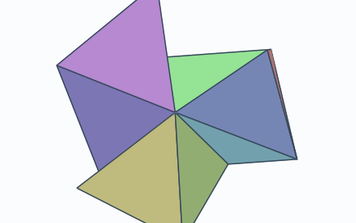
8-194



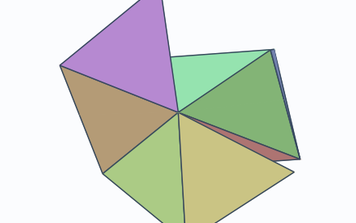
8-198



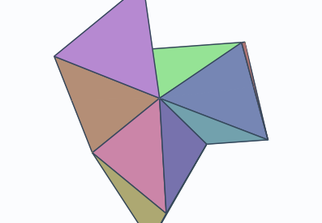
8-199



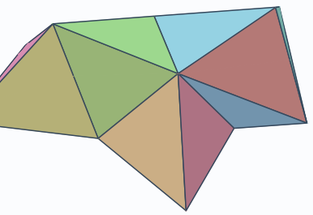
8-202



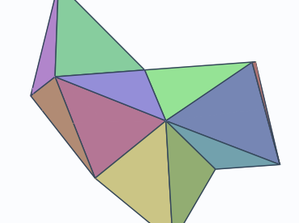
8-203



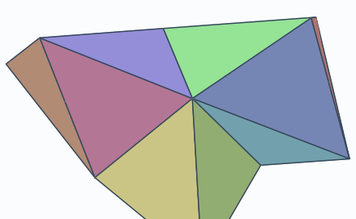
8-204



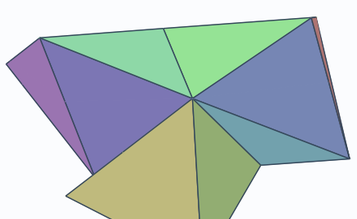
8-205



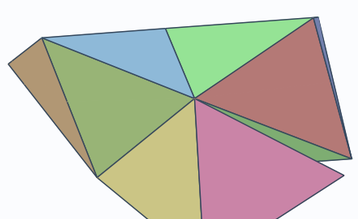
8-206



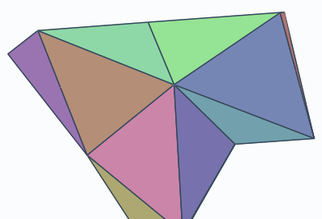
8-207



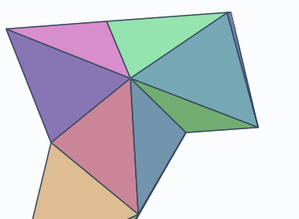
8-208



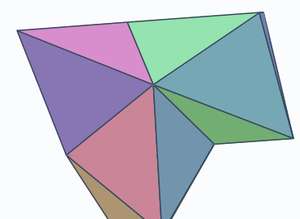
8-209



8-210



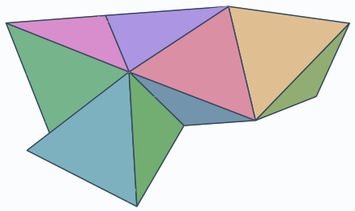
8-214



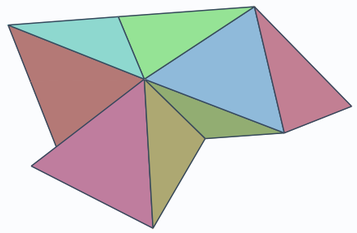
8-216

Solutions P8B - 8 tetrahedra / targets continued

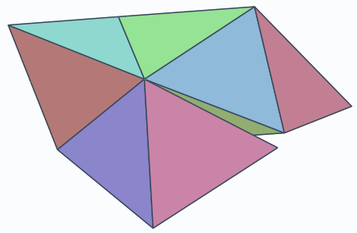
98 matching shapes out of 398. Matching colors identify the net panels on each target.



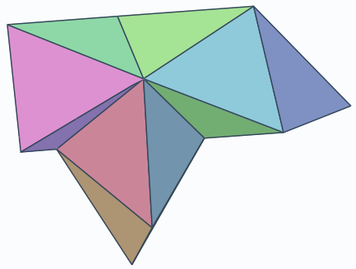
8-234



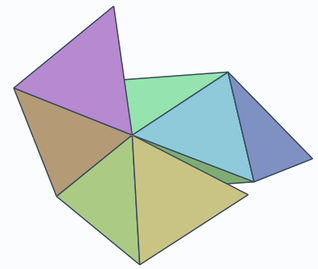
8-243



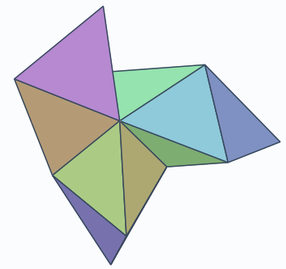
8-244



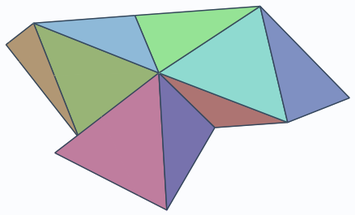
8-248



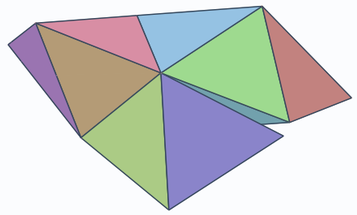
8-252



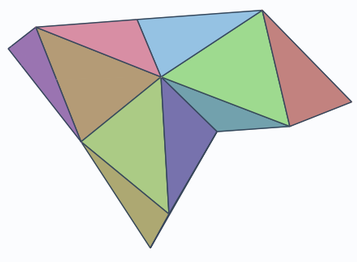
8-253



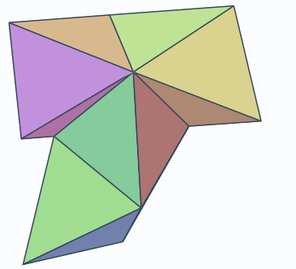
8-257



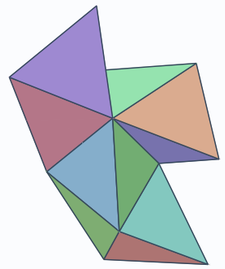
8-258



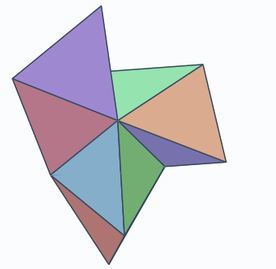
8-259



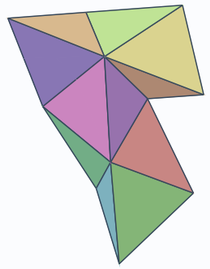
8-271



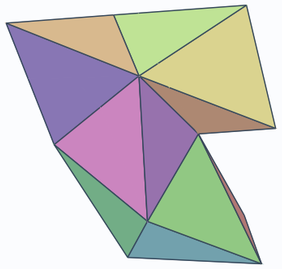
8-274



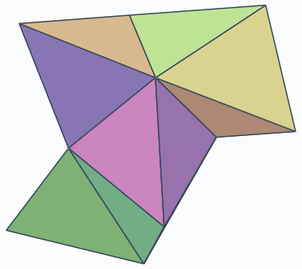
8-275



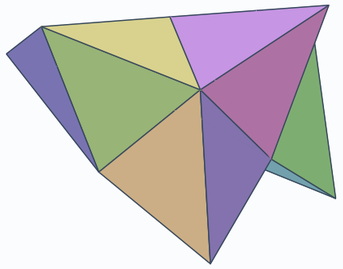
8-278



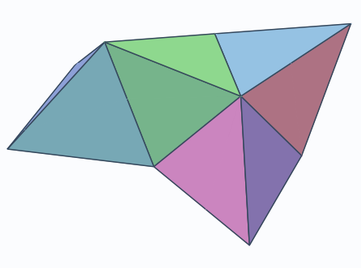
8-282



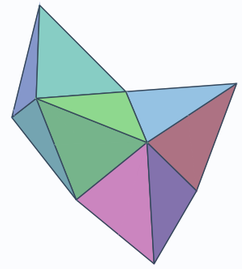
8-283



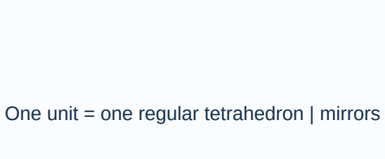
8-290



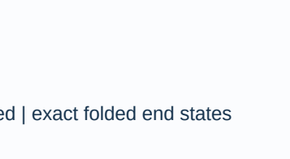
8-292



8-293



8-298

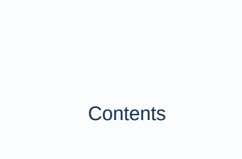


8-300

8-301

8-310

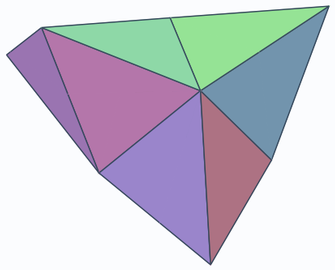
8-312



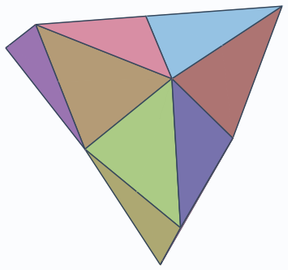
8-313

Solutions P8B - 8 tetrahedra / targets continued

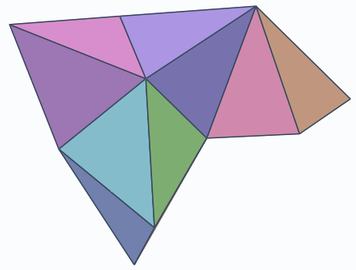
98 matching shapes out of 398. Matching colors identify the net panels on each target.



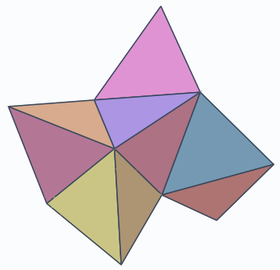
8-314



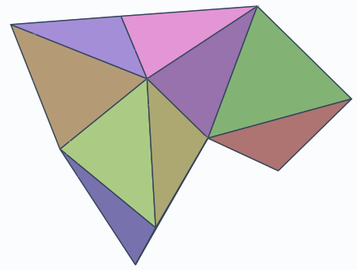
8-315



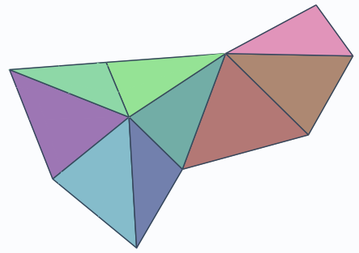
8-320



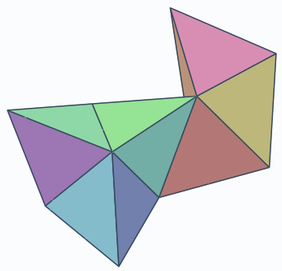
8-325



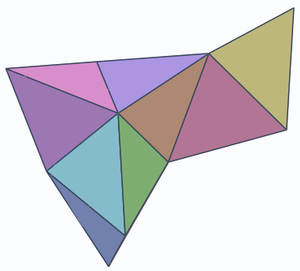
8-327



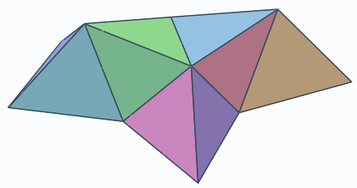
8-328



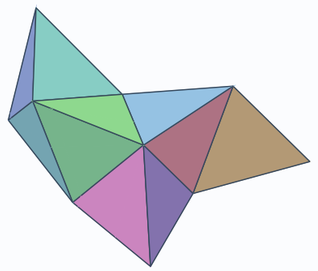
8-329



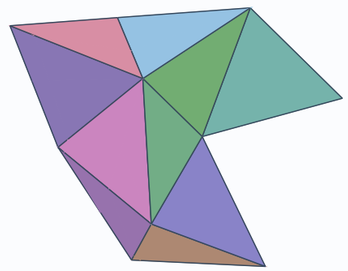
8-332



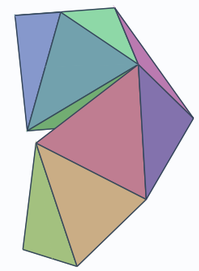
8-333



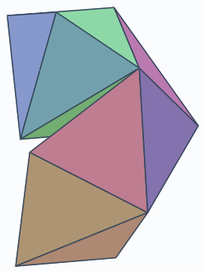
8-334



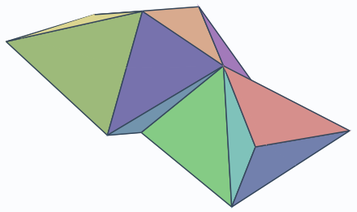
8-336



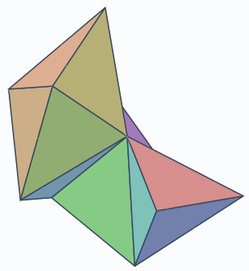
8-340



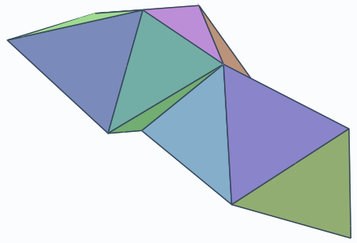
8-342



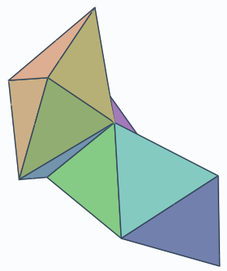
8-363



8-365



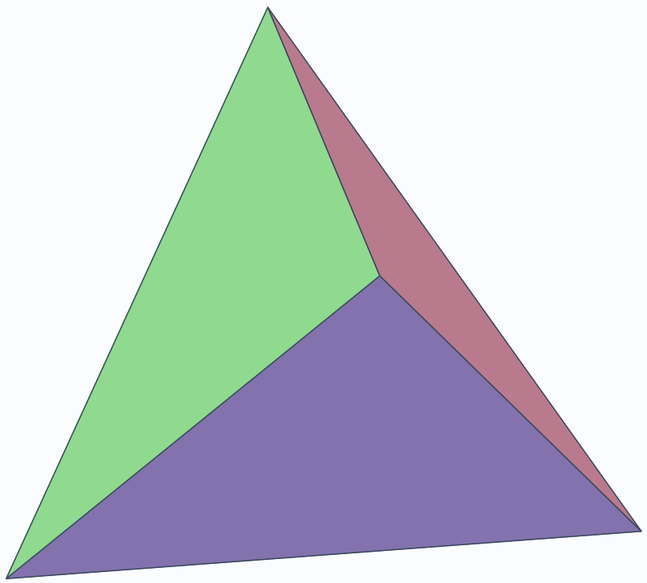
8-380



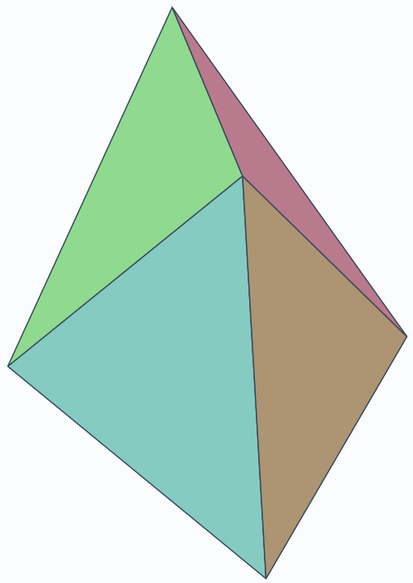
8-381

Part III - Complete catalog: one through four units

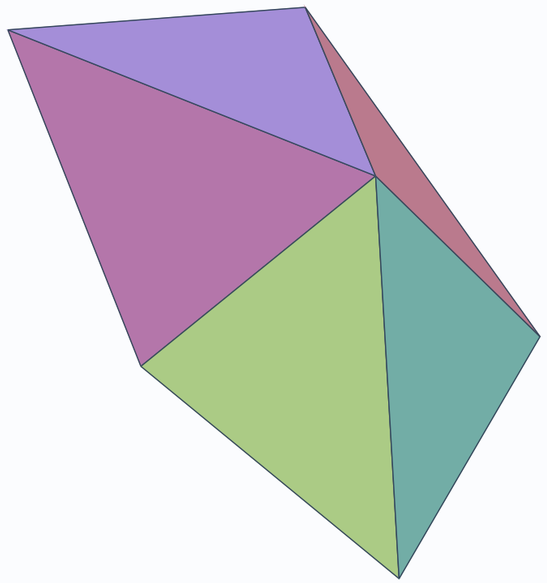
All six shapes at these volumes. IDs give the number of units followed by the shape number.



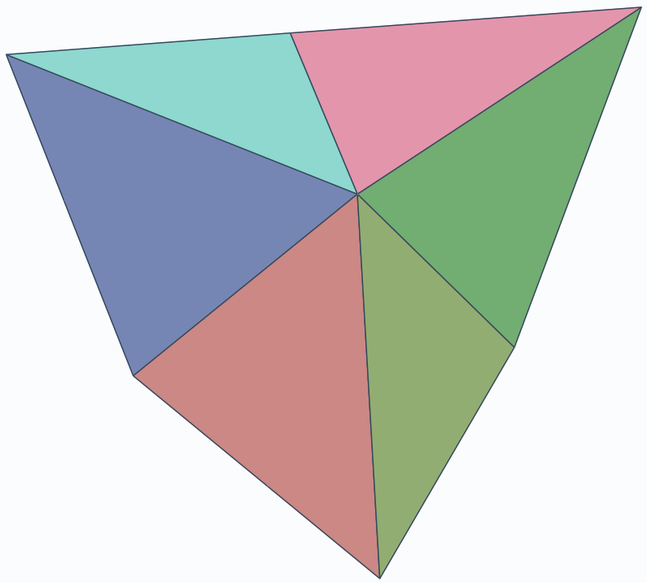
1-001



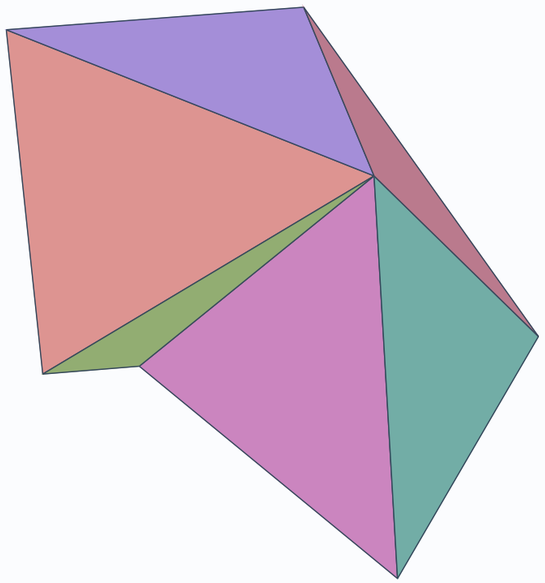
2-001



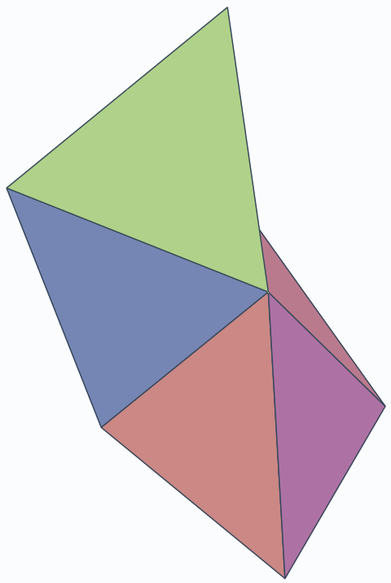
3-001



4-001



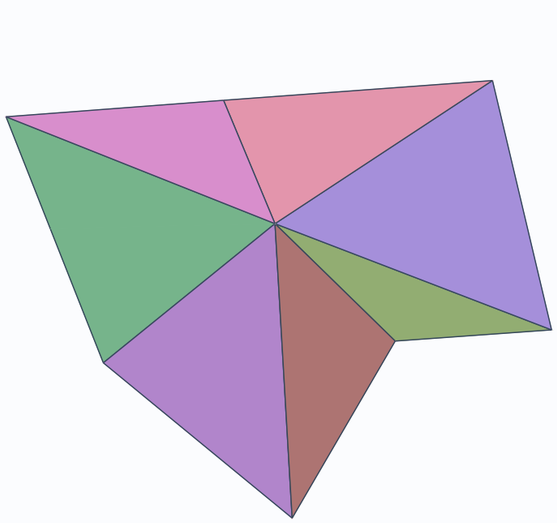
4-002



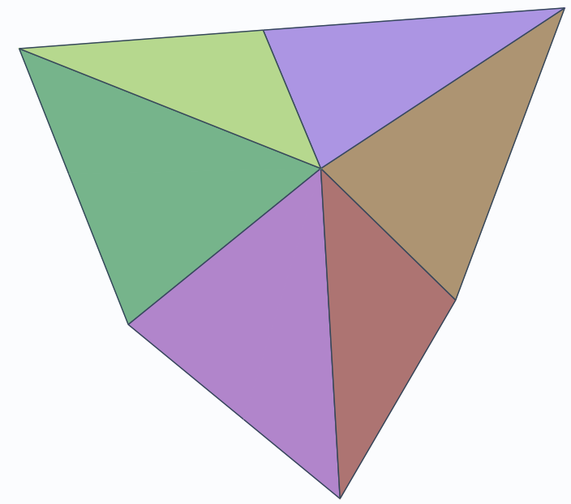
4-003

Complete catalog - 5 tetrahedra

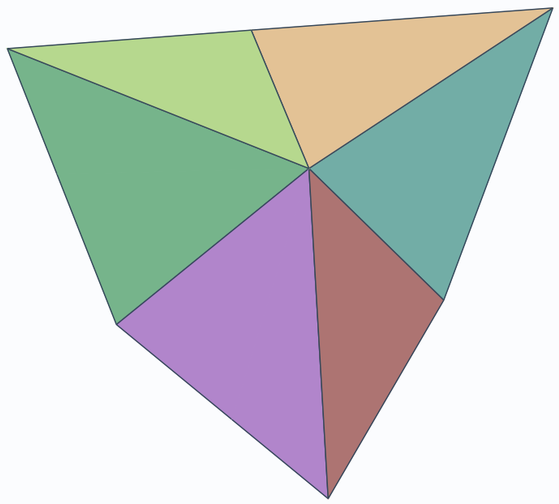
7 shapes in all; 12 triangular panels on each surface. Shapes 001-007.



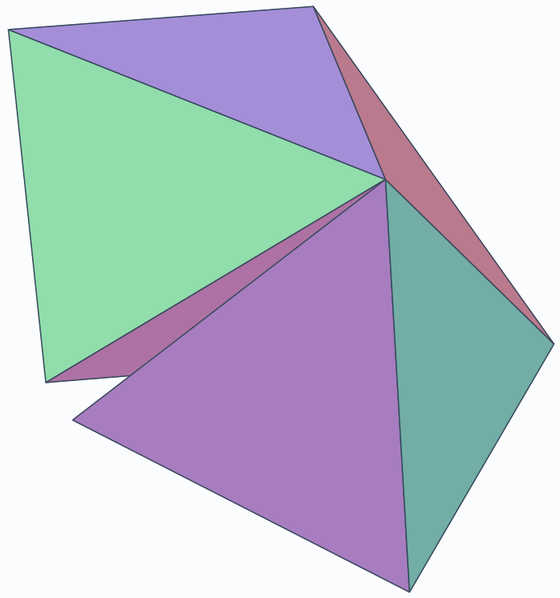
5-001



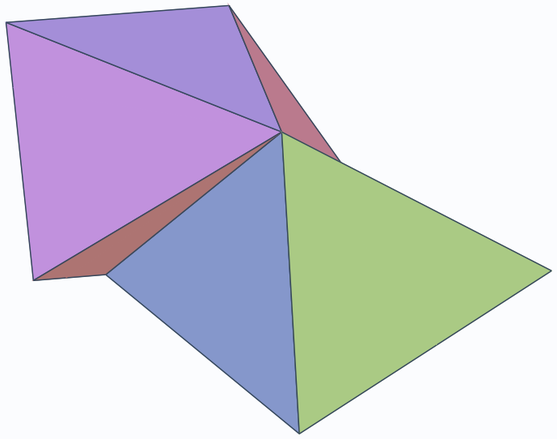
5-002



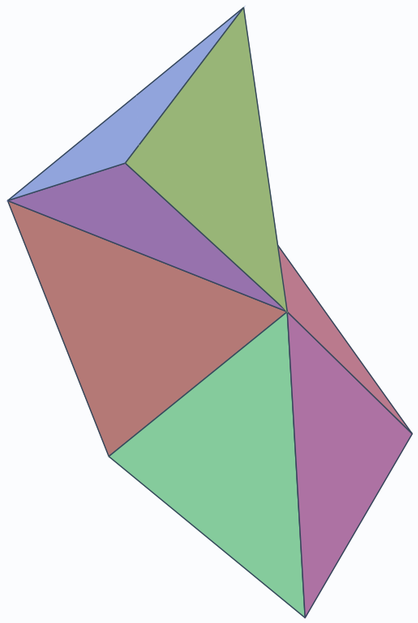
5-003



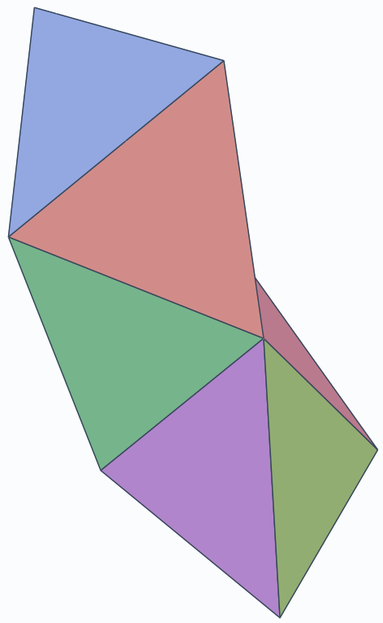
5-004



5-005



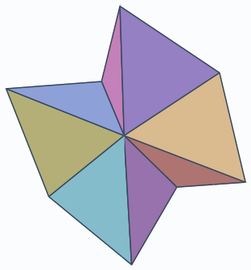
5-006



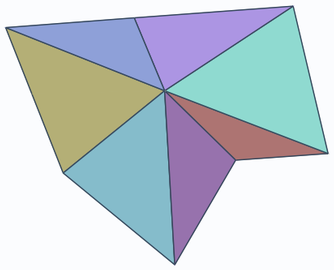
5-007

Complete catalog - 6 tetrahedra

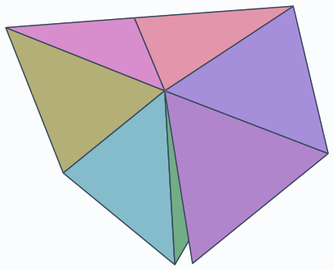
23 shapes in all; 14 triangular panels on each surface. Shapes 001-023.



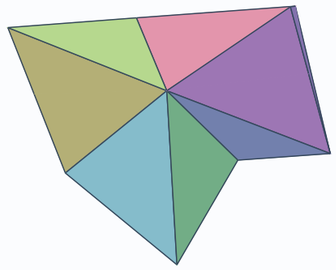
6-001



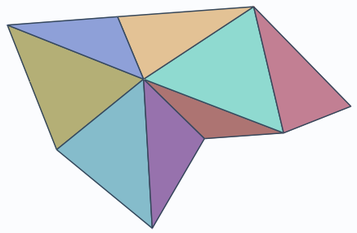
6-002



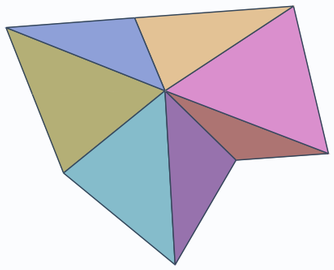
6-003



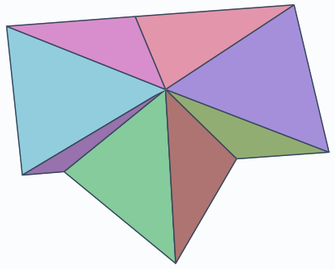
6-004



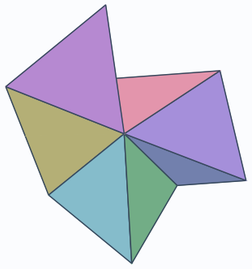
6-005



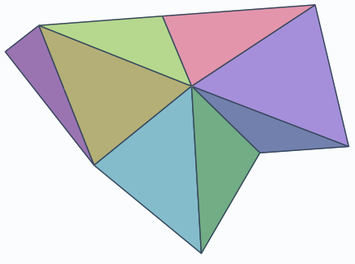
6-006



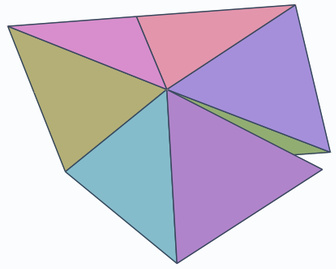
6-007



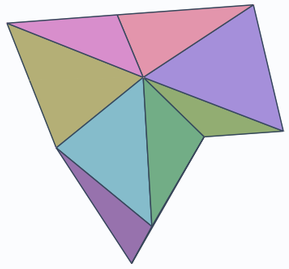
6-008



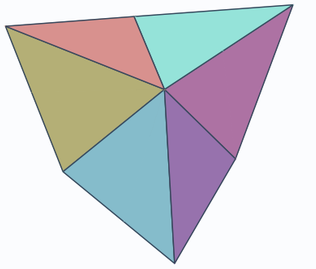
6-009



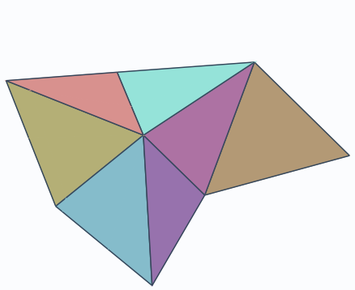
6-010



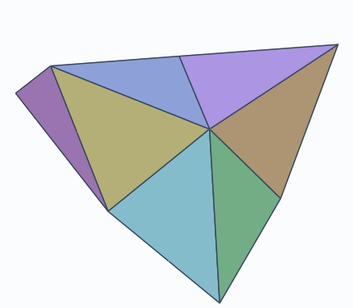
6-011



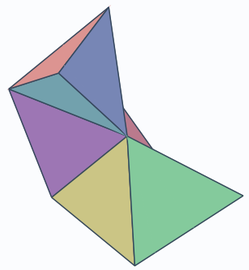
6-012



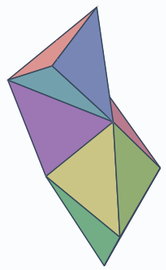
6-013



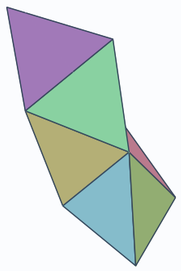
6-014



6-015

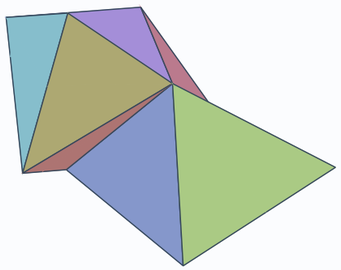


6-016

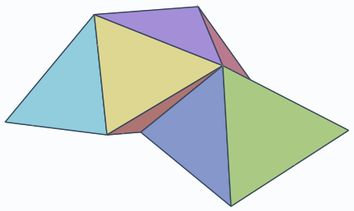


6-017

6-018



6-019



6-020

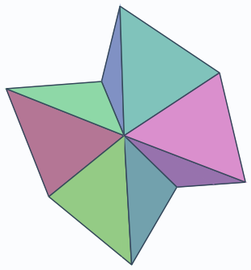
6-021

6-022

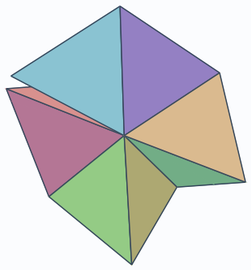
6-023

Complete catalog - 7 tetrahedra

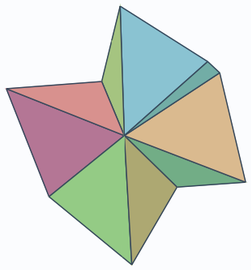
89 shapes in all; 16 triangular panels on each surface. Shapes 001-024.



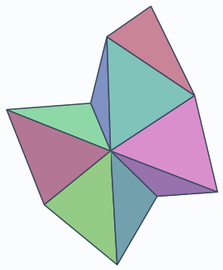
7-001



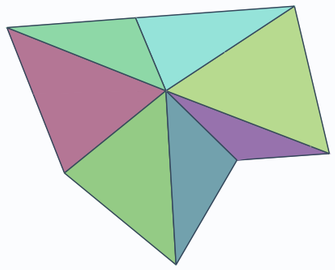
7-002



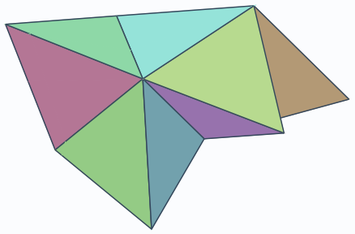
7-003



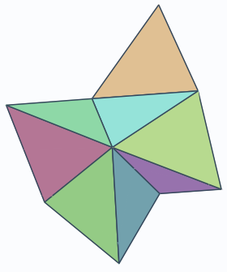
7-004



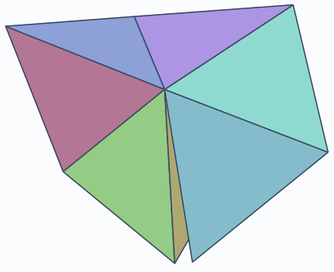
7-005



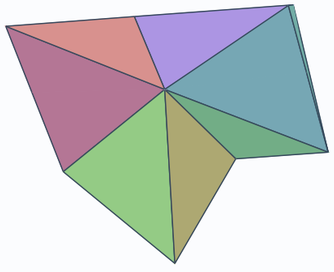
7-006



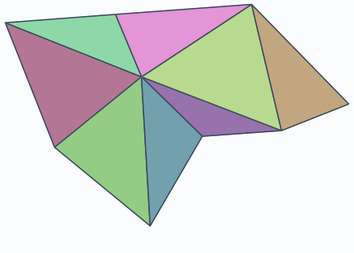
7-007



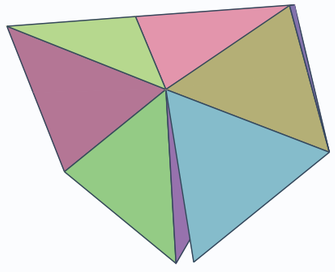
7-008



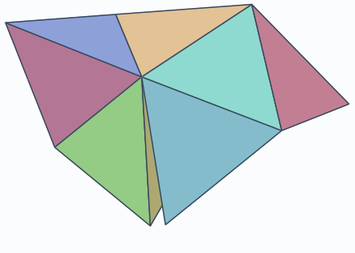
7-009



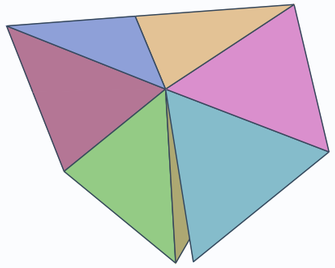
7-010



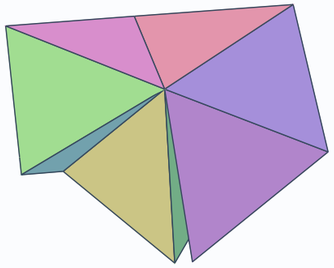
7-011



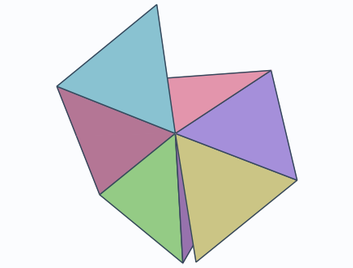
7-012



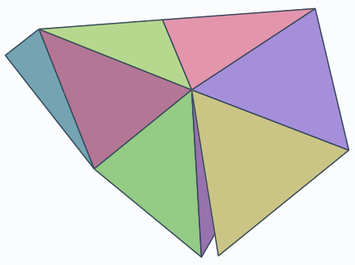
7-013



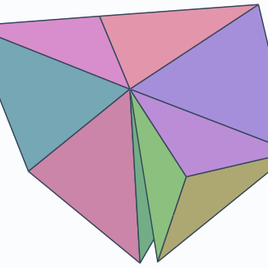
7-014



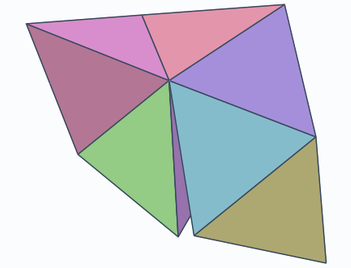
7-015



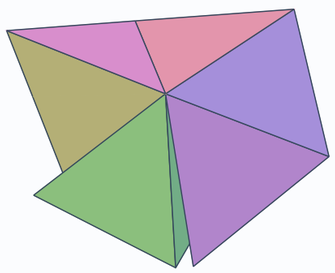
7-016



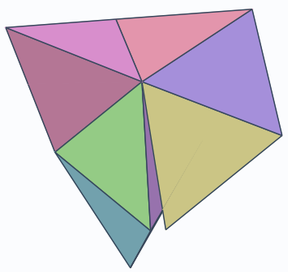
7-017



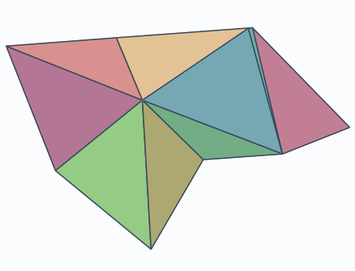
7-018



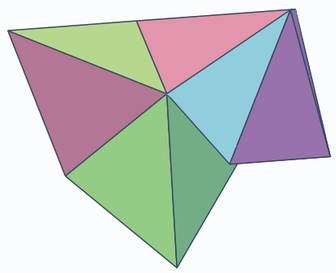
7-019



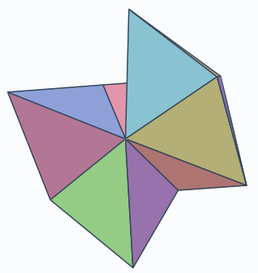
7-020



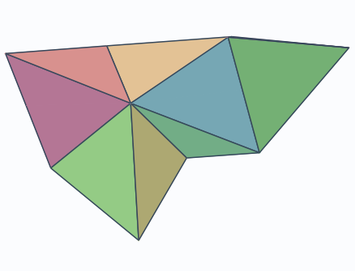
7-021



7-022



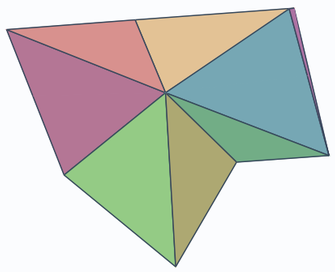
7-023



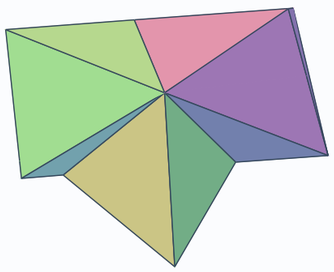
7-024

Complete catalog - 7 tetrahedra

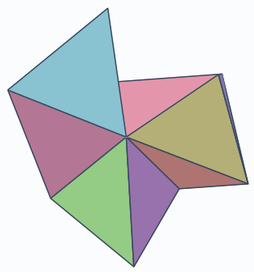
89 shapes in all; 16 triangular panels on each surface. Shapes 025-048.



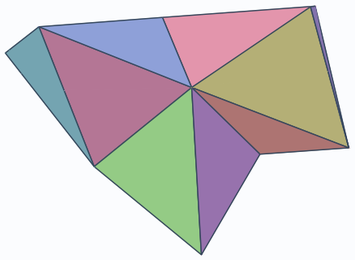
7-025



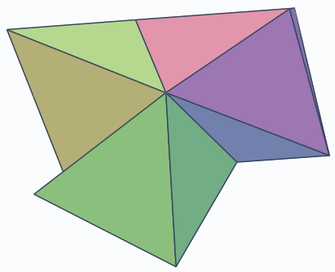
7-026



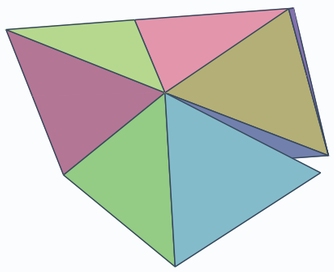
7-027



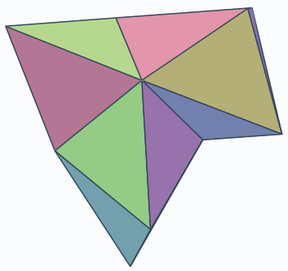
7-028



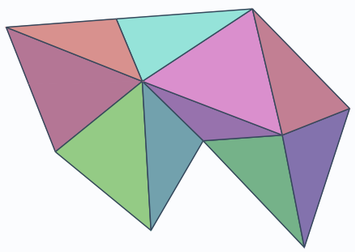
7-029



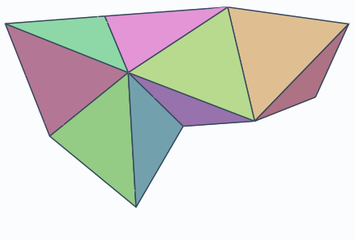
7-030



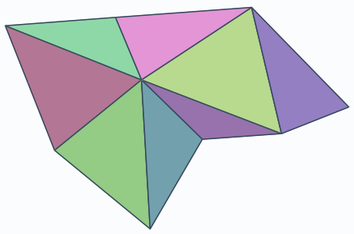
7-031



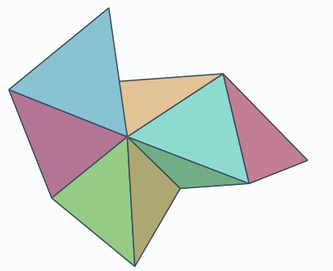
7-032



7-033

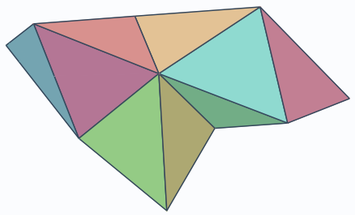


7-034

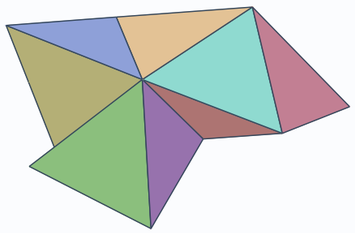


7-035

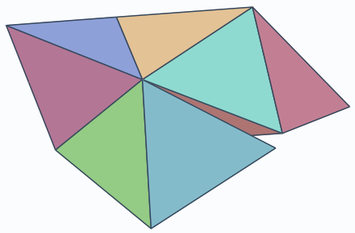
7-036



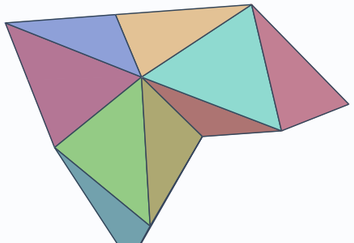
7-037



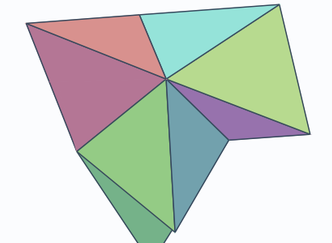
7-038



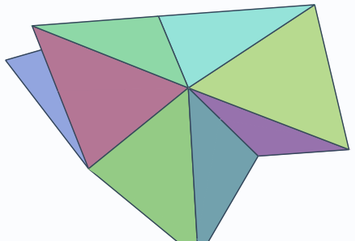
7-039



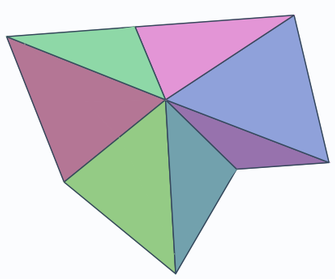
7-040



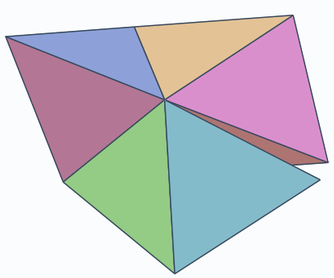
7-041



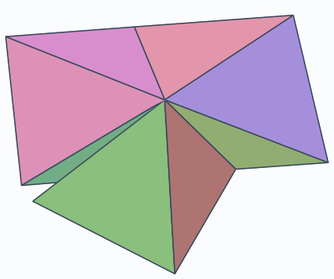
7-042



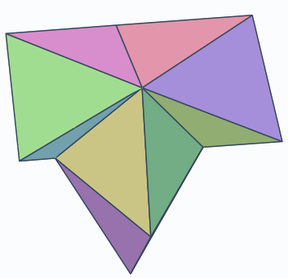
7-043



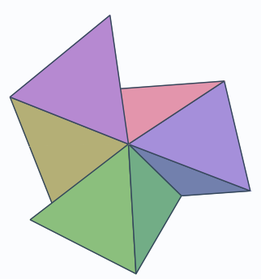
7-044



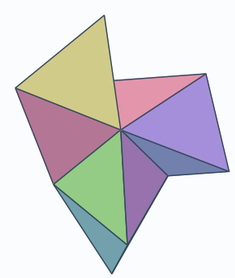
7-045



7-046



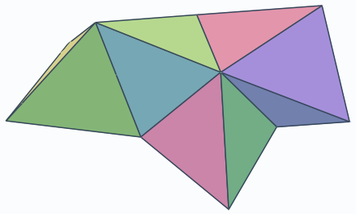
7-047



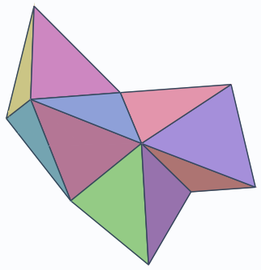
7-048

Complete catalog - 7 tetrahedra

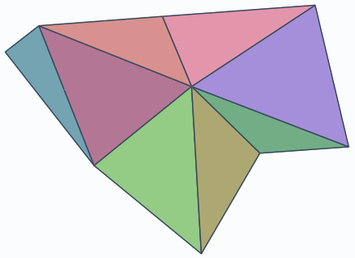
89 shapes in all; 16 triangular panels on each surface. Shapes 049-072.



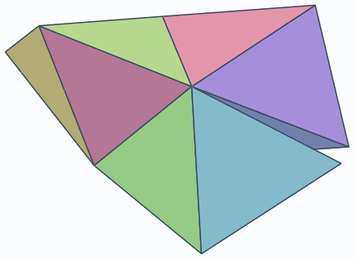
7-049



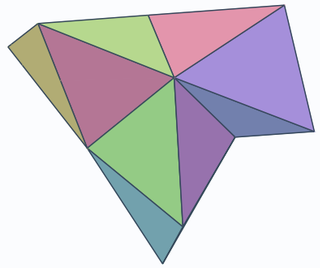
7-050



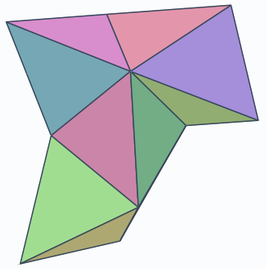
7-051



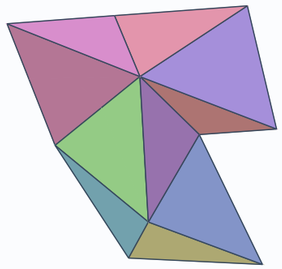
7-052



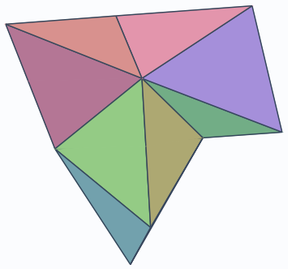
7-053



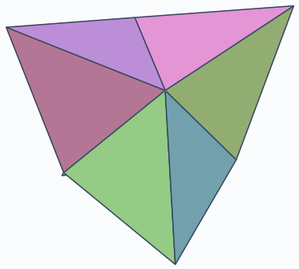
7-054



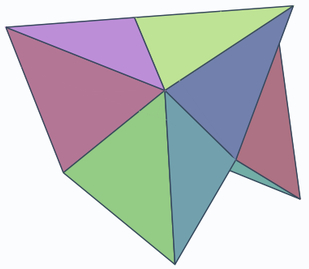
7-055



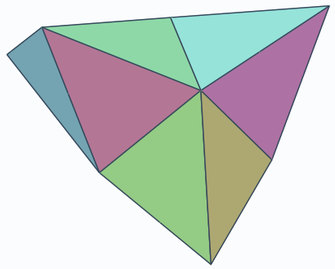
7-056



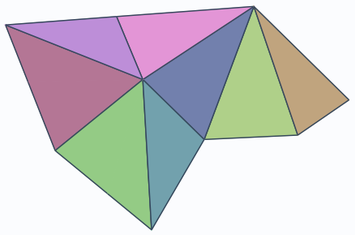
7-057



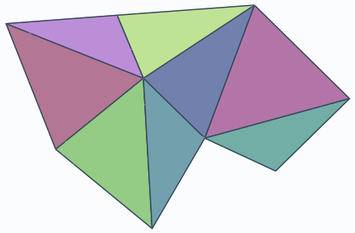
7-058



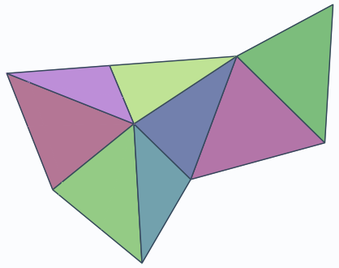
7-059



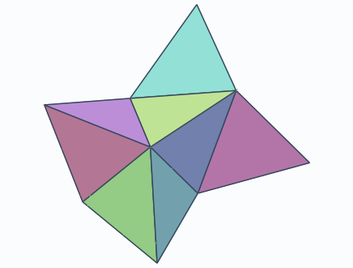
7-060



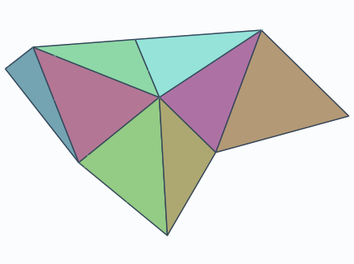
7-061



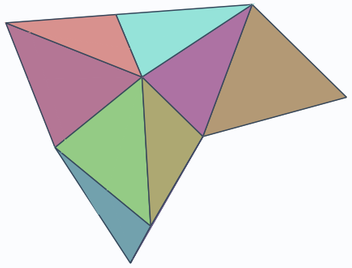
7-062



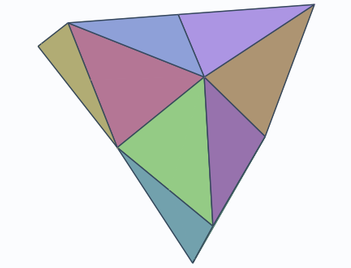
7-063



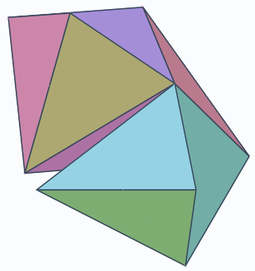
7-064



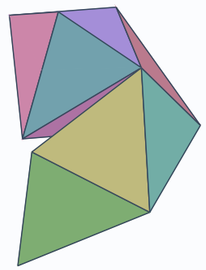
7-065



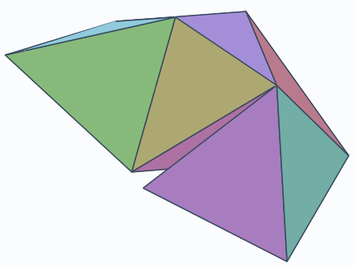
7-066



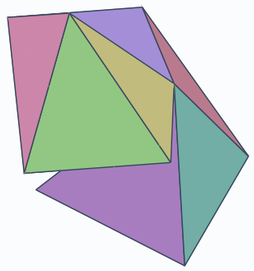
7-067



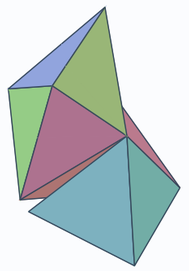
7-068



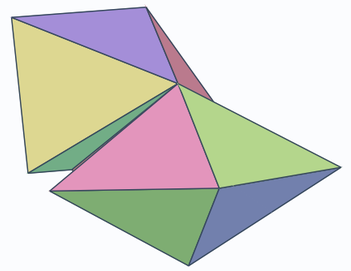
7-069



7-070



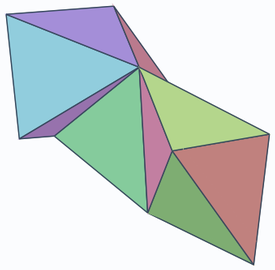
7-071



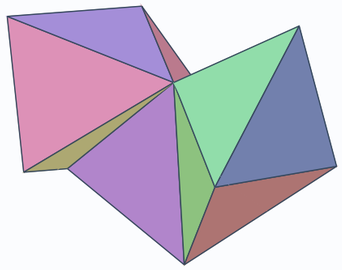
7-072

Complete catalog - 7 tetrahedra

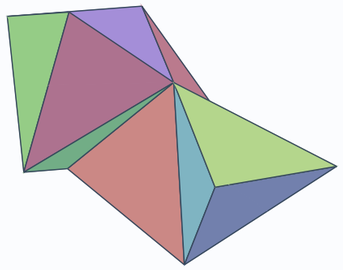
89 shapes in all; 16 triangular panels on each surface. Shapes 073-089.



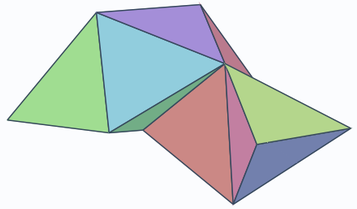
7-073



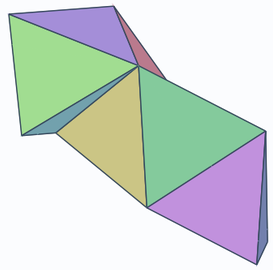
7-074



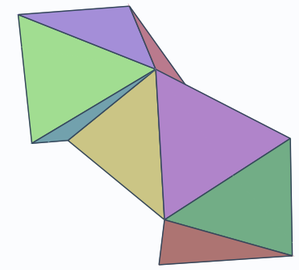
7-075



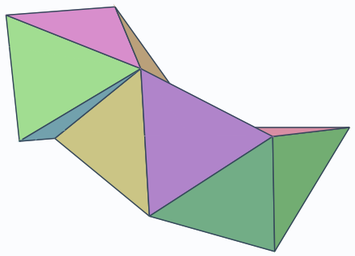
7-076



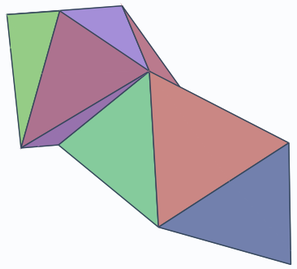
7-077



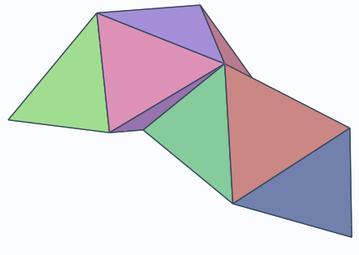
7-078



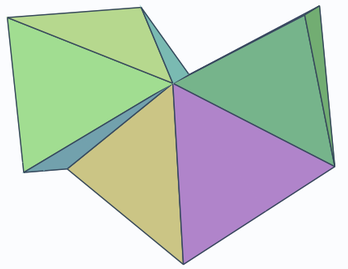
7-079



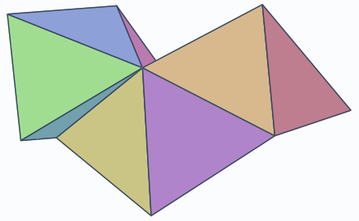
7-080



7-081

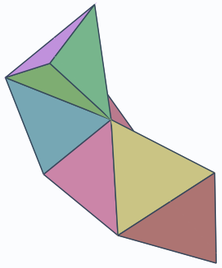


7-082

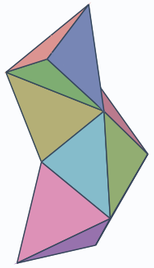


7-083

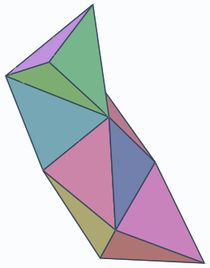
7-084



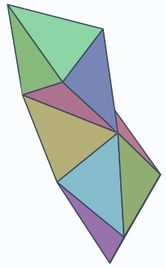
7-085



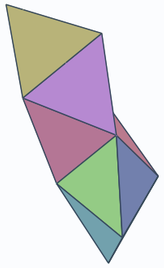
7-086



7-087



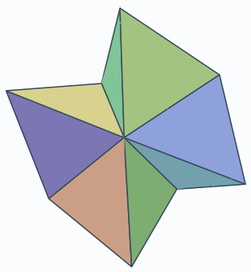
7-088



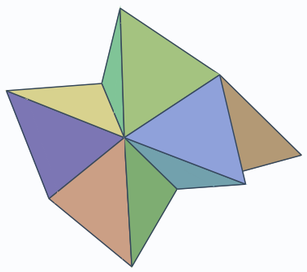
7-089

Complete catalog - 8 tetrahedra

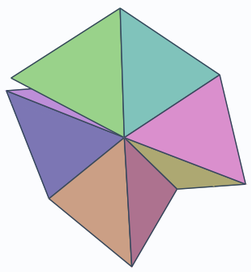
398 shapes in all; 18 triangular panels on each surface. Shapes 001-024.



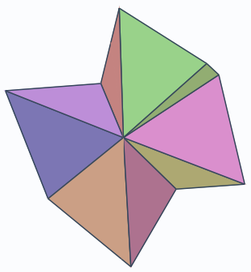
8-001



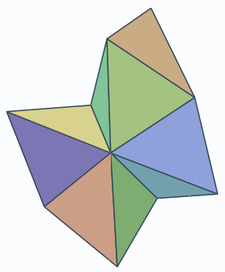
8-002



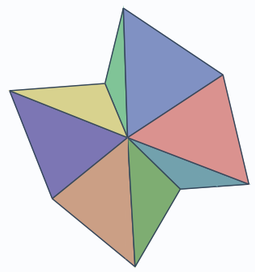
8-003



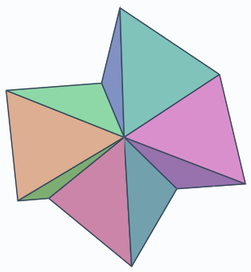
8-004



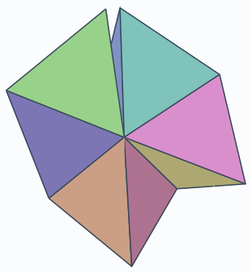
8-005



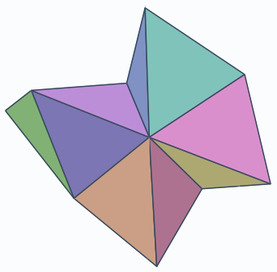
8-006



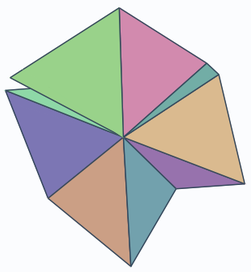
8-007



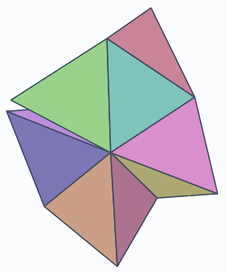
8-008



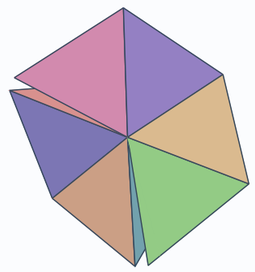
8-009



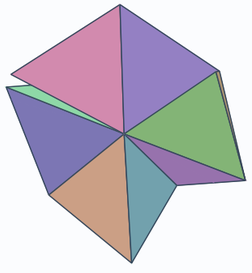
8-010



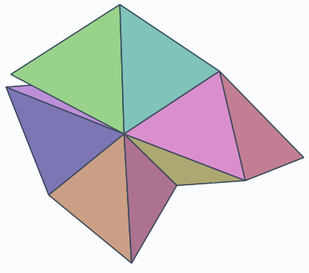
8-011



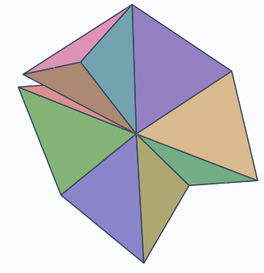
8-012



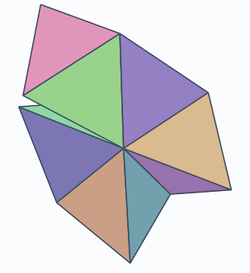
8-013



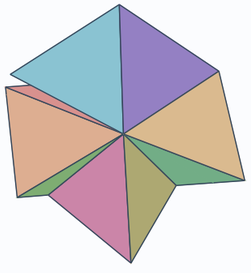
8-014



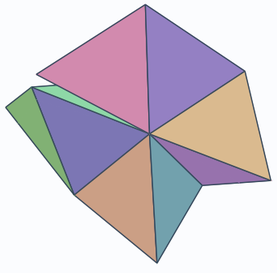
8-015



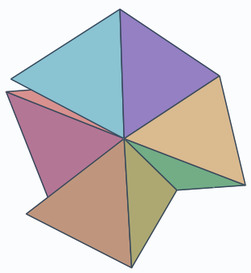
8-016



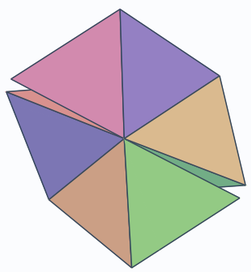
8-017



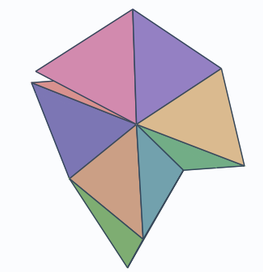
8-018



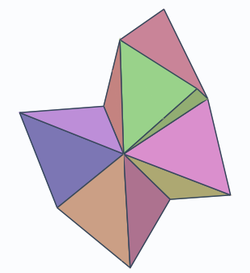
8-019



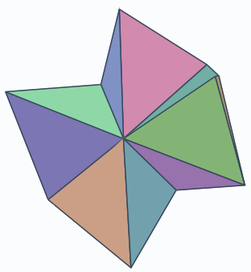
8-020



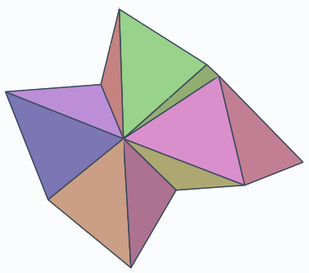
8-021



8-022



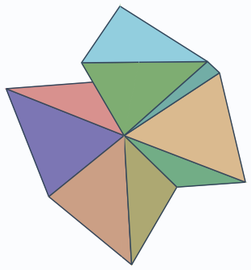
8-023



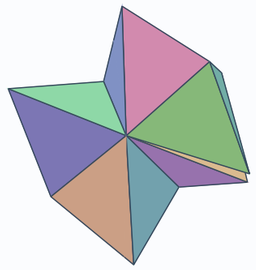
8-024

Complete catalog - 8 tetrahedra

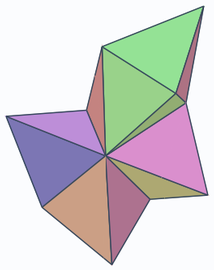
398 shapes in all; 18 triangular panels on each surface. Shapes 025-048.



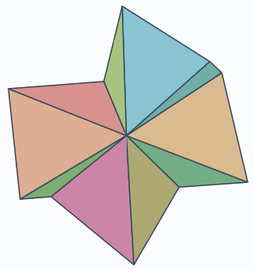
8-025



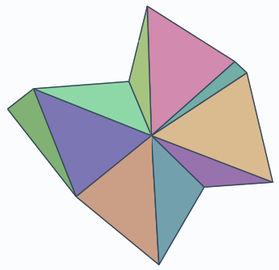
8-026



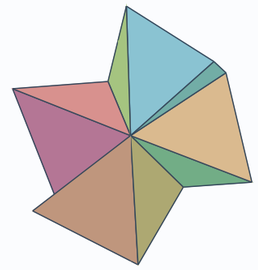
8-027



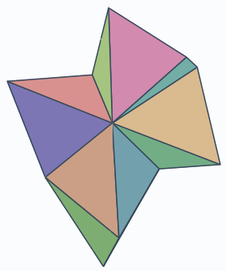
8-028



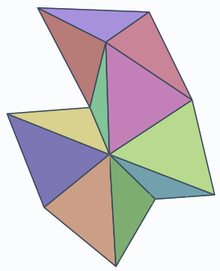
8-029



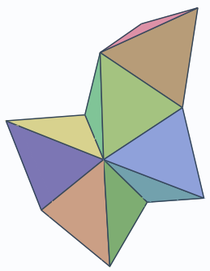
8-030



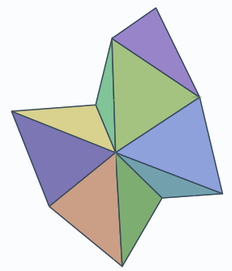
8-031



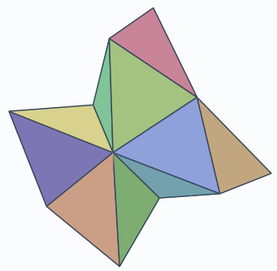
8-032



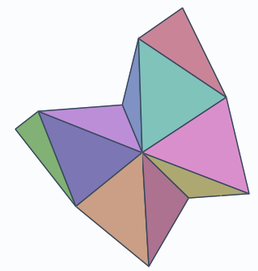
8-033



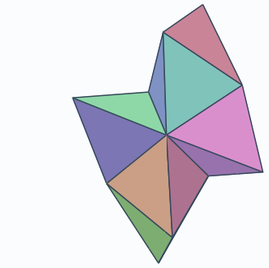
8-034



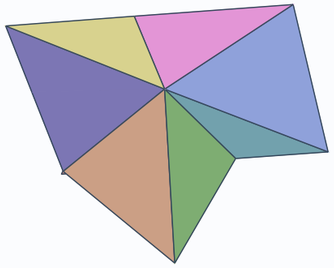
8-035



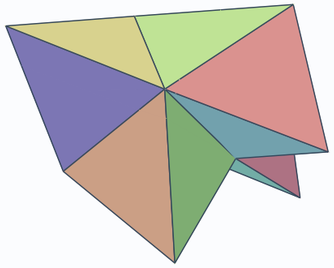
8-036



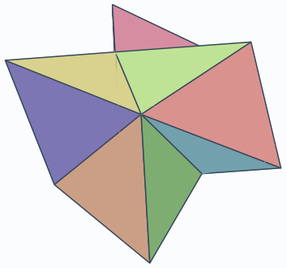
8-037



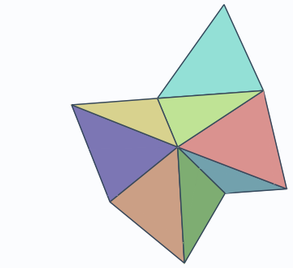
8-038



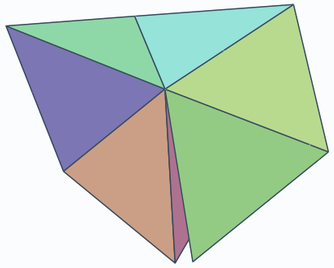
8-039



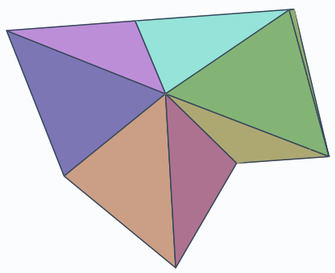
8-040



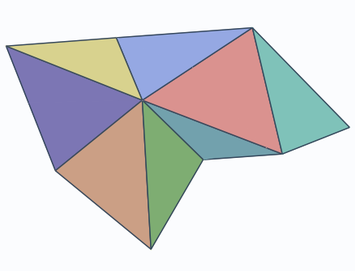
8-041



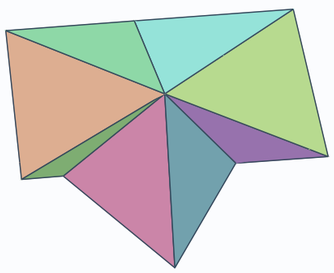
8-042



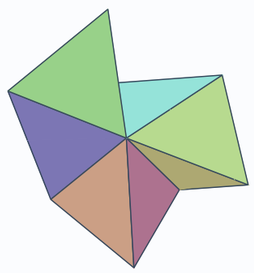
8-043



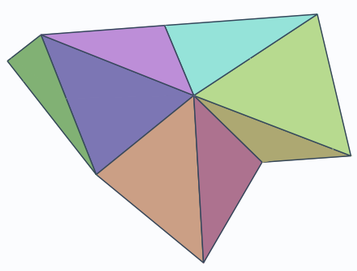
8-044



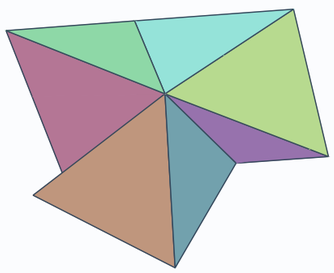
8-045



8-046



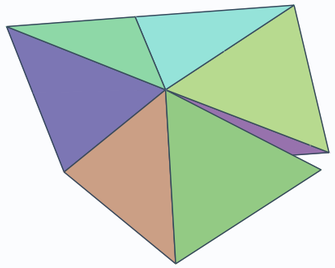
8-047



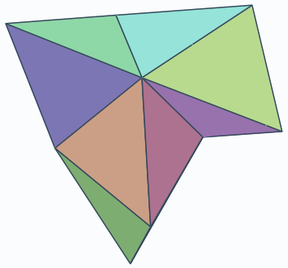
8-048

Complete catalog - 8 tetrahedra

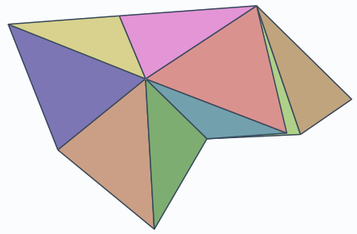
398 shapes in all; 18 triangular panels on each surface. Shapes 049-072.



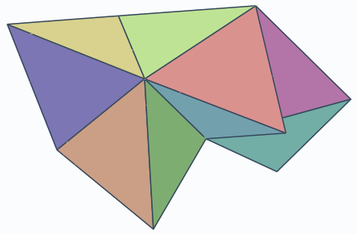
8-049



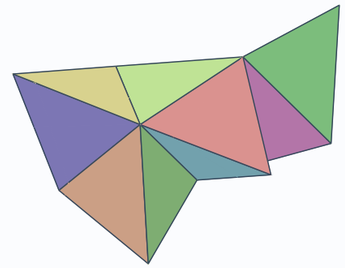
8-050



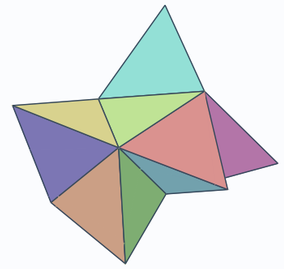
8-051



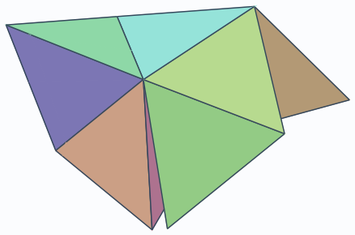
8-052



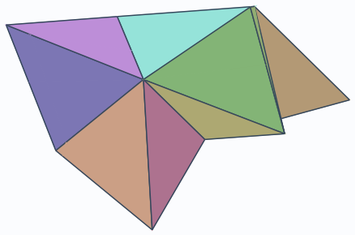
8-053



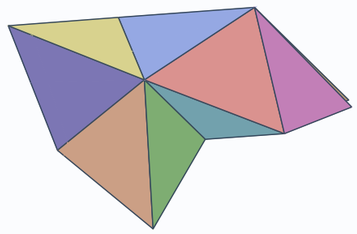
8-054



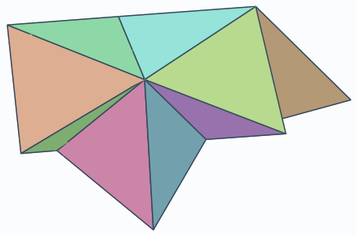
8-055



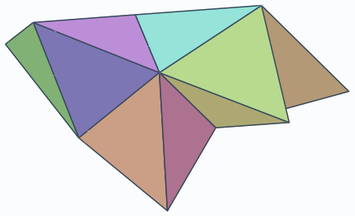
8-056



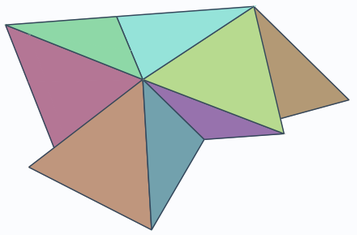
8-057



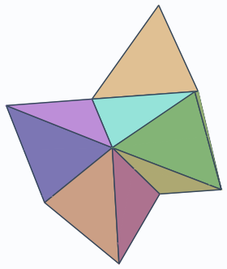
8-058



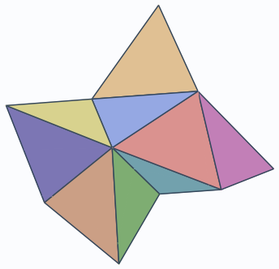
8-059



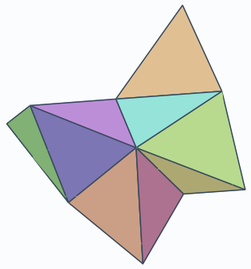
8-060



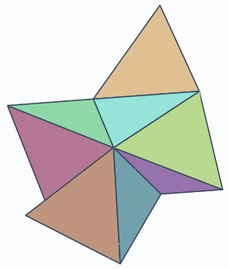
8-061



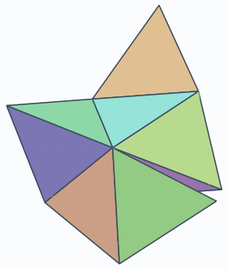
8-062



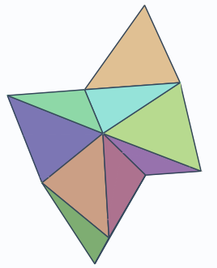
8-063



8-064



8-065



8-066

8-067

8-068

8-069

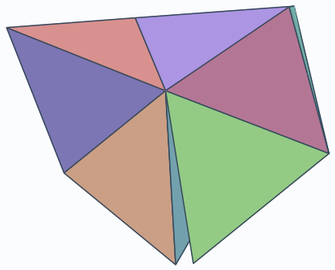
8-070

8-071

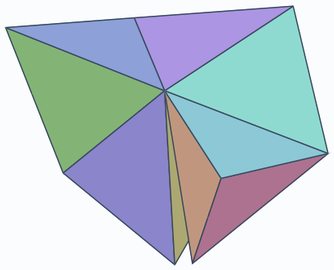
8-072

Complete catalog - 8 tetrahedra

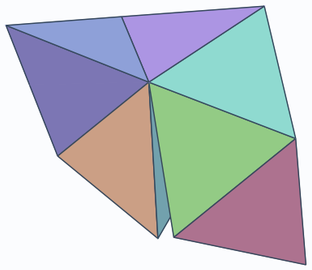
398 shapes in all; 18 triangular panels on each surface. Shapes 073-096.



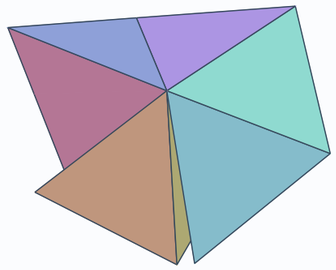
8-073



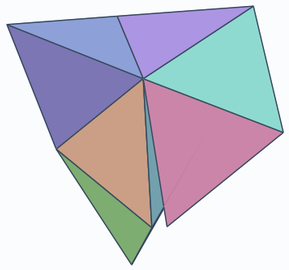
8-074



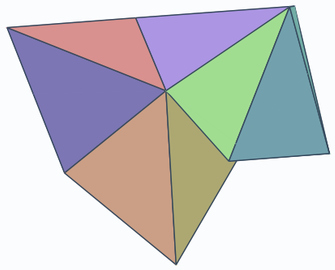
8-075



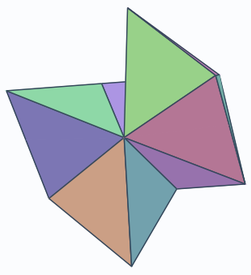
8-076



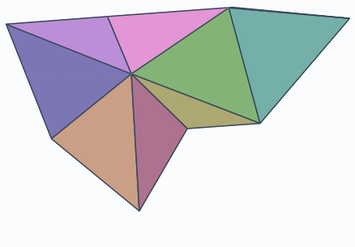
8-077



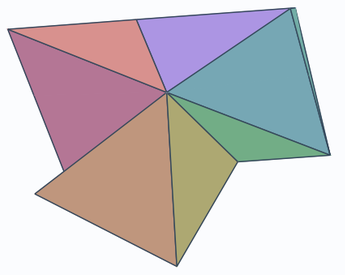
8-078



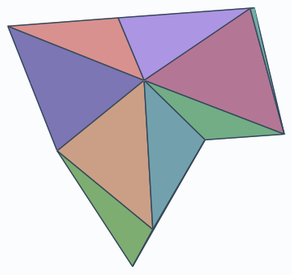
8-079



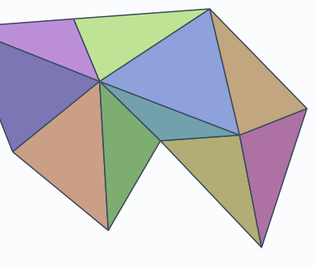
8-080



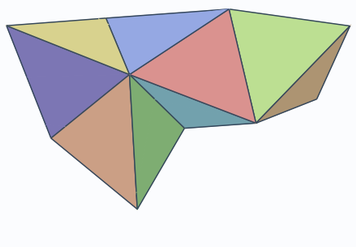
8-081



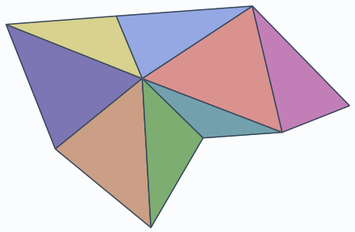
8-082



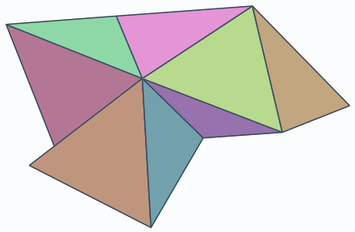
8-083



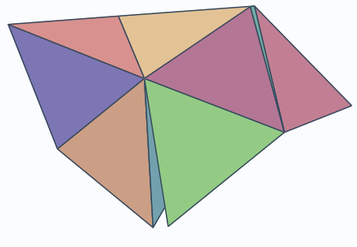
8-084



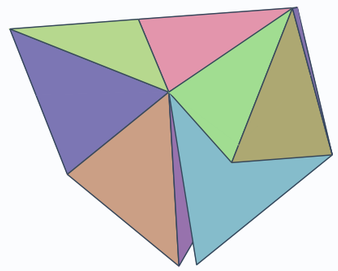
8-085



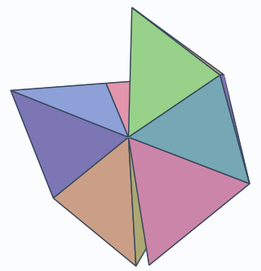
8-086



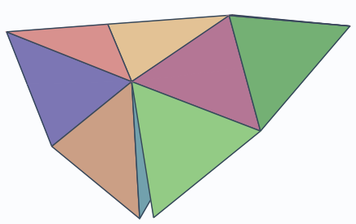
8-087



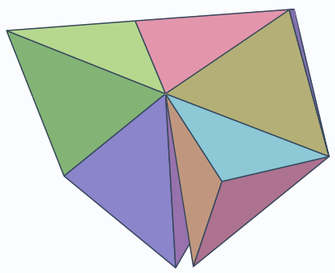
8-088



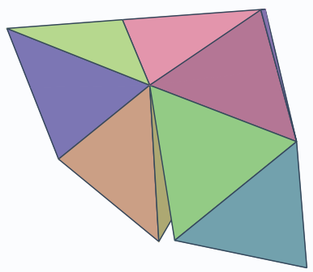
8-089



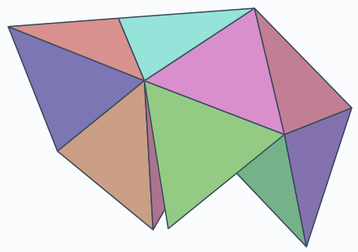
8-090



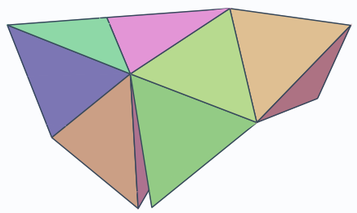
8-091



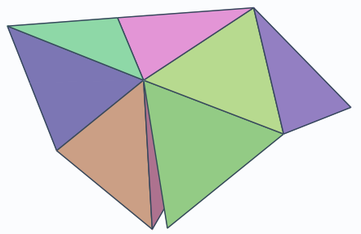
8-092



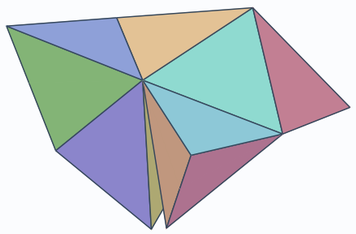
8-093



8-094



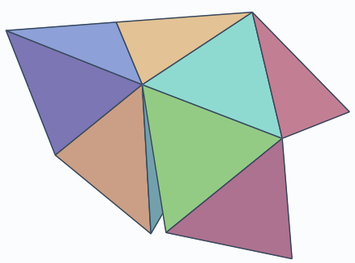
8-095



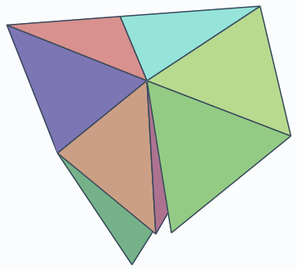
8-096

Complete catalog - 8 tetrahedra

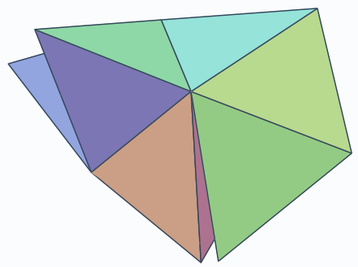
398 shapes in all; 18 triangular panels on each surface. Shapes 097-120.



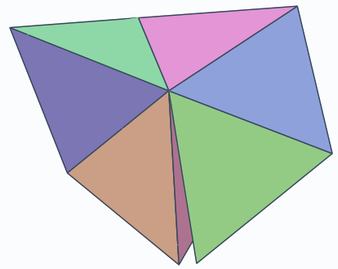
8-097



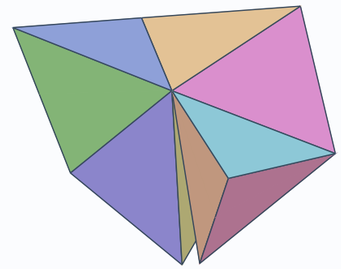
8-098



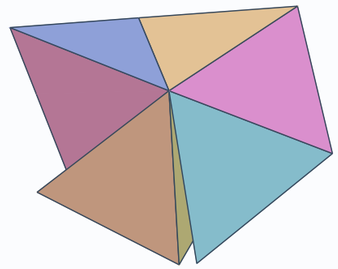
8-099



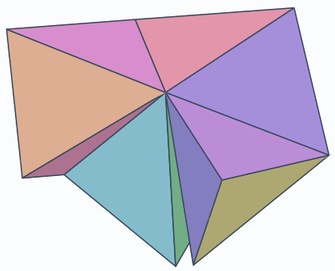
8-100



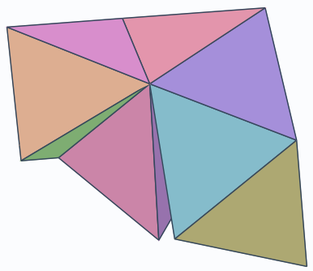
8-101



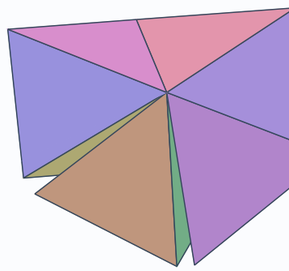
8-102



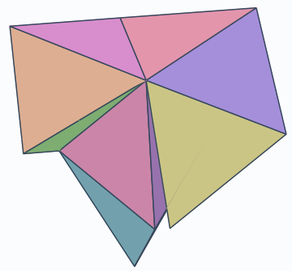
8-103



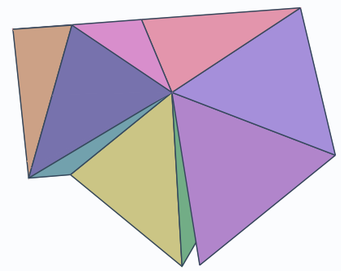
8-104



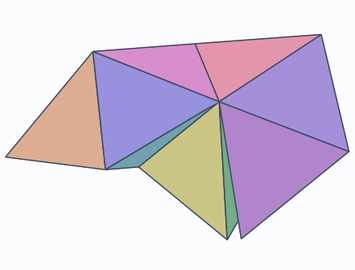
8-105



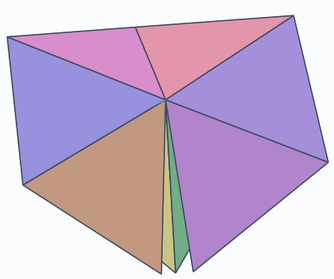
8-106



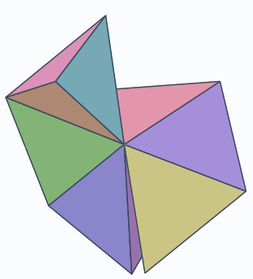
8-107



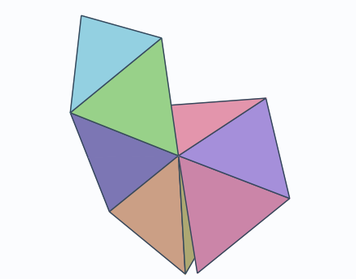
8-108



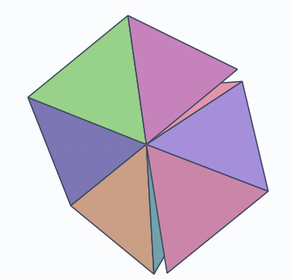
8-109



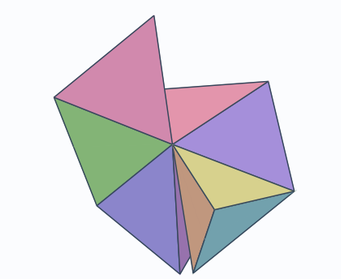
8-110



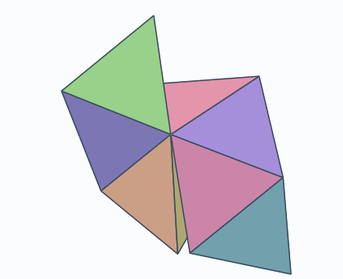
8-111



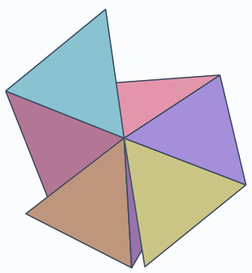
8-112



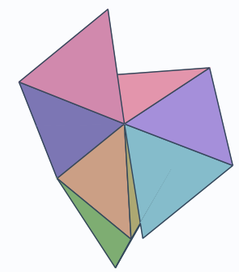
8-113



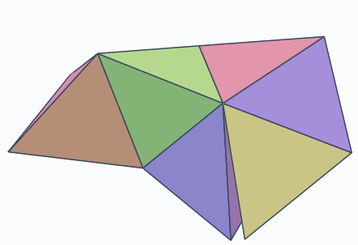
8-114



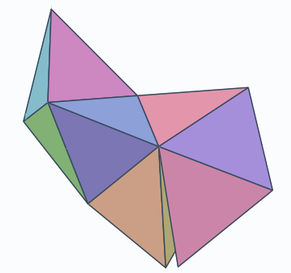
8-115



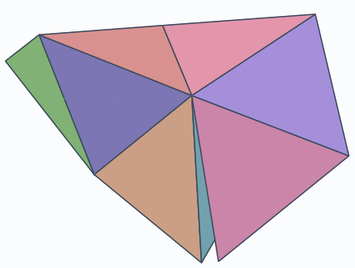
8-116



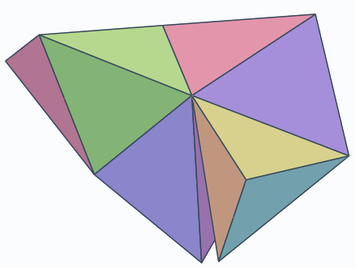
8-117



8-118



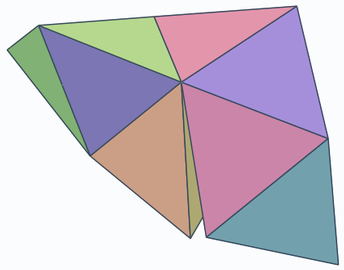
8-119



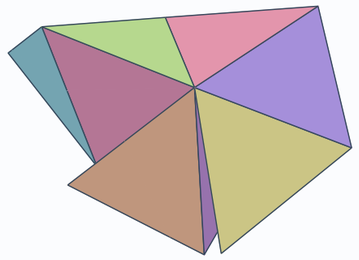
8-120

Complete catalog - 8 tetrahedra

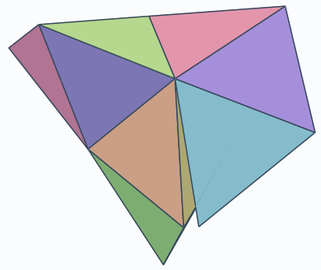
398 shapes in all; 18 triangular panels on each surface. Shapes 121-144.



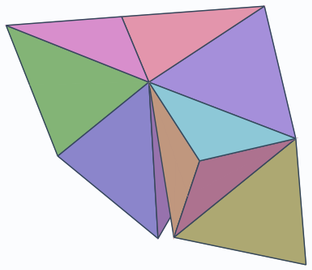
8-121



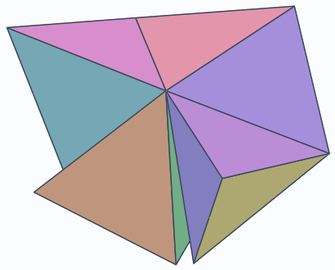
8-122



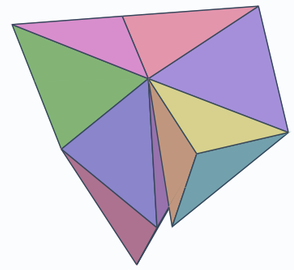
8-123



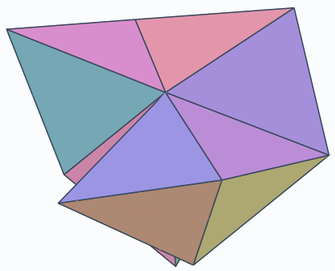
8-124



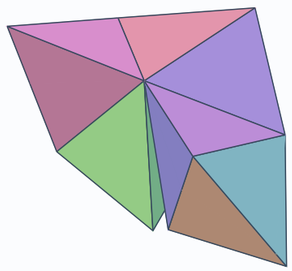
8-125



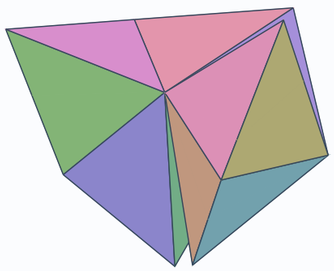
8-126



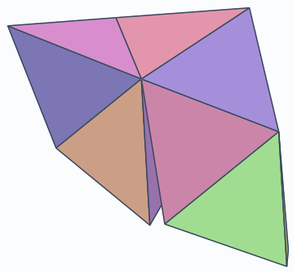
8-127



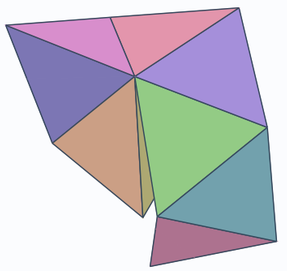
8-128



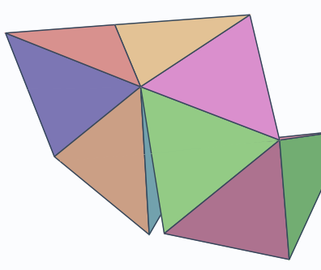
8-129



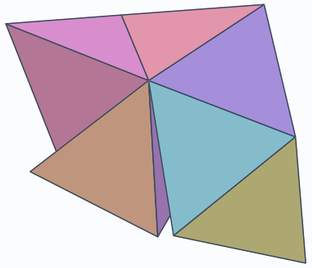
8-130



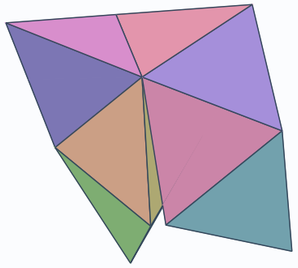
8-131



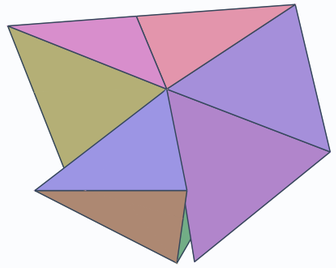
8-132



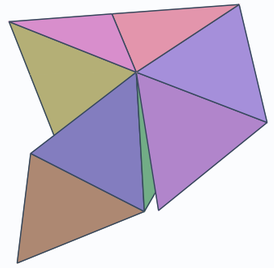
8-133



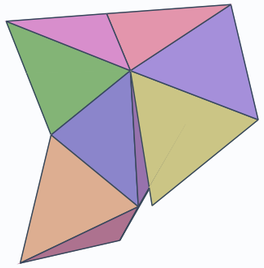
8-134



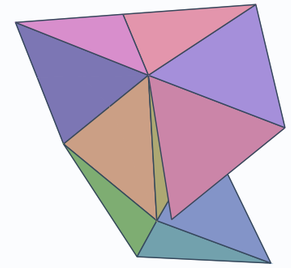
8-135



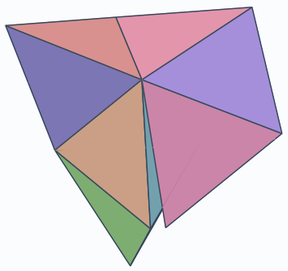
8-136



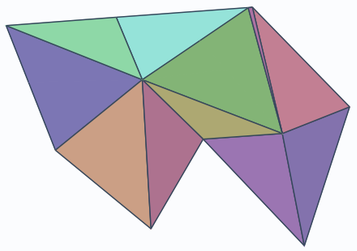
8-137



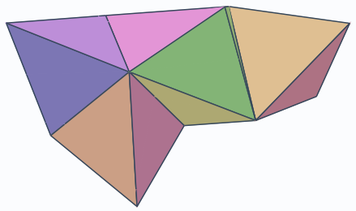
8-138



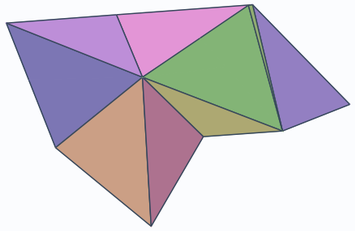
8-139



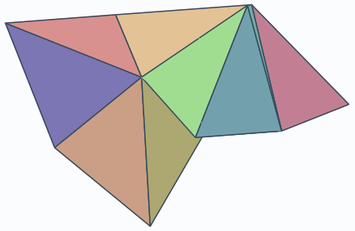
8-140



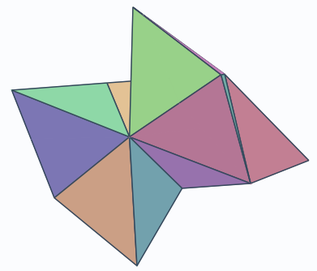
8-141



8-142



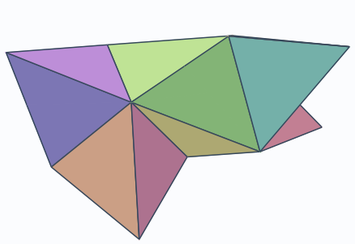
8-143



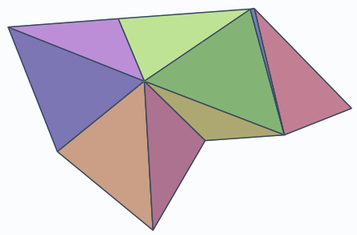
8-144

Complete catalog - 8 tetrahedra

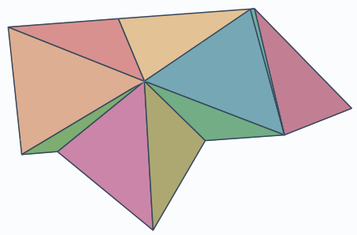
398 shapes in all; 18 triangular panels on each surface. Shapes 145-168.



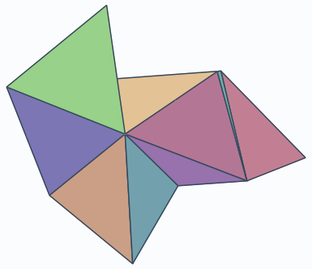
8-145



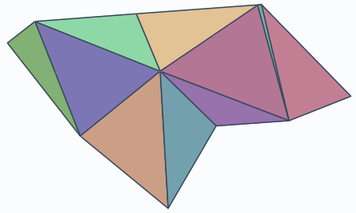
8-146



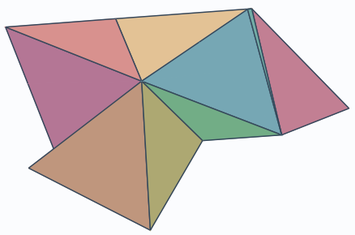
8-147



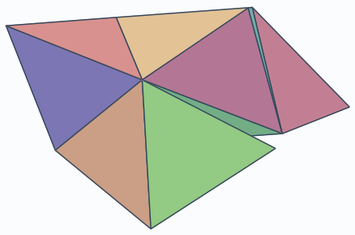
8-148



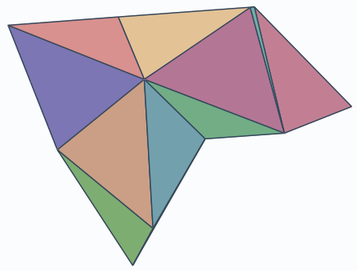
8-149



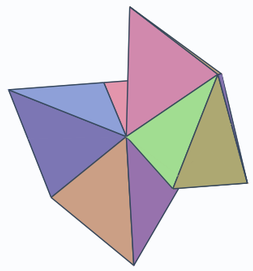
8-150



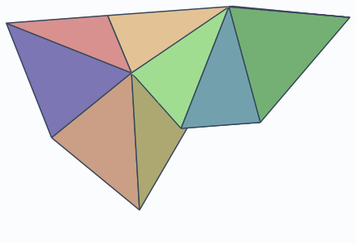
8-151



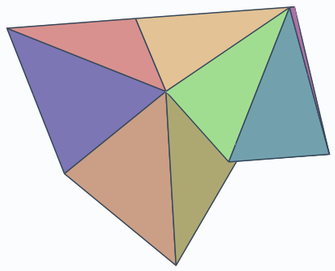
8-152



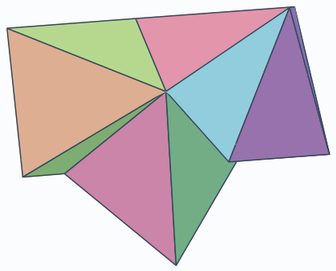
8-153



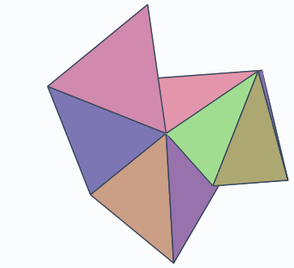
8-154



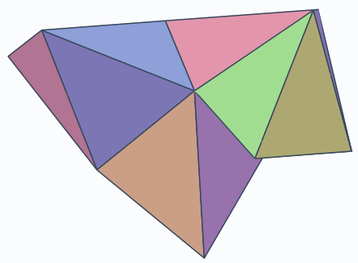
8-155



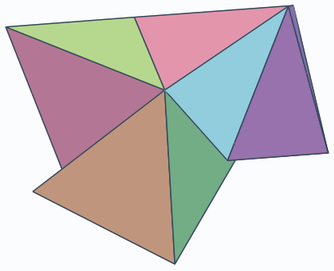
8-156



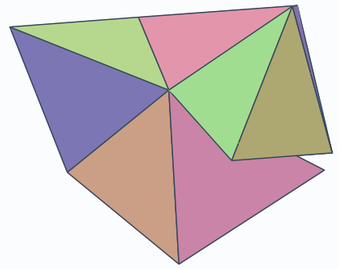
8-157



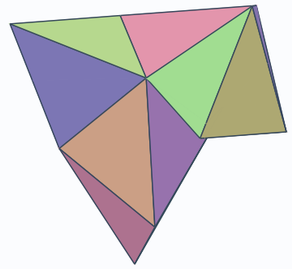
8-158



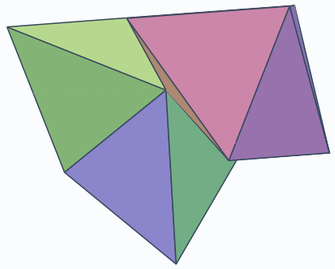
8-159



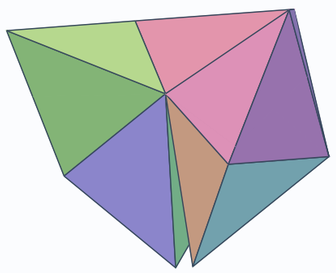
8-160



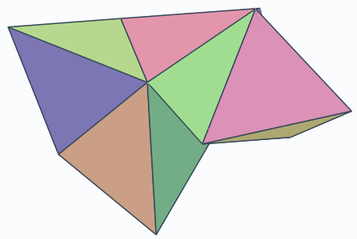
8-161



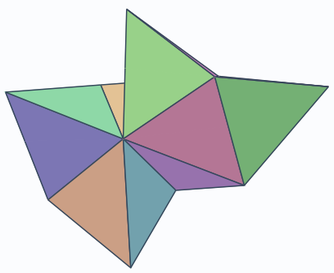
8-162



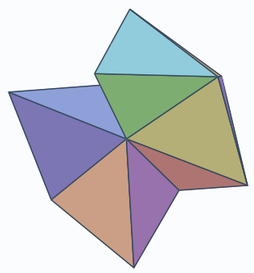
8-163



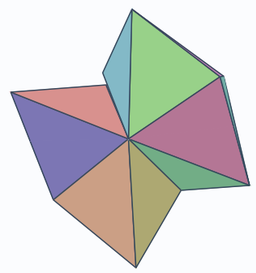
8-164



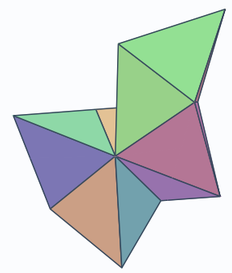
8-165



8-166



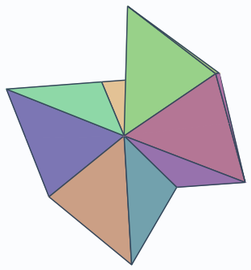
8-167



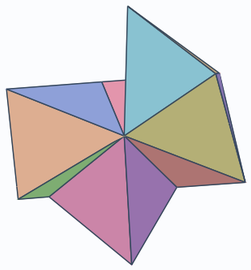
8-168

Complete catalog - 8 tetrahedra

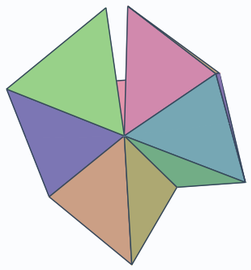
398 shapes in all; 18 triangular panels on each surface. Shapes 169-192.



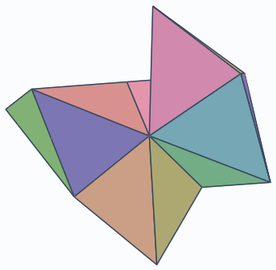
8-169



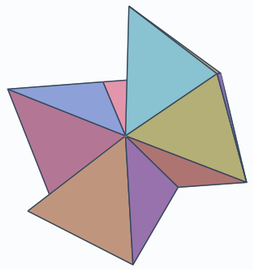
8-170



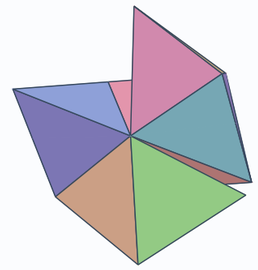
8-171



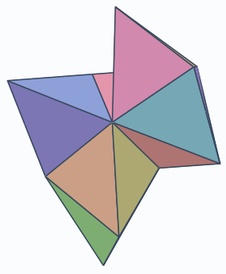
8-172



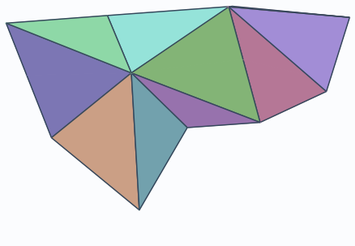
8-173



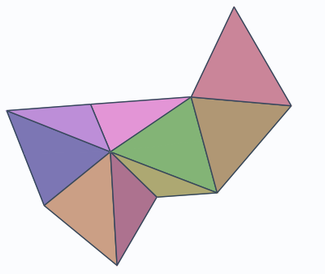
8-174



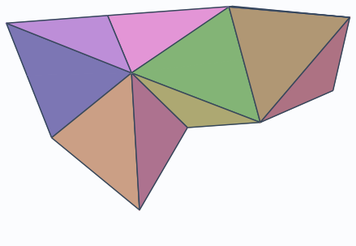
8-175



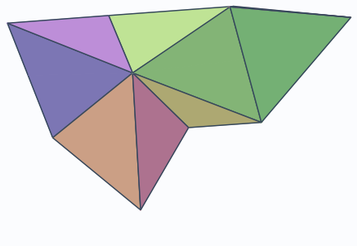
8-176



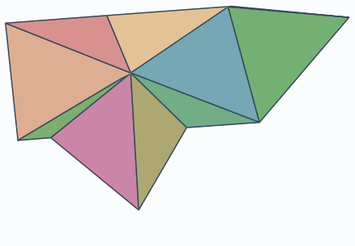
8-177



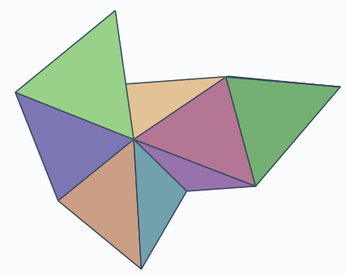
8-178



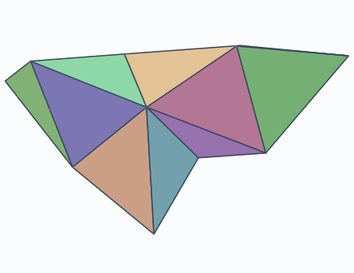
8-179



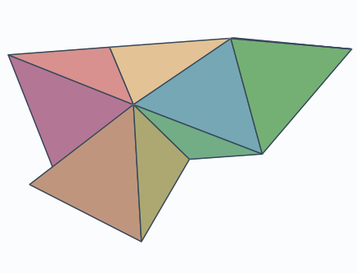
8-180



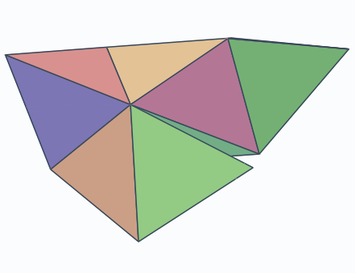
8-181



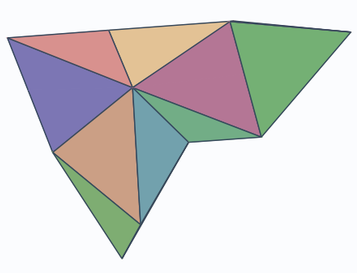
8-182



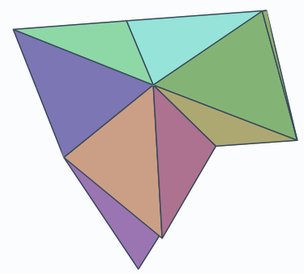
8-183



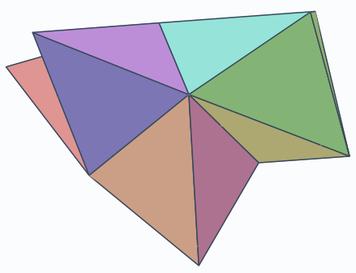
8-184



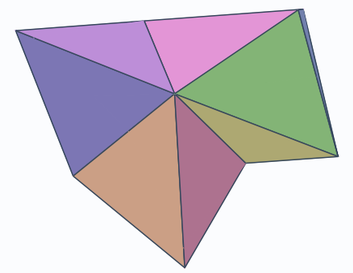
8-185



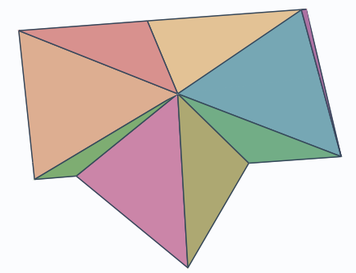
8-186



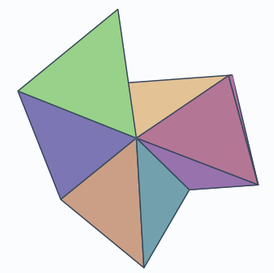
8-187



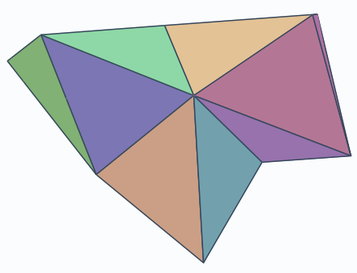
8-188



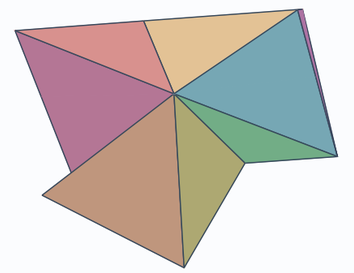
8-189



8-190



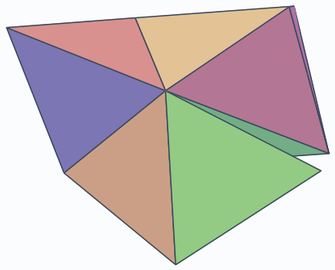
8-191



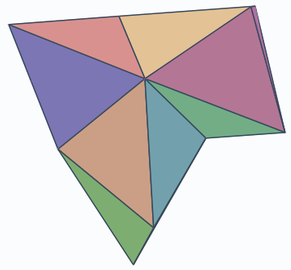
8-192

Complete catalog - 8 tetrahedra

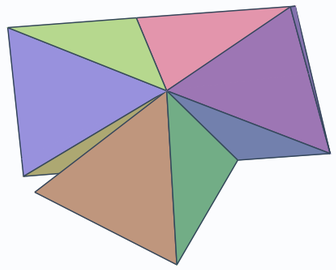
398 shapes in all; 18 triangular panels on each surface. Shapes 193-216.



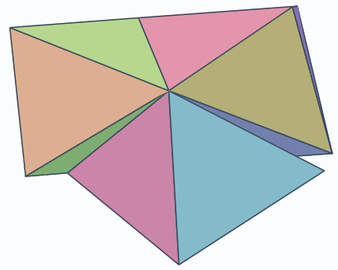
8-193



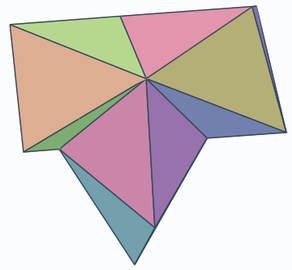
8-194



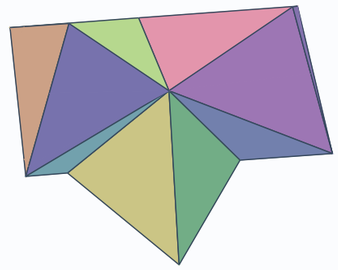
8-195



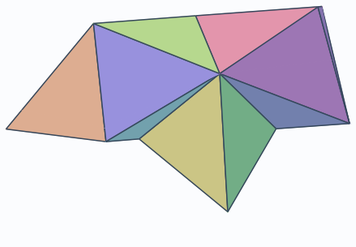
8-196



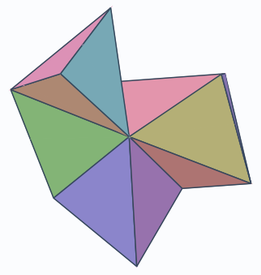
8-197



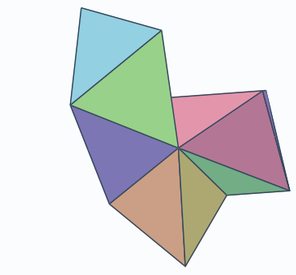
8-198



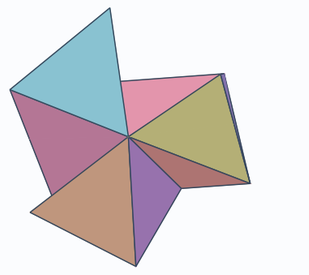
8-199



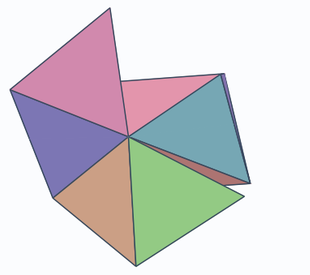
8-200



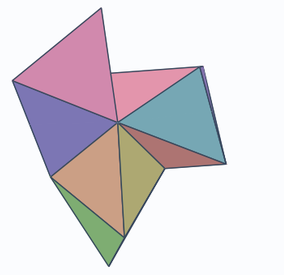
8-201



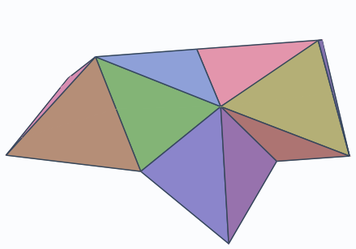
8-202



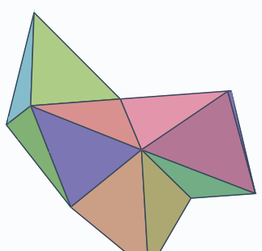
8-203



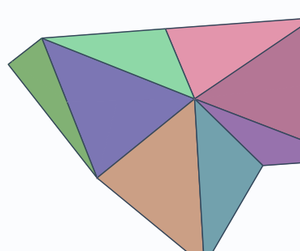
8-204



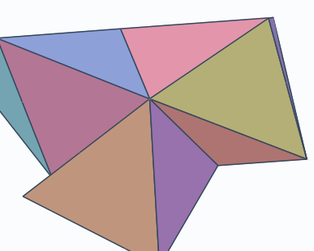
8-205



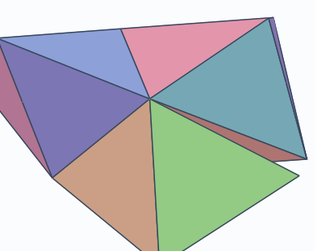
8-206



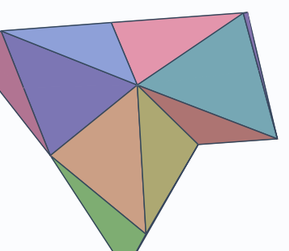
8-207



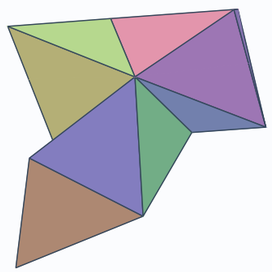
8-208



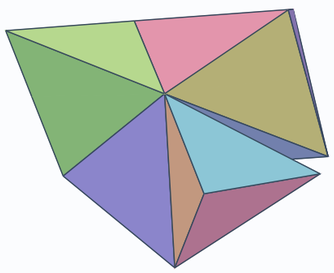
8-209



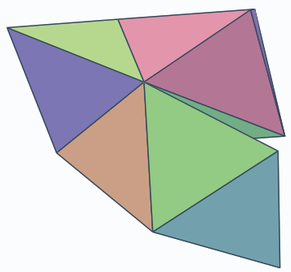
8-210



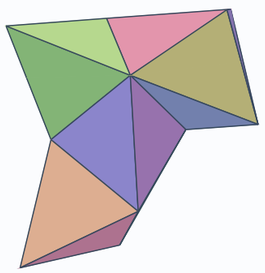
8-211



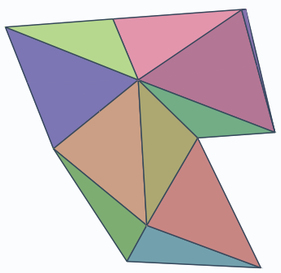
8-212



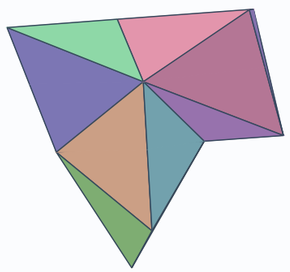
8-213



8-214



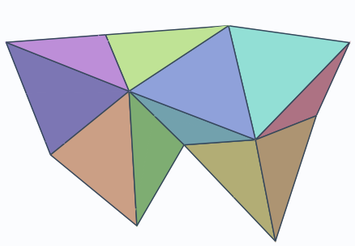
8-215



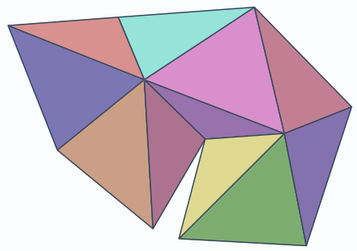
8-216

Complete catalog - 8 tetrahedra

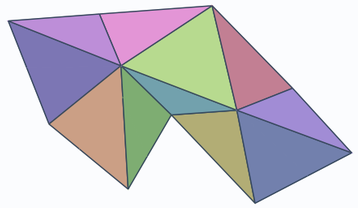
398 shapes in all; 18 triangular panels on each surface. Shapes 217-240.



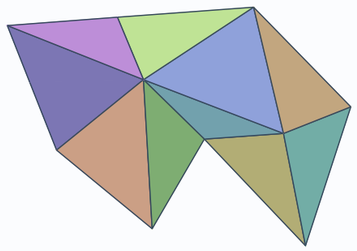
8-217



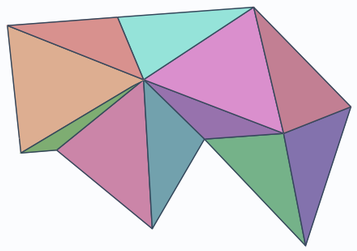
8-218



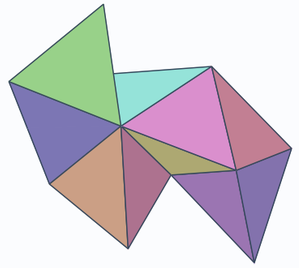
8-219



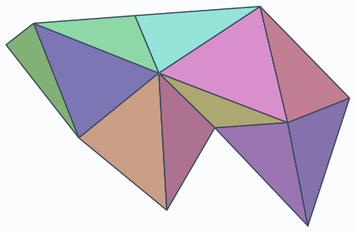
8-220



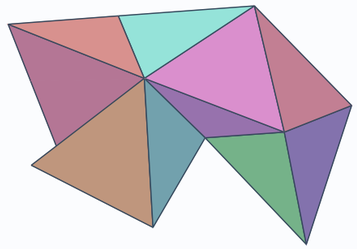
8-221



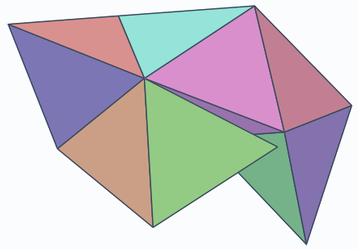
8-222



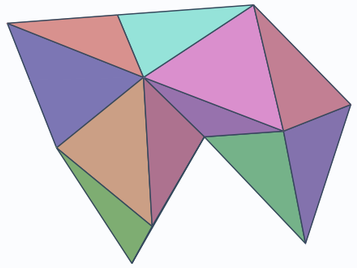
8-223



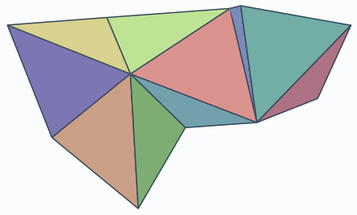
8-224



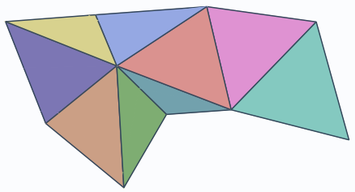
8-225



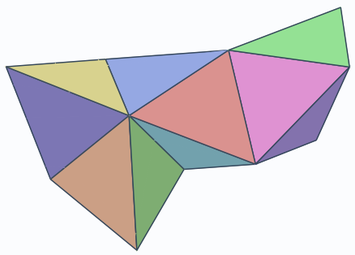
8-226



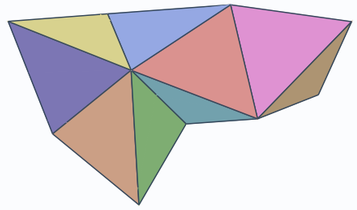
8-227



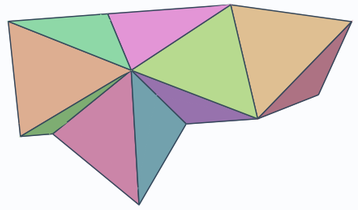
8-228



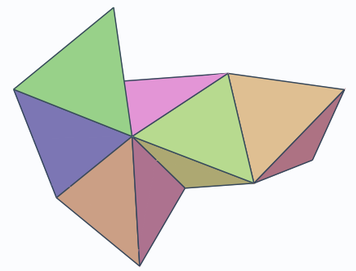
8-229



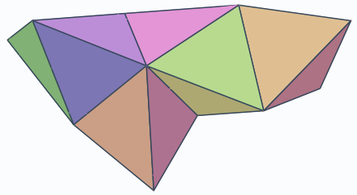
8-230



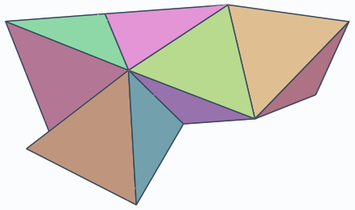
8-231



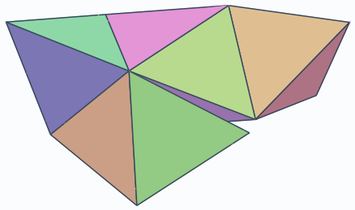
8-232



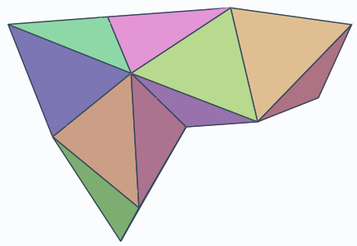
8-233



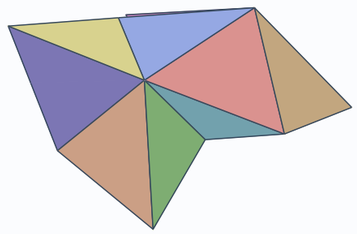
8-234



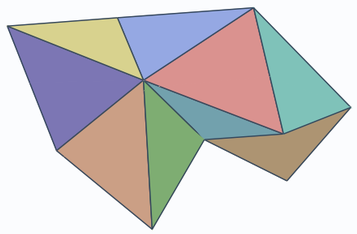
8-235



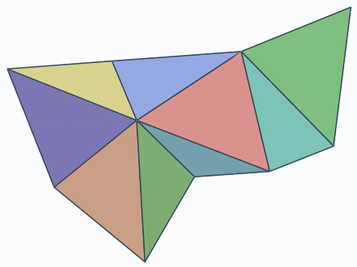
8-236



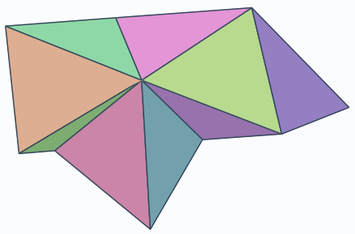
8-237



8-238



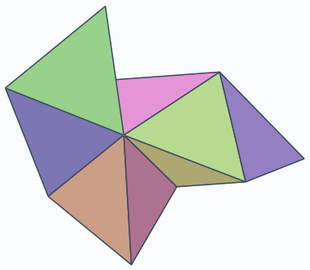
8-239



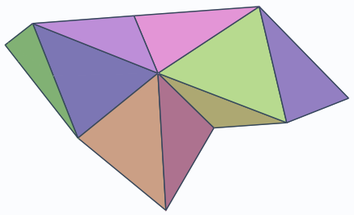
8-240

Complete catalog - 8 tetrahedra

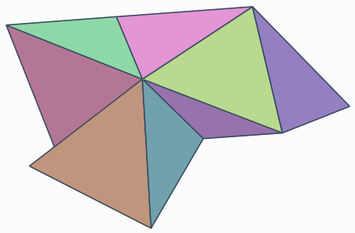
398 shapes in all; 18 triangular panels on each surface. Shapes 241-264.



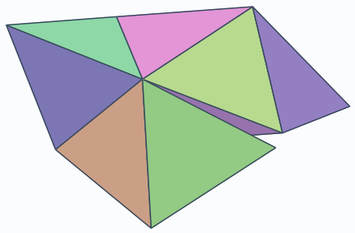
8-241



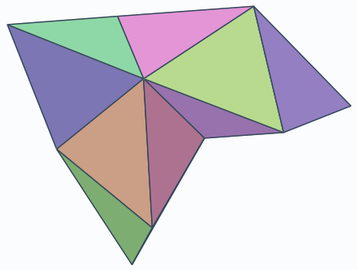
8-242



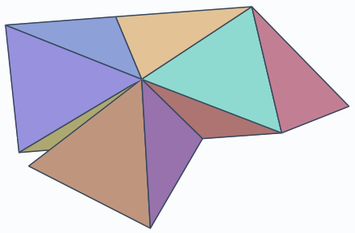
8-243



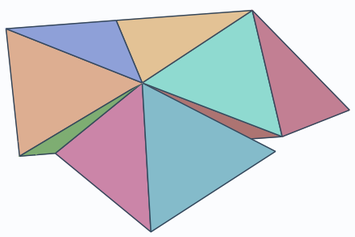
8-244



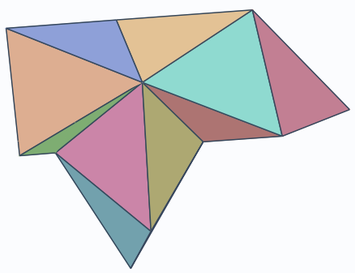
8-245



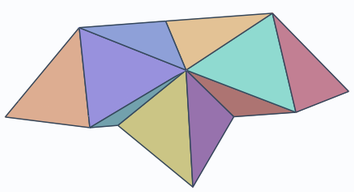
8-246



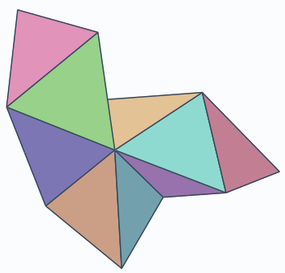
8-247



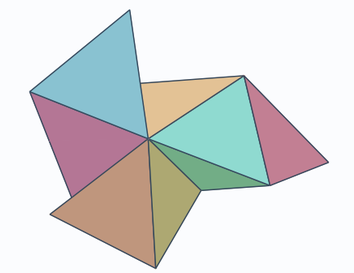
8-248



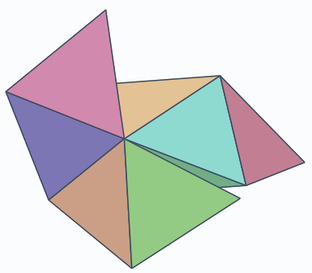
8-249



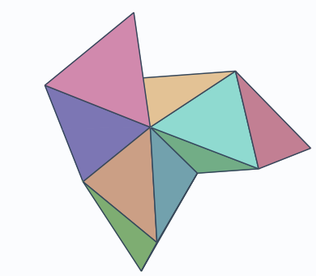
8-250



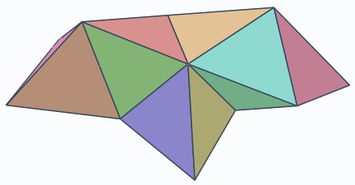
8-251



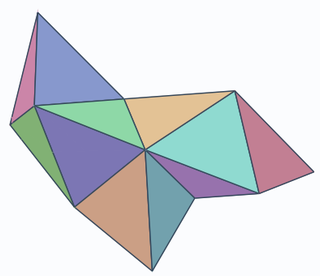
8-252



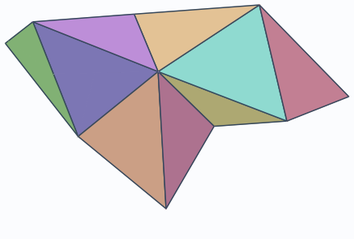
8-253



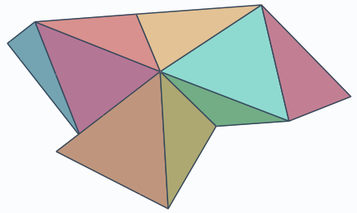
8-254



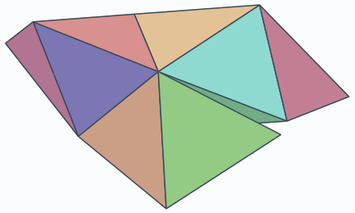
8-255



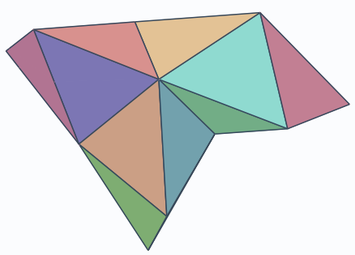
8-256



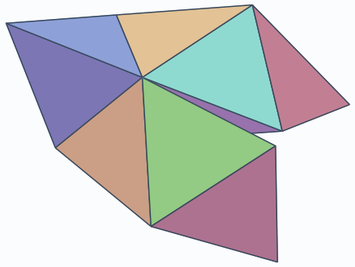
8-257



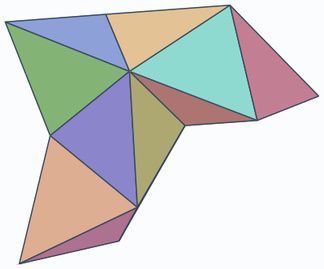
8-258



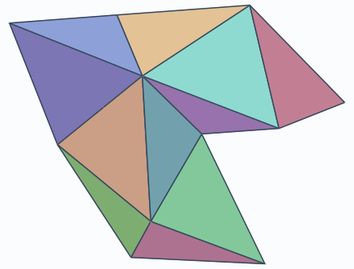
8-259



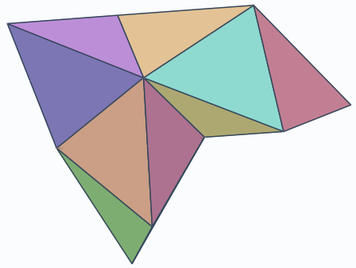
8-260



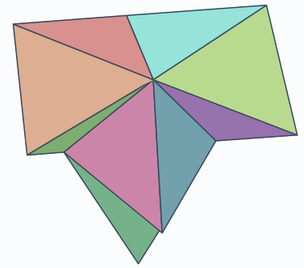
8-261



8-262



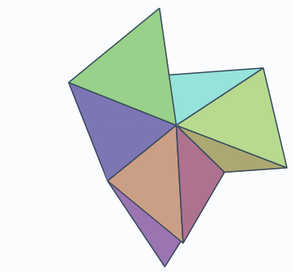
8-263



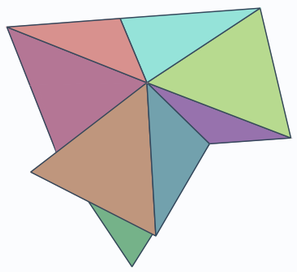
8-264

Complete catalog - 8 tetrahedra

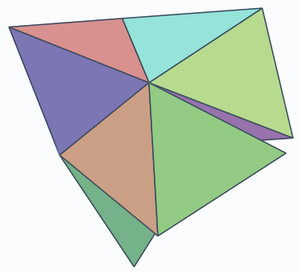
398 shapes in all; 18 triangular panels on each surface. Shapes 265-288.



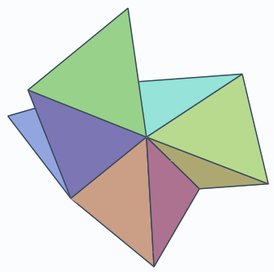
8-265



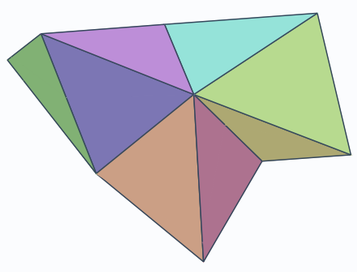
8-266



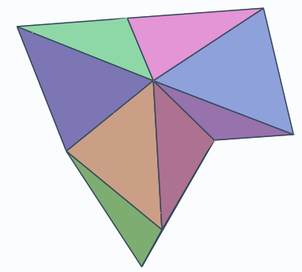
8-267



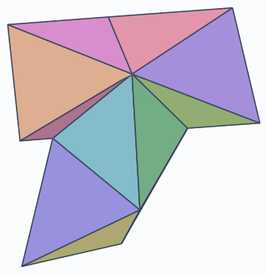
8-268



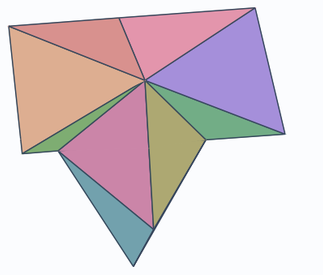
8-269



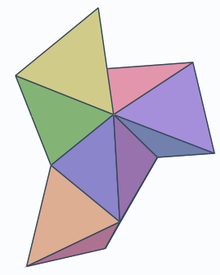
8-270



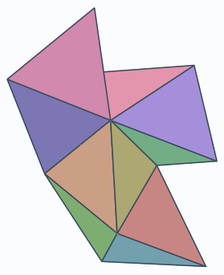
8-271



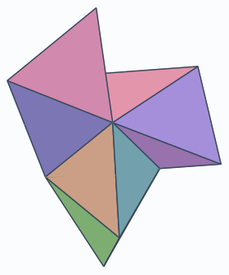
8-272



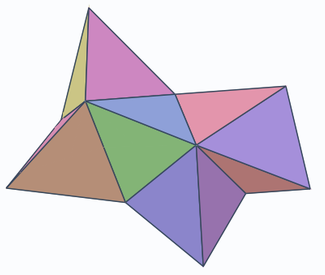
8-273



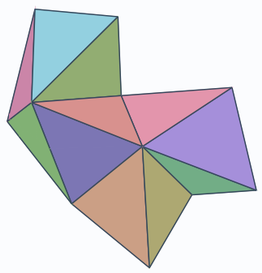
8-274



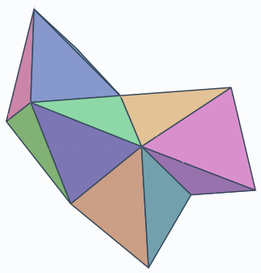
8-275



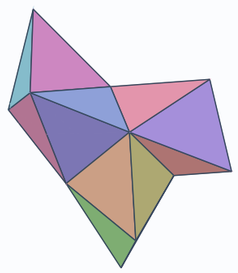
8-276



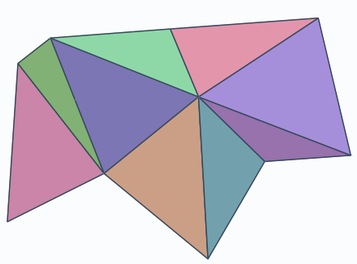
8-277



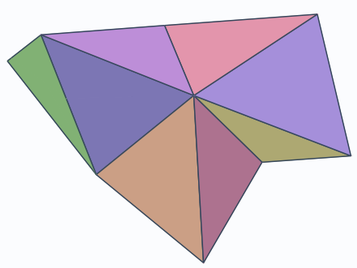
8-278



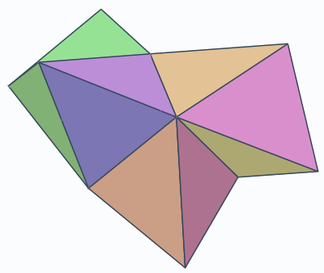
8-279



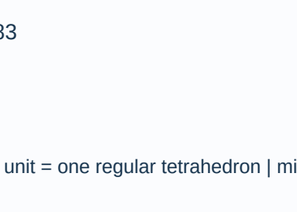
8-280



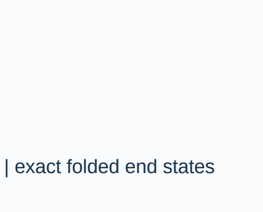
8-281



8-282



8-283



8-284

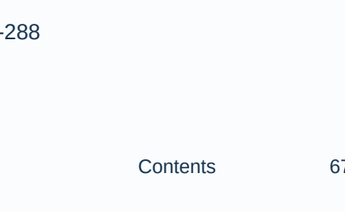
8-285



8-286



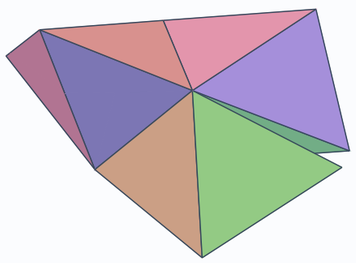
8-287



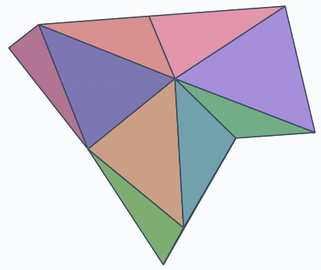
8-288

Complete catalog - 8 tetrahedra

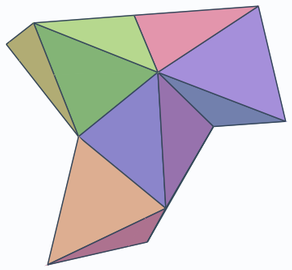
398 shapes in all; 18 triangular panels on each surface. Shapes 289-312.



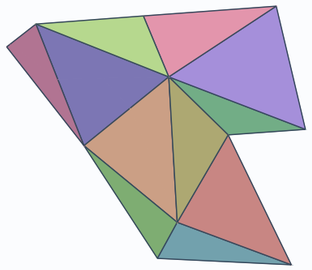
8-289



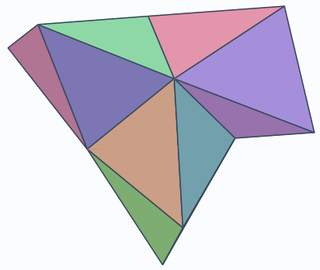
8-290



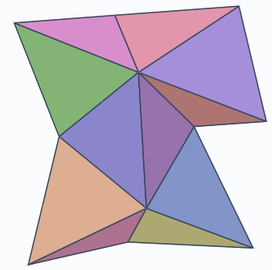
8-291



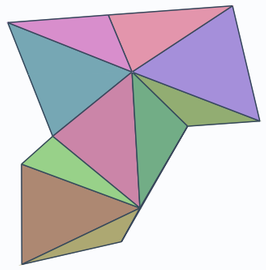
8-292



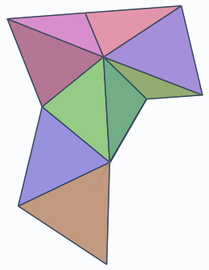
8-293



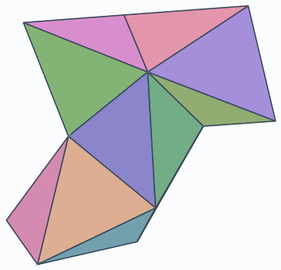
8-294



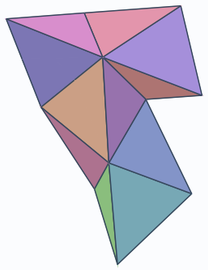
8-295



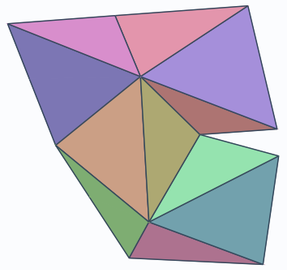
8-296



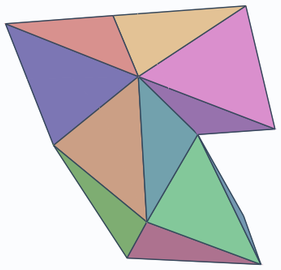
8-297



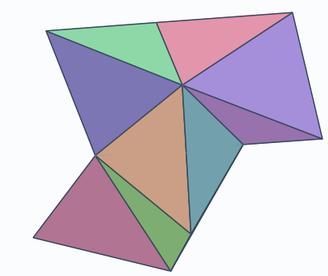
8-298



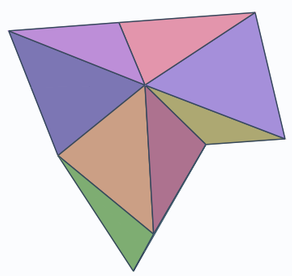
8-299



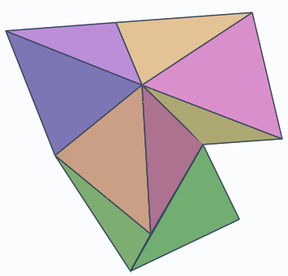
8-300



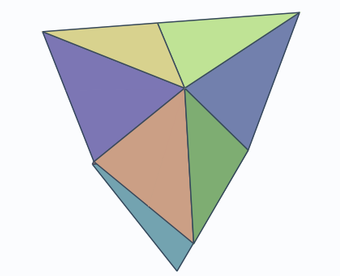
8-301



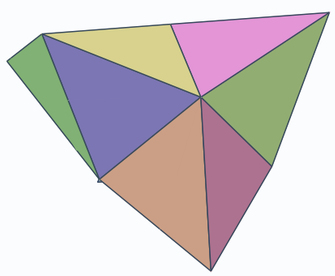
8-302



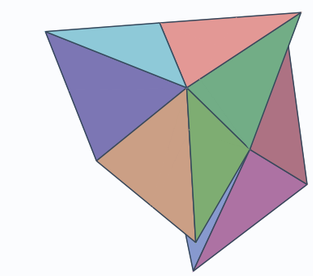
8-303



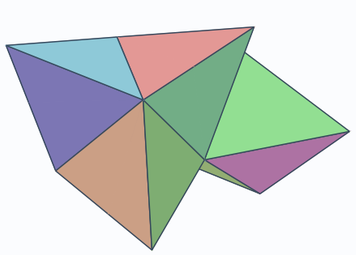
8-304



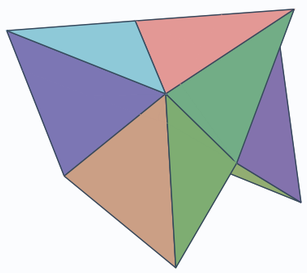
8-305



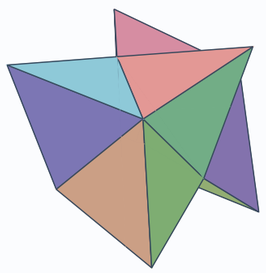
8-306



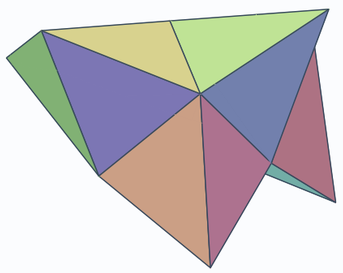
8-307



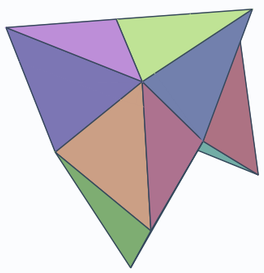
8-308



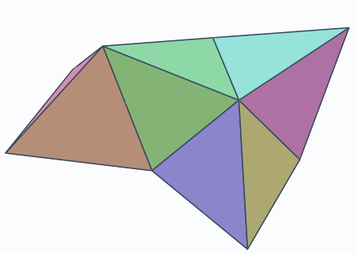
8-309



8-310



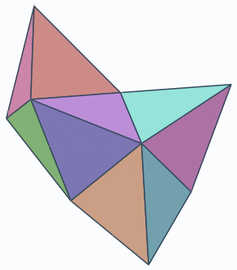
8-311



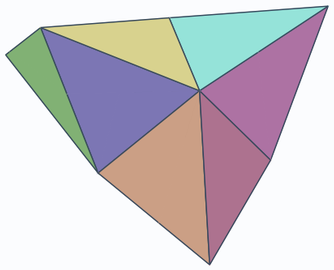
8-312

Complete catalog - 8 tetrahedra

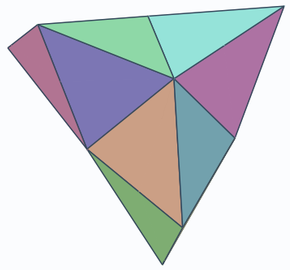
398 shapes in all; 18 triangular panels on each surface. Shapes 313-336.



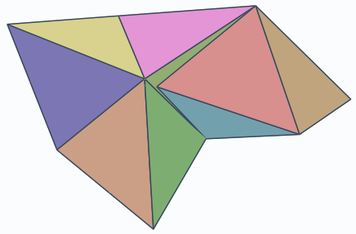
8-313



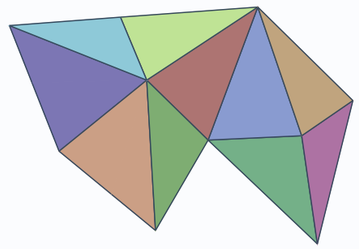
8-314



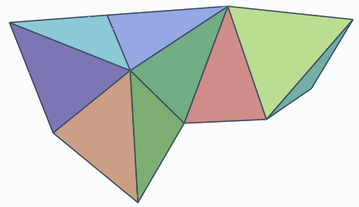
8-315



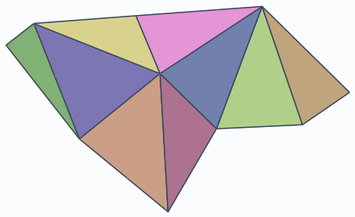
8-316



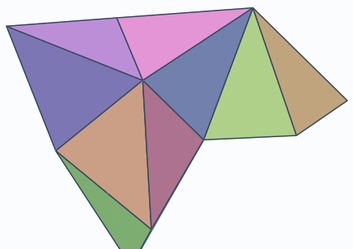
8-317



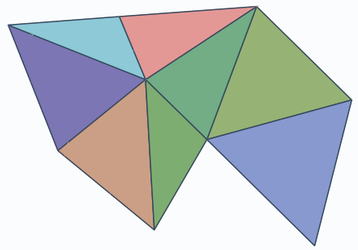
8-318



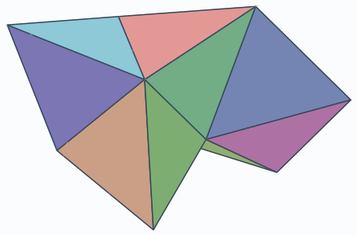
8-319



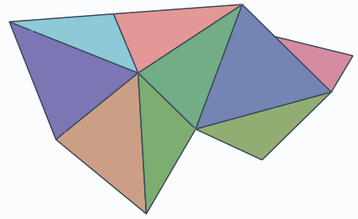
8-320



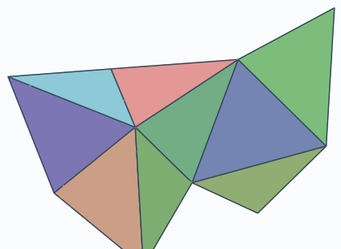
8-321



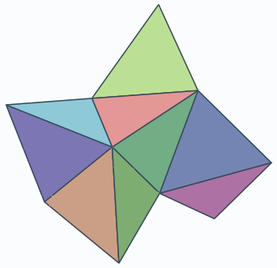
8-322



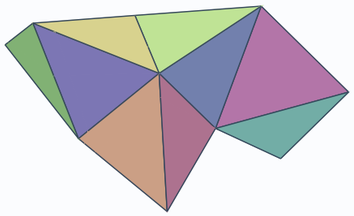
8-323



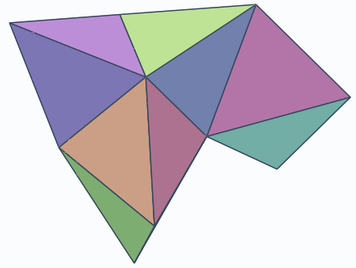
8-324



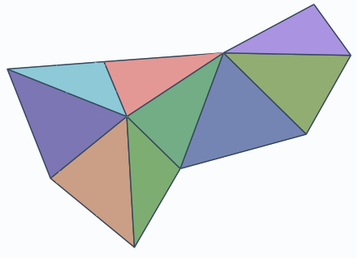
8-325



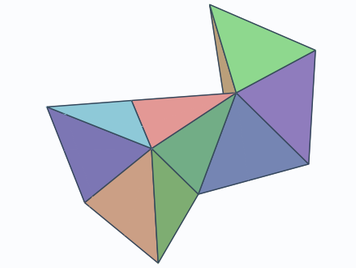
8-326



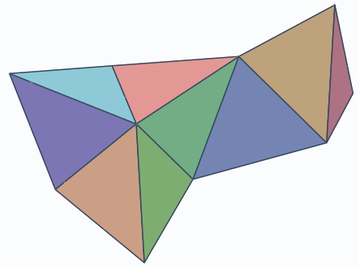
8-327



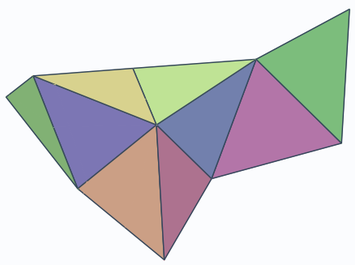
8-328



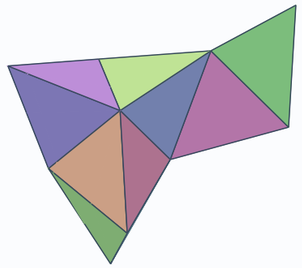
8-329



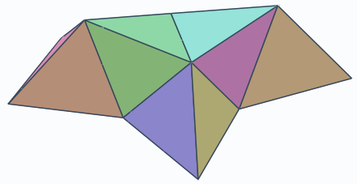
8-330



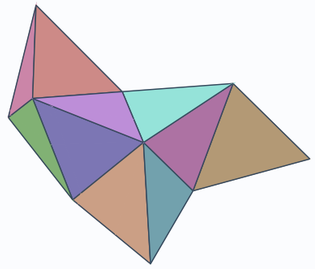
8-331



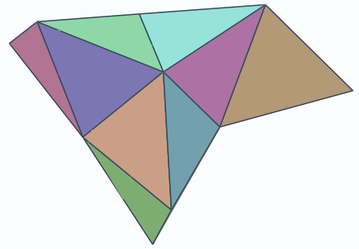
8-332



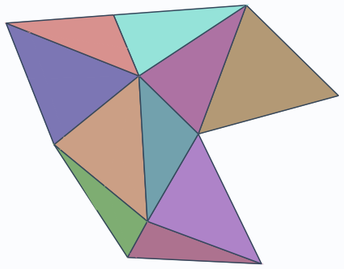
8-333



8-334



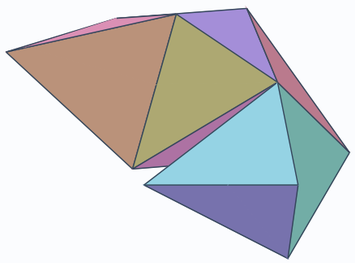
8-335



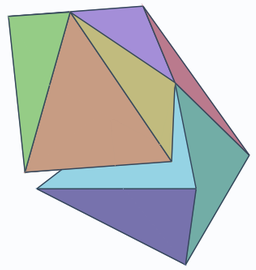
8-336

Complete catalog - 8 tetrahedra

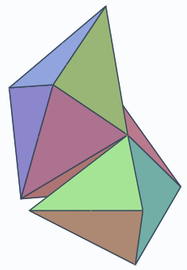
398 shapes in all; 18 triangular panels on each surface. Shapes 337-360.



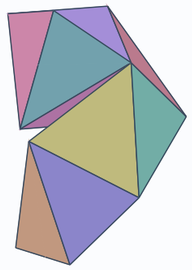
8-337



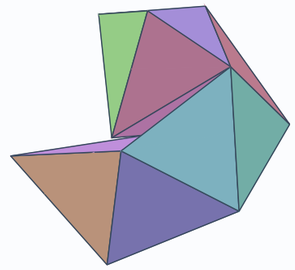
8-338



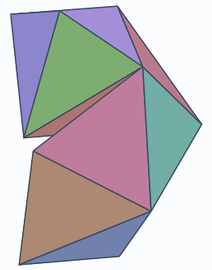
8-339



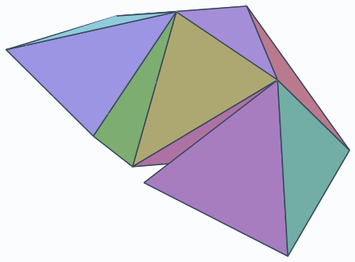
8-340



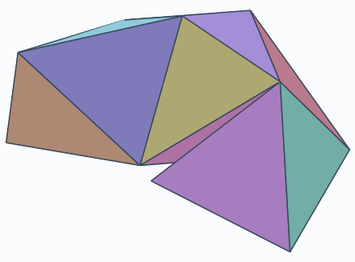
8-341



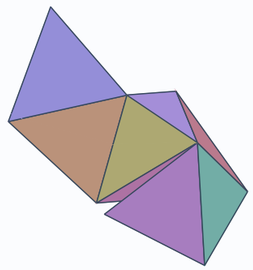
8-342



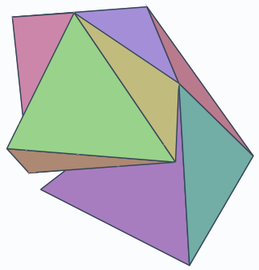
8-343



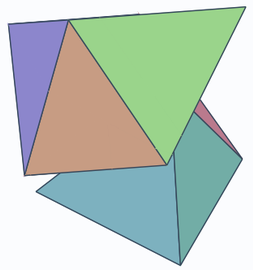
8-344



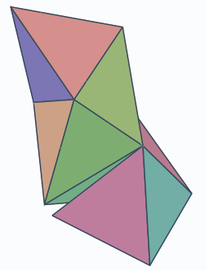
8-345



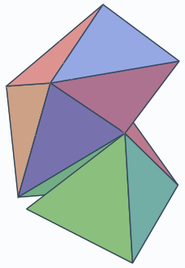
8-346



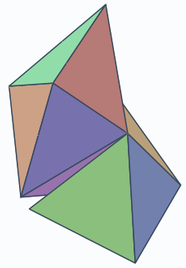
8-347



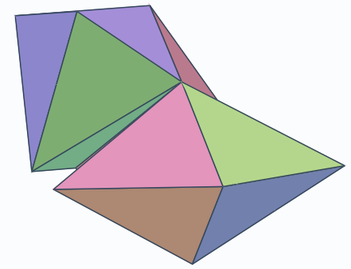
8-348



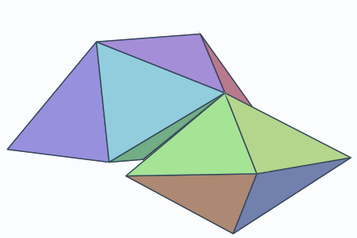
8-349



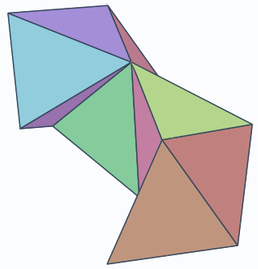
8-350



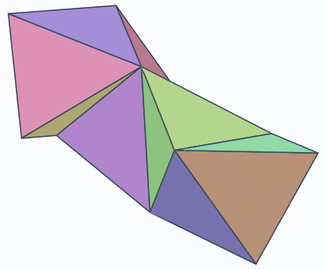
8-351



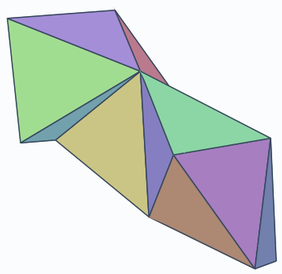
8-352



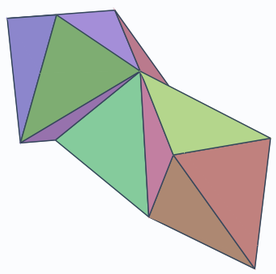
8-353



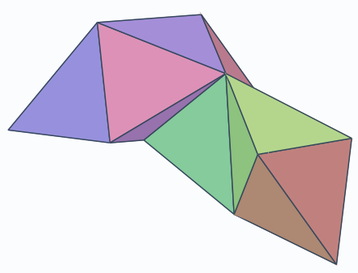
8-354



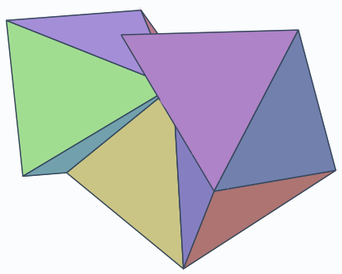
8-355



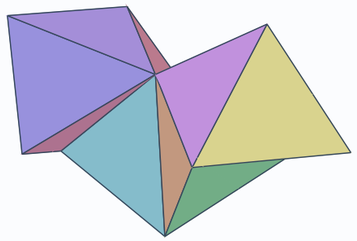
8-356



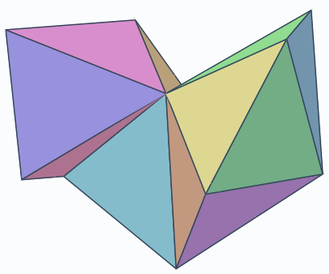
8-357



8-358



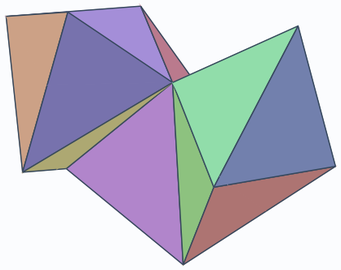
8-359



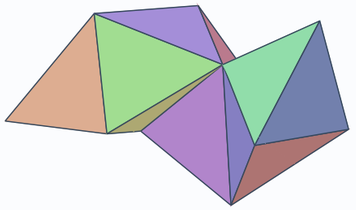
8-360

Complete catalog - 8 tetrahedra

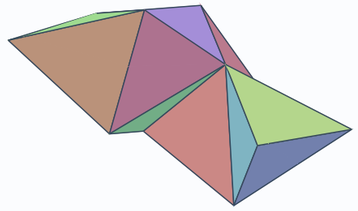
398 shapes in all; 18 triangular panels on each surface. Shapes 361-384.



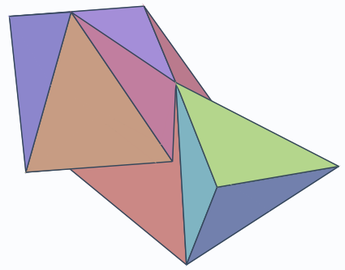
8-361



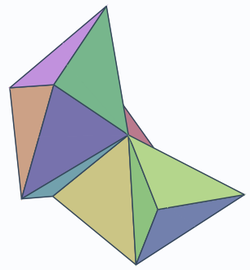
8-362



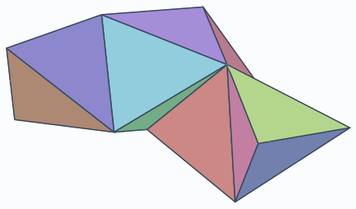
8-363



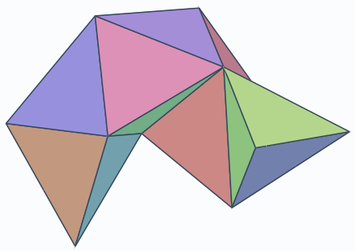
8-364



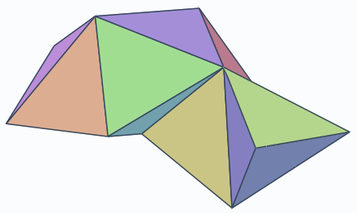
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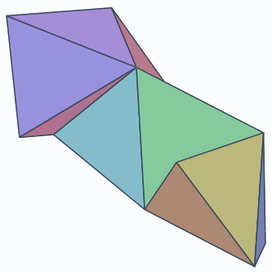
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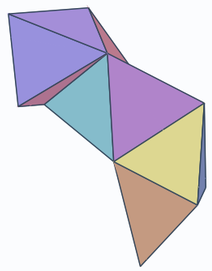
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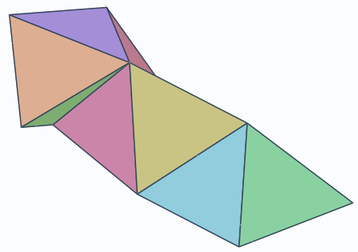
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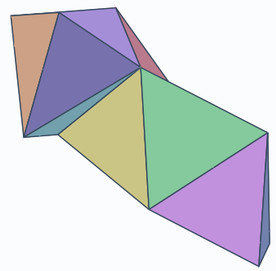
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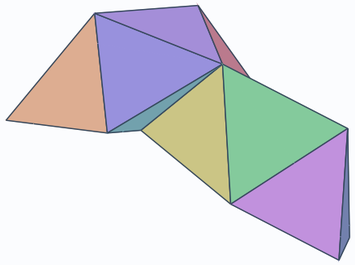
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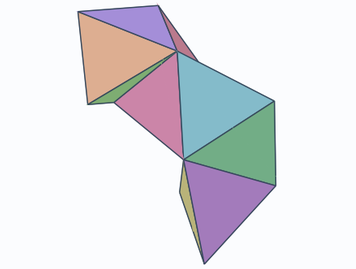
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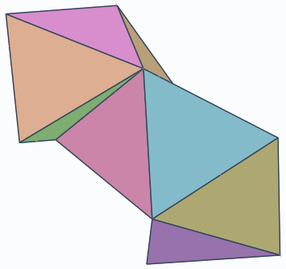
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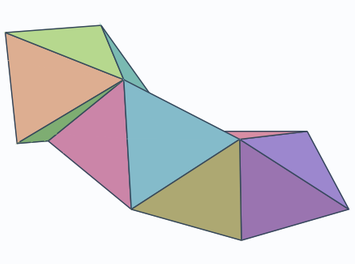
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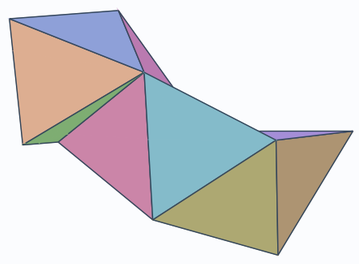
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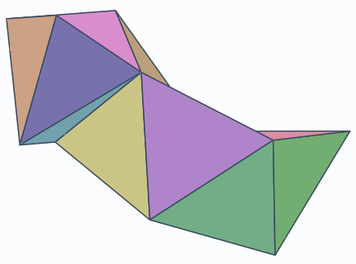
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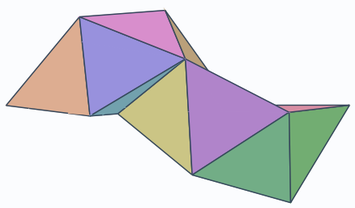
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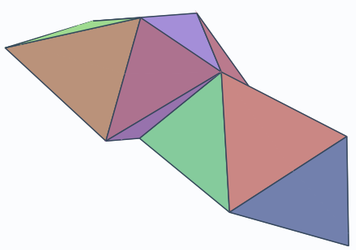
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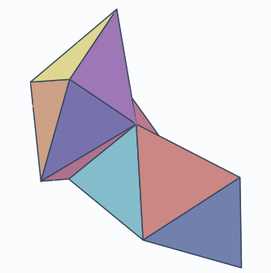
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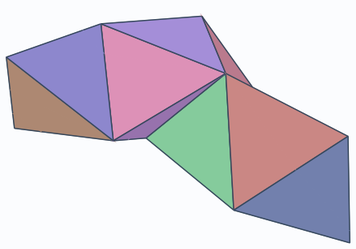
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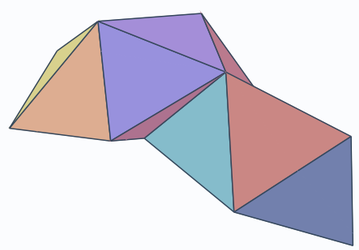
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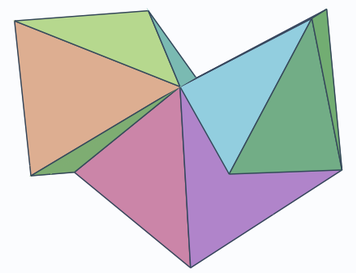
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8-382



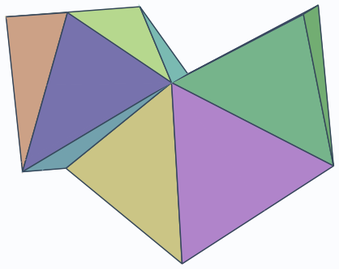
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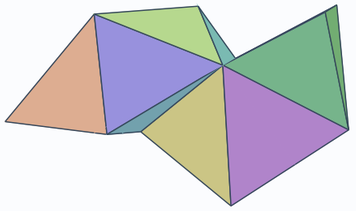
8-384

Complete catalog - 8 tetrahedra

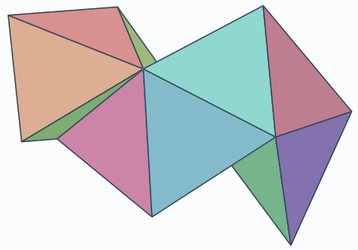
398 shapes in all; 18 triangular panels on each surface. Shapes 385-398.



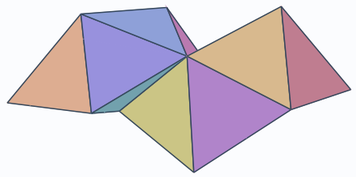
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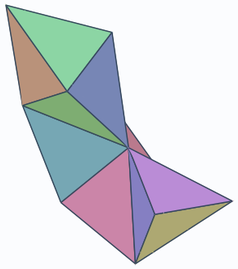
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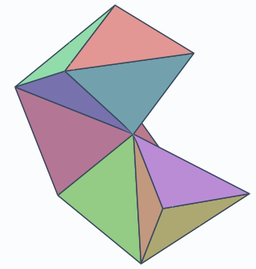
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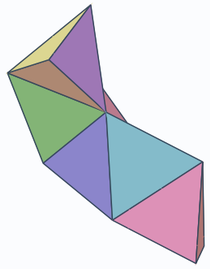
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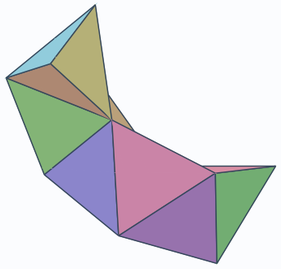
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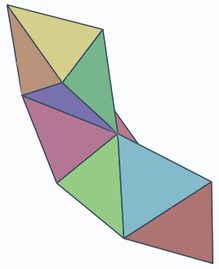
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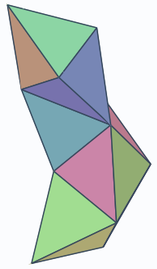
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8-392



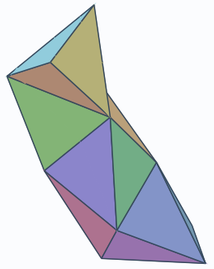
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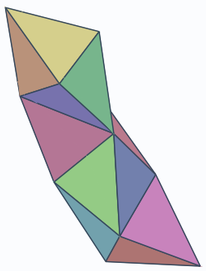
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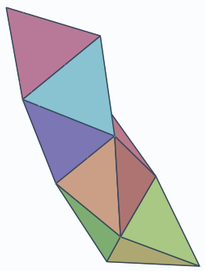
8-395



8-396



8-397



8-398

Checks, scope, and sources

A complete catalog is different from an exhaustive search for the best possible puzzle net.

Catalog. All 523 clusters through eight units are enumerated by attaching a regular tetrahedron to an exposed whole face. Integer arithmetic checks disjoint interiors and exact congruence, with mirrors identified. The counts 1,1,1,3,7,23,89,398 agree with the published polytet catalog. Each surface has 2n+2 exterior triangular panels at these volumes.

Puzzle checks. Each selected net is tested against every shape at its volume. A bijection to all exterior faces and equality of both endpoints of every hinge are checked exactly. A second, vertex-based folding implementation agrees. Only certified matching shapes appear in each puzzle's list.

Search scope. The six-unit 19-target maximum has a finite computational upper-bound certificate in this fixed-cut model. Seven- and eight-unit examples come from a deterministic bounded sample of source trees. Their complete coverage lists are exact, but those lists are not claims of optimality. Other interesting nets may cover different or larger collections.

Physical scope. Fixed internal slits and either side of the paper are allowed. There are no extra creases through triangle interiors, hidden layers, stretching, or panel overlaps. The checks establish folded end states, not continuous collision-free assembly. Manufacturing clearances, tabs, magnets and material thickness are outside this mathematical template.

Prior work. Rick Mabry's four-unit common nets and five-unit cobra net predate this experiment. The complete four-unit catalog agrees with our 37 outlines / 47 fixed-cut variants; the pictured 16 are a selected subset. No publication priority is claimed for the larger results.

Sources. George Sicherman: Catalogue of Polytets. Rick Mabry: four-unit common nets. Rick Mabry: twelve-face common nets.

Files. The PDF embeds the exact catalog, selected puzzle certificates, and a page manifest. The companion offline gallery lets you rotate every shape and download individual SVG nets. Open index.html after unzipping. No network connection is required.